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Dental Caries: Etiology, Clinical Characteristics, Risk Assessment, and Management

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This chapter presents basic definitions and information on dental caries, clinical characteristics of the caries lesion, caries risk assessment, and caries management, in the context of clinical operative dentistry.

What Is Dental Caries?

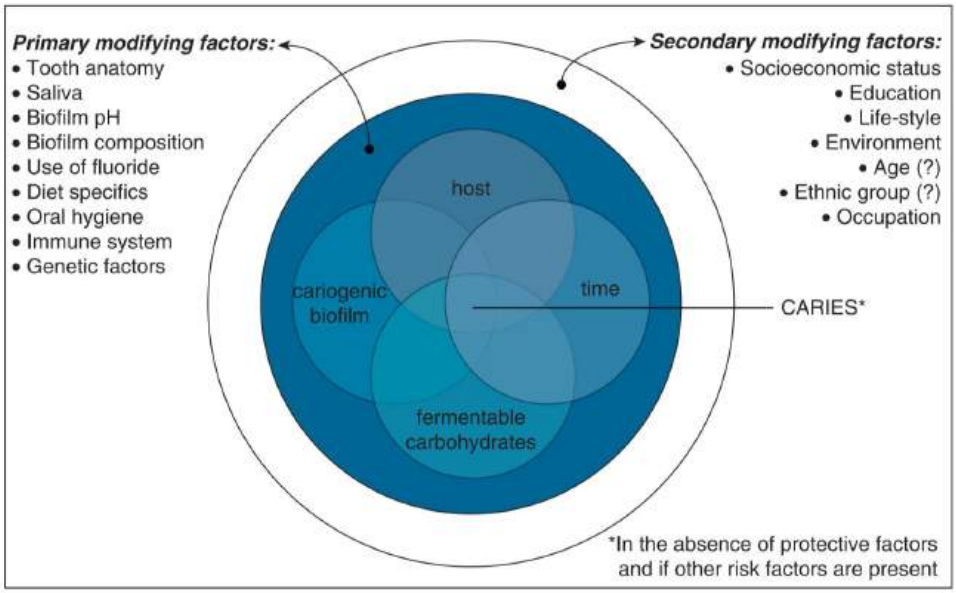
Dental caries is a preventable, chronic, and biofilm-mediated disease modulated by diet. This multifactorial, oral disease is caused primarily by an imbalance of the oral flora (biofilm) due to the presence of fermentable dietary carbohydrates on the tooth surface over time. Traditionally, this tooth-biofilm-carbohydrate interaction has been illustrated by the classical Keyes-Jordan diagram (Fig. 2.1).[1] However, dental caries onset and activity are in fact much more complex than this three-way interaction, as not all persons with teeth, biofilm, and consuming carbohydrates will have caries lesions over time. Several modifying risk and protective factors influence the dental caries process, as will be discussed later in this chapter.

At the tooth level, dental caries activity is characterized by localized demineralization and loss of tooth structure, the *caries lesion* (Figs. 2.2 and 2.3). In health, the microbiome is in symbiosis, the oral commensals striving in a neutral pH. Some bacteria in the biofilm metabolize refined carbohydrates for energy and produce organic acid by-products. These organic acids, if present in the biofilm ecosystem for extended periods, can lower the pH in the biofilm to below a critical level (5.5 for enamel, 6.2 for dentin). This low pH has effects both on the biofilm composition and at the tooth surface level (Fig. 2.4).[132] With extended periods of low pH there is a shift in the microbiome to bacteria that are acidogenic and acidophilic, causing a dysbiosis in the microbiome. This change in turn will lead to further acidification of the environment. The low pH drives calcium and phosphate from the tooth to the biofilm in an attempt to reach equilibrium, hence resulting in a net loss of minerals by the tooth, or *demineralization*. When the pH in the biofilm returns to neutral and the concentration of soluble calcium and phosphate is supersaturated relative to that in the tooth, mineral can then be added back to partially demineralized enamel in a process called *remineralization*. Thus at the tooth surface and subsurface level, dental caries lesions result from a dynamic process of damage (demineralization) and restitution (remineralization) of the tooth matter. These events take place several times a day over the life of the tooth and are modulated by many factors including number and type of microbial flora in the biofilm, diet, oral hygiene, genetics, dental anatomy, dentin and enamel composition, use of fluorides and other chemotherapeutic agents, saliva composition, salivary flow, and buffering capacity. These factors are highly individual and tooth specific, and will differ from person to person, tooth to tooth in the same individual, and site to site on a same tooth. The balance between demineralization and remineralization has been illustrated in terms of pathologic factors (i.e., those favoring demineralization) and protective factors (i.e., those favoring remineralization) (Fig. 2.5).[2] Individuals in whom the balance tilts predominantly toward protective factors (remineralization) are much less likely to develop dental caries lesions than those in which the balance is tilted toward pathologic factors (demineralization). *Understanding the balance between demineralization and remineralization is key to caries management.*

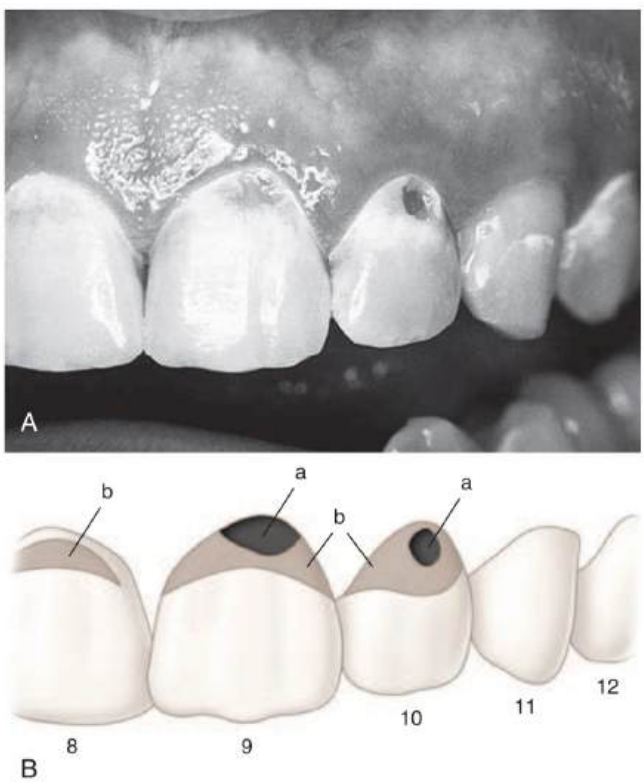
Repeated demineralization events may result from a predominantly pathologic environment causing the localized dissolution and destruction of the calcified dental tissues, evidenced as a caries lesion. The ongoing demineralization at the enamel subsurface level leads to a collapse of the surface and hence the formation of a cavitation in the enamel surface. Severe demineralization of dentin results in the exposure of the protein matrix, which is denatured initially by host matrix metalloproteinases (MMPs) and subsequently degraded by MMPs and other bacterial proteases. Demineralization of the inorganic phase (dentin mineral), and denaturation and degradation of the organic phase (primarily dentin collagen) result in dentin cavitation.[3]

It is essential to understand that caries lesions, or cavitations in teeth, are signs of an underlying condition, an imbalance between protective and pathologic factors favoring the latter. In clinical practice it is very easy to lose sight of this fact and focus entirely on the restorative treatment of caries lesions, failing to treat the underlying cause of the disease (Table 2.1). Although symptomatic treatment is important for many reasons, failure to identify and

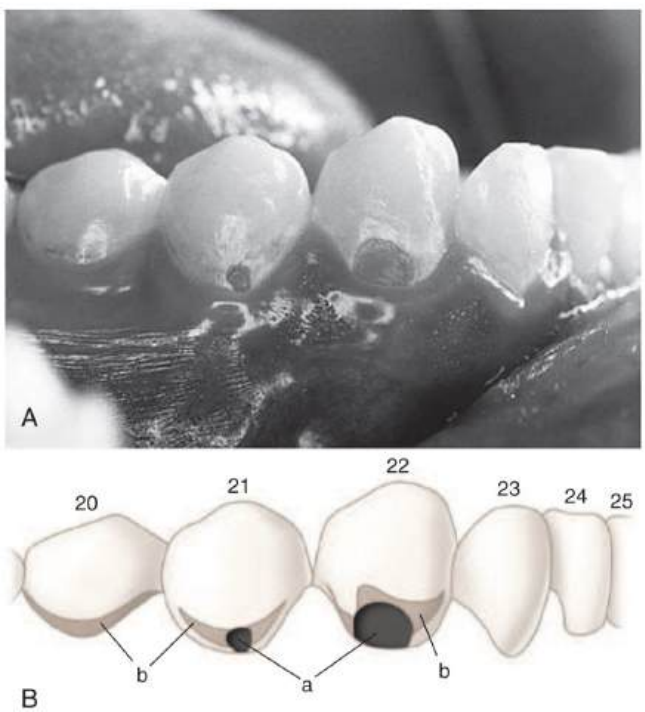
[a]Dr. Eidson was an inactive author in this edition.



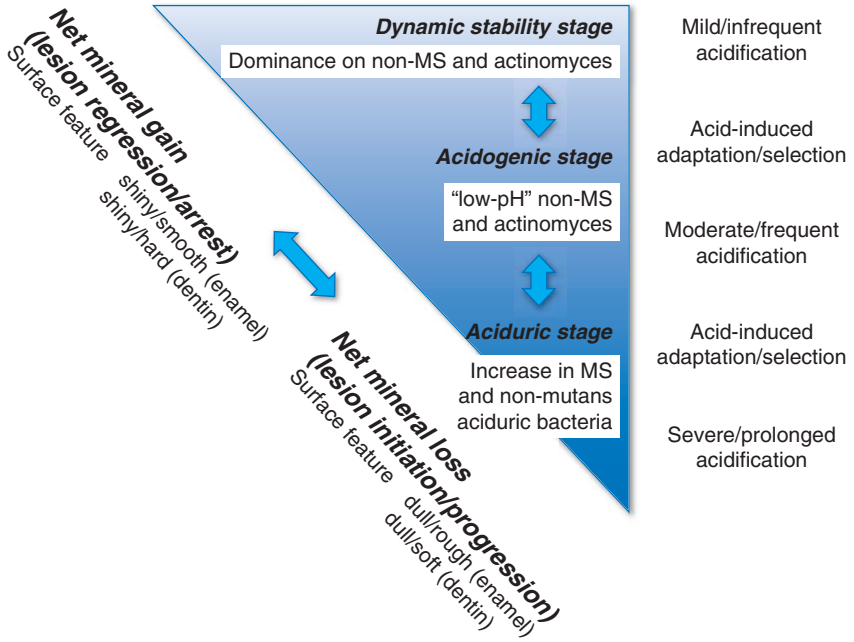
• **Fig. 2.1** Modified Keyes-Jordan diagram. As a simplified description, dental caries is a result of the interaction of cariogenic oral flora (biofilm) with fermentable dietary carbohydrates on the tooth surface (host) over time. However, dental caries onset and activity are, in fact, much more complex, as not all persons with teeth, biofilm, and who are consuming carbohydrates will have caries over time. Several modifying risk factors and protective factors influence the dental caries process. (Modified from Keyes PH, Jordan HV: Factors influencing initiation, transmission and inhibition of dental caries. In Harris RJ, editor: *Mechanisms of hard tissue destruction*, New York, 1963, Academic Press.)



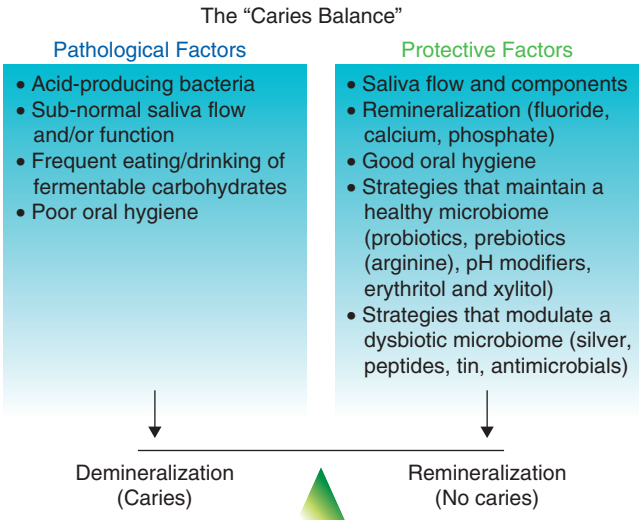
• **Fig. 2.2** A, Young adult with multiple active initial and cavitated caries lesions involving teeth No. 8 through 10. B, Cavitated areas *(a)* are surrounded by areas of extensive demineralization that are chalky and opaque *(b)*. Some areas of initial (noncavitated) caries have superficial stain.



• **Fig. 2.3** Extensive active caries in a young adult (same patient as in Fig. 2.2). A, Mirror view of teeth No. 20 through 22. B, Cavitated lesions *(a)* are surrounded by extensive areas of chalky, opaque demineralized areas *(b)*. The presence of smooth-surface lesions such as these is associated with rampant caries. Occlusal and interproximal smooth-surface caries usually occur in advance of facial smooth-surface lesions. The presence of these types of lesions should alert the dentist to the possibility of a high caries risk patient and possibly extensive caries activity elsewhere in the mouth. The interproximal gingiva is swollen red and would bleed easily on probing. These gingival changes are the consequence of long-standing irritation from the biofilm adherent to the teeth.



• **Fig. 2.4** This diagram illustrates how the diet affects the biofilm and consequences on the tooth surface. In health, the biofilm is in symbiosis. A person who has only occasional consumption of fermentable carbohydrates will have a resident microbiome dominated by nonmutans streptococci and actinomyces in a dynamic stability stage that is adapting to the relatively minor changes in pH. As the diet changes to more frequent exposures to fermentable carbohydrates, the microbiome adapts to the more acidic environment in an acidogenic stage, where bacteria that favor low pH flourish. In diets with frequent exposures to fermentable carbohydrates this constant acid-induced adaptation and selection of the microbiome continues and leads to an aciduric stage where there is an increase in MS and nonmutans aciduric bacteria, thus leading to a dysbiosis of the microbiome. At one end of the spectrum, in an environment with severe and prolonged acidification and aciduric bacteria, there is a net mineral loss at the tooth surface, enamel, or dentin, causing lesion initiation or lesion progression, while at the other end of the spectrum with only mild acidification there is a net mineral gain at the tooth surface and consequently lesion regression or arrest. Certainly this is a simplified explanation of the changes in the biofilm as the microbiome adapts to the acidic environment due to cariogenic diets, which does not consider the many other factors that are at play to either halt the process or boost the process. (From Takahashi N1, Nyvad B. The role of bacteria in the caries process: ecological perspectives, *J Dent Res* 90(3):294–303, 2011. doi:10.1177/0022034510379602)



• **Fig. 2.5** The caries balance. The balance between demineralization and remineralization is illustrated in terms of pathologic factors (i.e., those favoring demineralization) and protective factors (i.e., those favoring remineralization). (Modified from Featherstone JDB: Prevention and reversal of dental caries: role of low level fluoride, *Community Dent Oral Epidemiol* 27:31–40, 1999.)

TABLE 2.1 Caries Management Based on the Medical Model

Primary Etiology	Cariogenic Biofilm Resultant From a Cariogenic Diet
Symptoms	Demineralization lesions in teeth
Treatment, therapeutic	Improvement of host resistance by (1) biofilm modification, (2) elevating biofilm pH, and (3) enhancing remineralization
Treatment, symptomatic	Remineralization/arrest of lesions, restoration of cavitated lesions
Posttreatment assessment, therapeutic	Reevaluation of etiologic conditions and primary and secondary risk factors; and continuous management based on findings
Posttreatment assessment, symptomatic	Examination of teeth for new lesions, assessment of lesion activity, assessment of lesion progression

manage the underlying causative factors of dental caries allows the disease to continue and increases the chance of treatment failure. Caries management efforts must be directed not only at the tooth level (traditional or surgical treatment) but also at the total-patient level (caries management by risk assessment). *Restorative treatment does not cure the caries process*. Instead, identifying and managing the risk factors for caries must be the primary focus, in addition to the restorative repair of damage caused by caries.

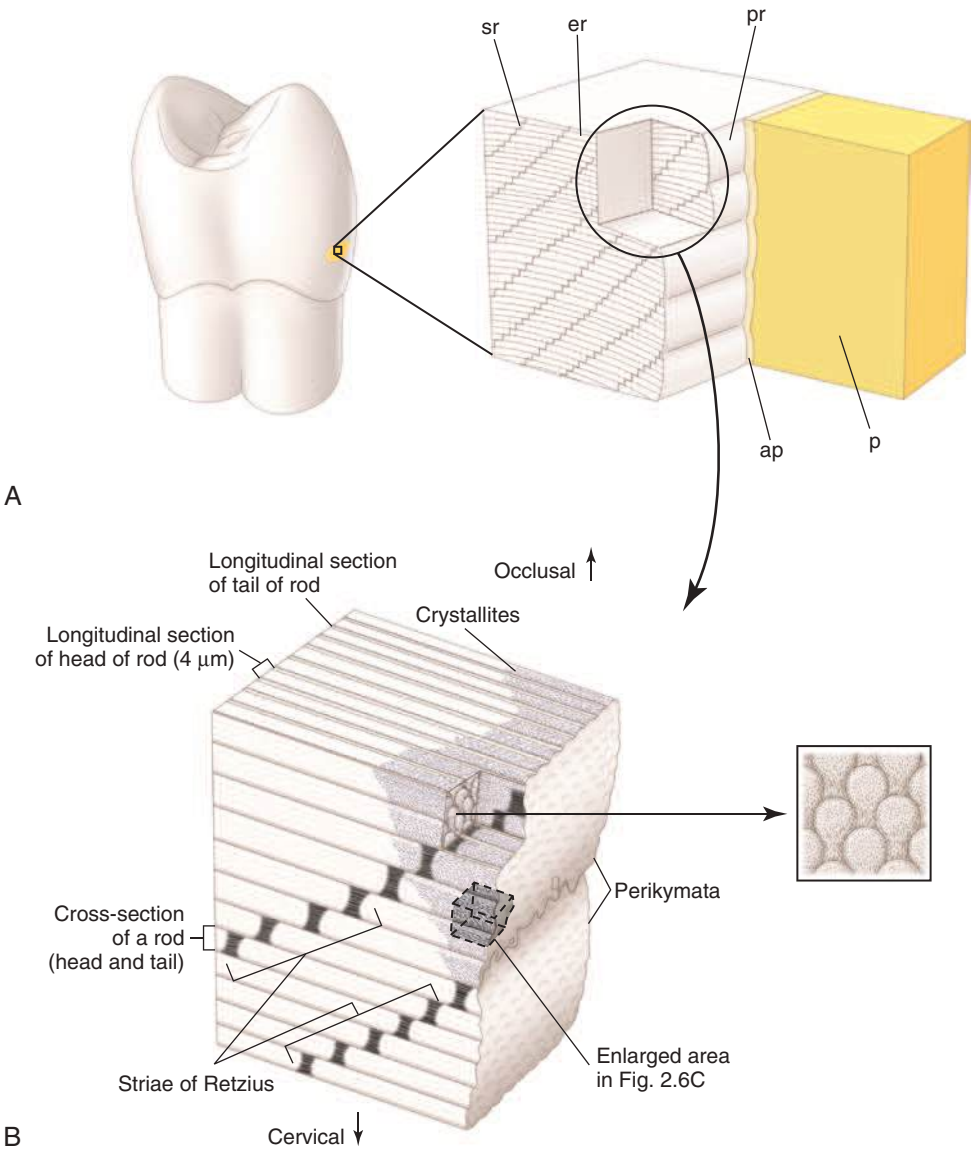
This chapter emphasizes the components of a caries management program that is based first on risk assessment and then on modifying the biofilm ecology to enhance protective factors and minimize pathologic factors.[4] This chapter also presents information on clinical characteristics of the caries lesion as they relate to clinical operative dentistry. Use of correct and consistent terms when referring to caries and caries lesions is important. Box 2.1 summarizes the most common terms used in this textbook to define caries lesions based on their location, cavitation status, and activity status. More in-depth information on the dental caries process can be found elsewhere.[133]

Ecologic Basis of Dental Caries: The Role of the Biofilm

Dental plaque is a term historically used to describe the soft, tenacious film accumulating on the surface of teeth. Dental plaque has been more recently referred to as the *dental biofilm* or simply the *biofilm*, which is a more complete and accurate description of its composition (bio) and structure (film).[5] The biofilm is composed mostly of bacteria, their by-products, extracellular matrix, and water (Figs. 2.6, 2.7, 2.8, 2.9, and 2.10). Biofilm is not adherent food debris, as was widely and erroneously thought, nor does it result from the haphazard collection of opportunistic microorganisms. The accumulation of the biofilm on teeth is a highly organized

• BOX 2.1 Caries Lesion Definitions

- **Caries lesion.** Tooth demineralization as a result of the caries process. Other texts may use the term *carious lesion*. Laypeople may use the term *cavity*.
- **Smooth-surface caries.** A caries lesion on a smooth tooth surface.
- **Pit-and-fissure caries.** A caries lesion on a pit-and-fissure area.
- **Occlusal caries.** A caries lesion on an occlusal surface.
- **Proximal caries.** A caries lesion on a proximal surface.
- **Enamel caries.** A caries lesion in enamel, typically indicating that the lesion has not penetrated into dentin. (Note that many lesions detected clinically as enamel caries may very well have extended into dentin histologically.)
- **Dentin caries.** A caries lesion into dentin.
- **Coronal caries.** A caries lesion in any surface of the anatomic tooth crown.
- **Root caries.** A caries lesion in the root surface.
- **Primary caries.** A caries lesion not adjacent to an existing restoration or crown.
- **Secondary caries.** A caries lesion adjacent to an existing restoration, crown, or sealant. Other term used is *caries adjacent to restorations and sealants* (CARS). Also referred to as recurrent caries, implying that a primary caries lesion was restored but that the lesion reoccurred.
- **Residual caries.** Refers to carious tissue that was not completely excavated prior to placing a restoration. Sometimes residual caries can be difficult to differentiate from secondary caries.
- **Cavitated caries lesion.** A caries lesion that results in the breaking of the integrity of the tooth, or a cavitation.
- **Initial caries lesion.** A caries lesion that has not been cavitated. In enamel caries, initial caries lesions are also referred to as noncavitated or "white spot" lesions. (Clinically, the distinction between a cavitated and a noncavitated caries lesion is not as simple as it may seem. Although historically any roughness detectable with a sharp explorer has been considered a cavitated lesion, more recent caries detection guidelines establish that only lesions in which a blunt probe penetrates without pressure are to be considered cavitated. This distinction has important implications on lesion management, because most initial caries lesions can be arrested or remineralized, while most cavitated lesions require restorative intervention.)
- **Moderate caries lesion.** A caries lesion that may or may not have cavitated but that has not reached the inner one-third of dentin. This can be observed clinically by microcavitations in the enamel or a gray shadow.
- **Advanced (deep) caries lesion.** A definitely cavitated lesion exposing dentin. For the purposes of this protocol any lesion that has reached the inner one-third of the dentin will be considered an advanced lesion.
- **Active caries lesion.** A caries lesion that is considered to be biologically active—that is, a lesion in which tooth demineralization is in frank activity at the time of examination.
- **Inactive caries lesion.** A caries lesion that is considered to be biologically inactive at the time of examination—that is, in which tooth demineralization caused by caries may have happened in the past but has stopped and is currently stalled. Also referred to as *arrested caries,* meaning that the caries process has been arrested but that the clinical signs of the lesion itself are still present.
- **Rampant caries.** Term used to describe the presence of extensive and multiple cavitated and active caries lesions in the same person. Typically used in association with "baby bottle caries," "radiation therapy caries," or "meth-mouth caries." These terms refer to the etiology of the condition.
- **Primary dentin.** Sound, normal dentin that forms during tooth development; it is usually completed 3 years after tooth eruption. Histologically, primary dentin has tubules with smooth odontoblastic processes, with no intratubular crystals. The intertubular dentin has normal cross-banded collagen and normal dense apatite crystals. Clinically, primary dentin is hard, cannot be easily penetrated with a blunt explorer, and can only be cut by a bur or a sharp cutting instrument.
- **Secondary dentin.** Sound, normal dentin that forms physiologically on all internal aspects of the pulp cavity throughout the life of the tooth. Histologically, secondary dentin resembles primary dentin. Clinically, secondary dentin is similar to primary dentin.
- **Tertiary dentin.** Dentin that forms in response to stimuli such as caries, attrition, and operative procedures. Also known as reparative or reactive dentin. Usually appears as a localized dentin deposit on the wall of the pulp space immediately subjacent to the area of the tooth that has received the injury. Tertiary dentin is less mineralized than primary and secondary dentin, and contains irregular dentinal tubules. Clinically, tertiary dentin is not as hard as primary dentin.
- **Sclerotic dentin.** Dentin that forms in response to stimuli such as aging or mild irritation (slow advancing caries). When responding to initial caries demineralization events, crystalline material precipitates in intratubular and intertubular dentin. Sclerotic dentin walls off a lesion by blocking (sealing) the dentinal tubules. This zone can be seen even before the demineralization reaches the dentin and it may not be present in rapidly advancing lesions. Clinically, sclerotic dentin is dark and harder than normal dentin.
- **Leathery dentin.** Term used to describe the clinical presentation of the transition zone between soft and firm dentin (see next section). Technically leathery dentin is part of the firm dentin zone.

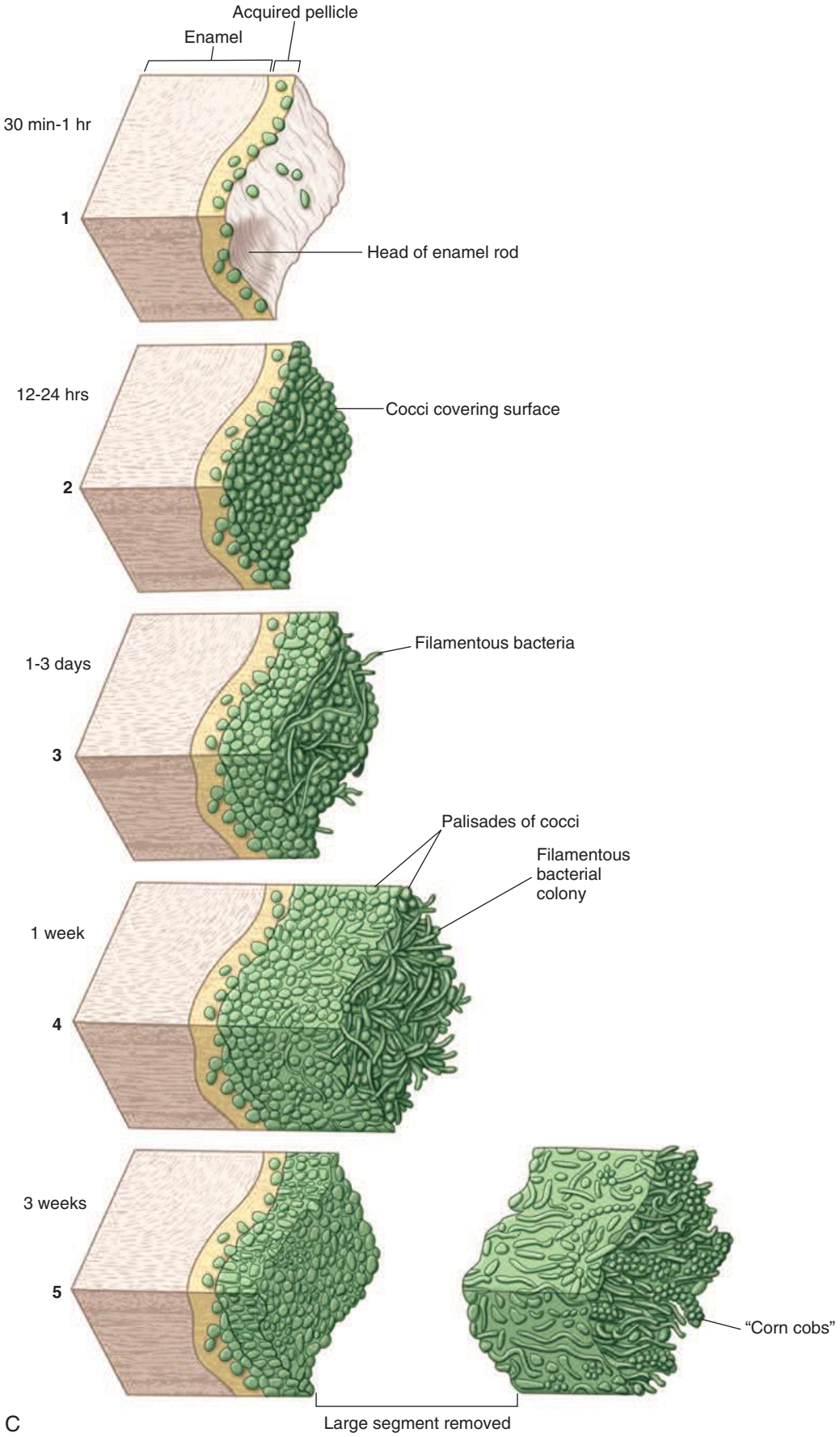


• **Fig. 2.6** A, Composite diagram illustrating the relationship of biofilm *(p)* to the enamel in a smooth-surface initial (noncavitated) lesion. A relatively cell-free layer of precipitated salivary protein material, the acquired pellicle *(ap)* covers the perikymata ridges *(pr)*. The biofilm bacteria attach to the pellicle. Overlapping perikymata ridges can be seen on the surface of enamel (see Fig. 2.7). (Figs. 2.9 and 2.10 are photomicrographs of cross sections of biofilm.) The enamel is composed of rodlike structures *(er)* that course from the inner dentinoenamel junction (DEJ) to the surface of the crown. Striae of Retzius *(sr)* can be seen in cross sections of enamel. B, Higher power view of the cutout portion of enamel in *A*. Enamel rods interlock with each other in a head-to-tail orientation. The rod heads are visible on the surface as slight depressions on the perikymata ridges. The enamel rods comprise tightly packed crystallites. The orientation of the crystallites changes from being parallel to the rod in the head region to being perpendicular to the rod axis in the tail end. Striae of Retzius form a descending diagonal line, descending cervically.

and ordered sequence of events. Bacteria seem to occupy the same spatial niche on most individuals. A "hedgehog" formation has been recently characterized[209] because of the spine of radially oriented filaments. The filaments are a mass of *Corynebacterium* filaments with *Streptococcus* at the periphery. Actinomyces are usually found at the base of the biofilm suggesting that *Corynebacterium* attaches to a preexisting biofilm containing *Actinomyces*. In any case it is notable that each taxon is localized in a precise and well-defined spatial zone indicating that the microbes in the oral biofilm have a precise and well-tuned interaction[209] (see Fig. 2.6D).

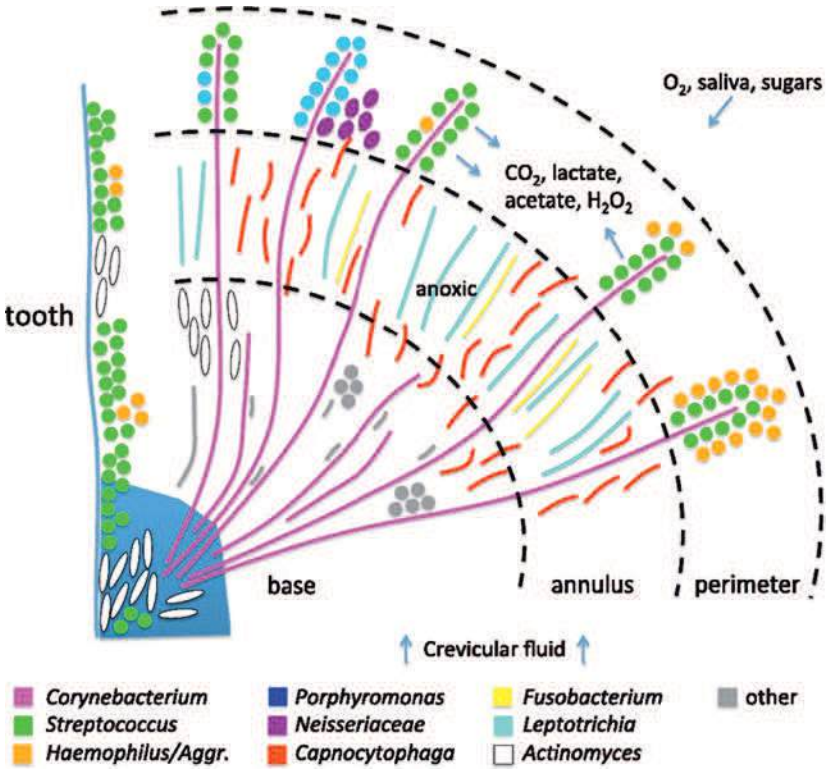
Many of the organisms found in the mouth are not found elsewhere in nature. Survival of microorganisms in the oral environment depends on their ability to adhere to a surface. Free-floating organisms are cleared rapidly from the mouth by salivary flow and frequent swallowing. Although a few specialized organisms, primarily streptococci, are particularly able to adhere to oral surfaces such

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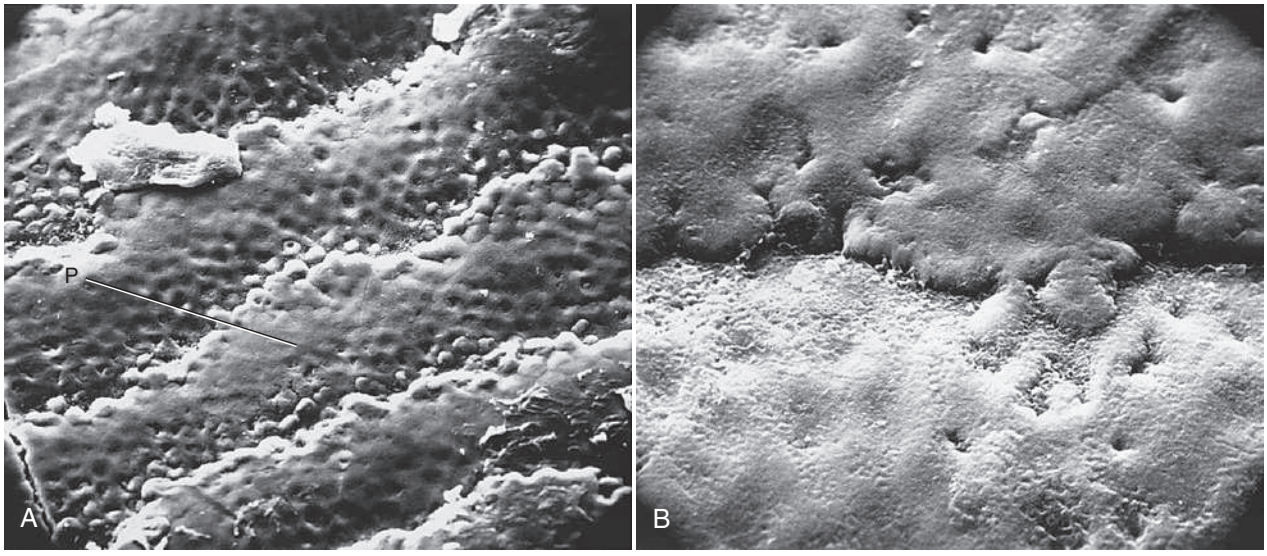


• **Fig. 2.6, cont'd** C, Drawings 1 through 5 illustrate the various stages in colonization during plaque formation on the shaded enamel block shown in *B*. The accumulated mass of bacteria on the tooth surface may become so thick that it is visible to the unaided eye. Such plaques are gelatinous and tenaciously adherent; they readily take up disclosing dyes, aiding in their visualization for oral hygiene instruction. Thick plaque biofilms (*4* and *5*) are capable of great metabolic activity when sufficient nutrients are available. The gelatinous nature of the plaque limits outward diffusion of metabolic products and serves to prolong the retention of organic acid metabolic by-products.

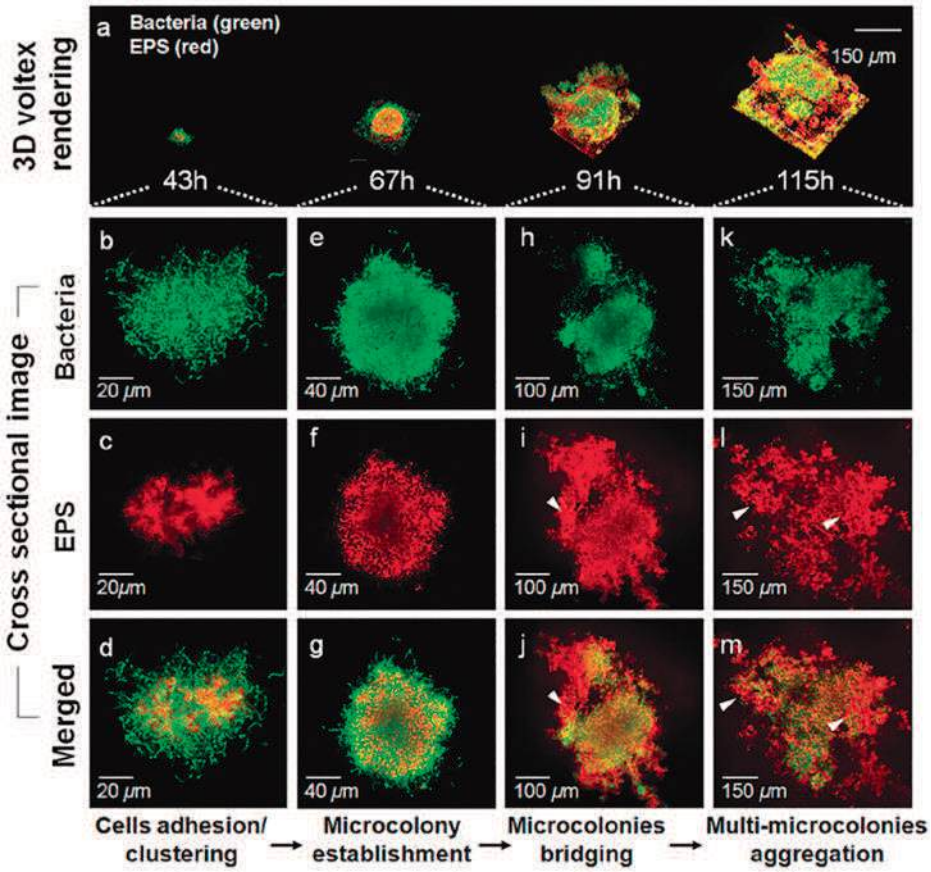
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• **Fig. 2.6, cont'd** D, This illustrates how different taxons inhabit specific niches on the biofilm creating microenvironments. There is a fine-tuned synergy among the cells in the oral microbial communities. The environment and the biochemical gradients drive the selection process. This can be exemplified by the the role of *Streptococcus*. Where *Streptococcus* predominate they create an environment rich in CO_2, lactate, and acetate, containing peroxide and having low oxygen. This environment is advantageous for the growth of bacteria such as *Fusobacterium* and *Leptotrichia*. (From Welch JL, Rossetti BJ, Riekem CW, et al: Biogeography of a human oral microbiome at the micron scale, *Proc Natl Acad Sci USA* 9;113(6):E791–E800, 2016. doi:10.1073/pnas.1522149113)



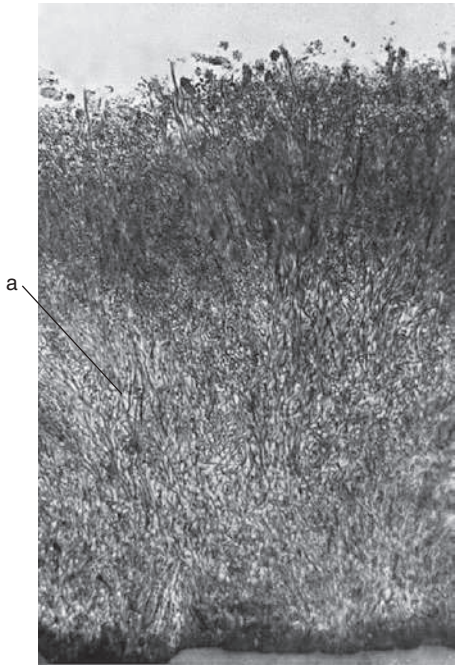
• **Fig. 2.7** A, Scanning electron microscope view (600×) of overlapping perikymata *(P)* in sound enamel from unerupted molar. B, Higher power view (2300×) of overlapped site rotated 180 degrees. Surface of noncavitated enamel lesions has "punched-out" appearance. (From Hoffman S: Histopathology of caries lesions. In Menaker L, editor: *The biologic basis of dental caries*, New York, 1980, Harper & Row.)



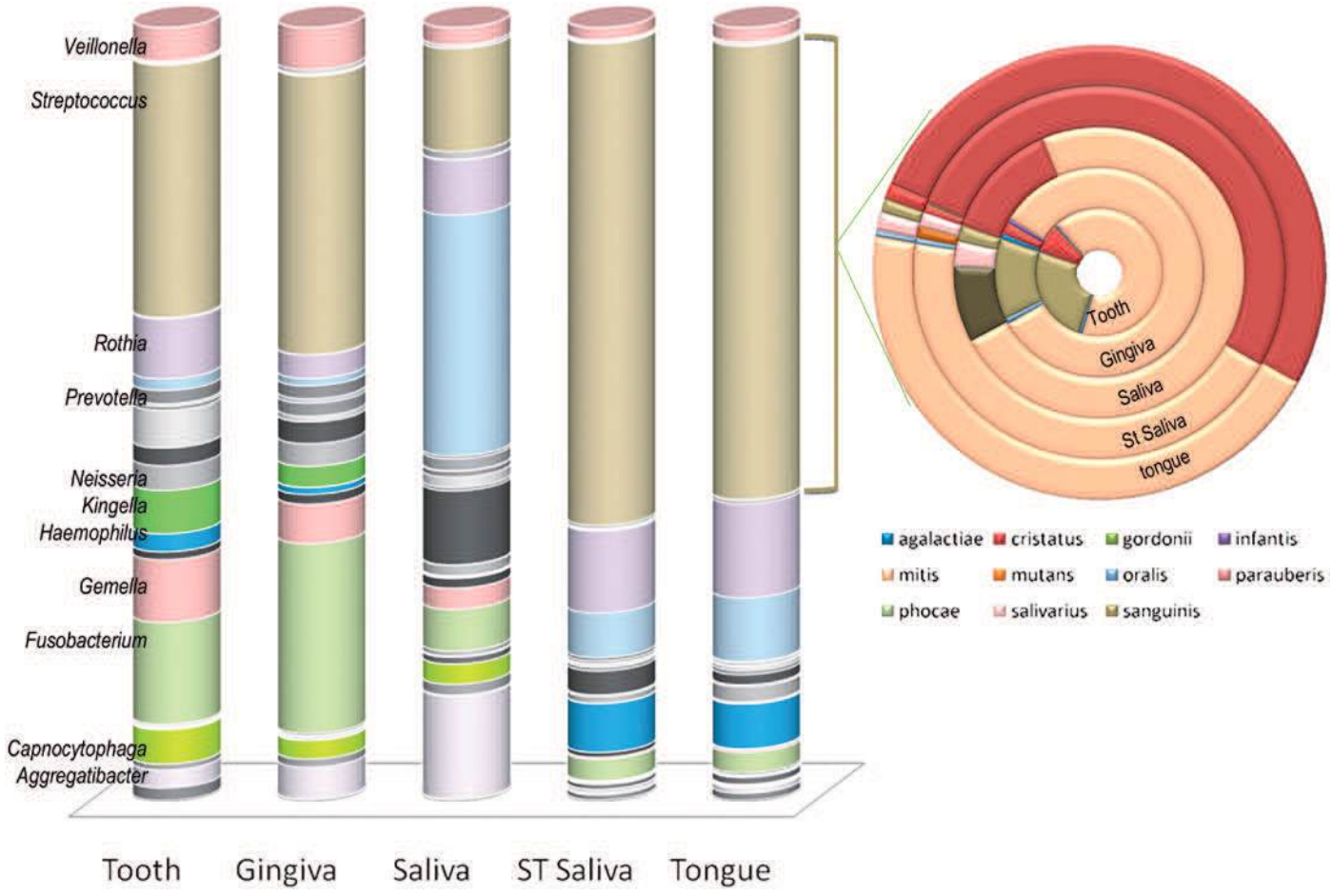
• **Fig. 2.8** Representative 3-D rendering images of mixed-species biofilms in an environment with 1% (W/V) sucrose. The images show the evolution of the microcolonies over time and the arrangement with the EPS matrix. (From Xiao J, Klein MI, Falsetta ML, et al: The exopolysaccharide matrix modulates the interaction between 3D architecture and virulence of a mixed-species oral biofilm, *PLoS Pathog* 8(4):e1002623, 2012. https://doi.org/10.1371/journal.ppat.1002623)



• **Fig. 2.9** Plaque biofilm formation at 1 week. Filamentous bacteria *(f)* appear to be invading cocci microcolonies. Plaque near gingival sulcus has fewer coccal forms and more filamentous bacteria (860×). (From Listgarten MA, Mayo HE, Tremblay R: Development of dental plaque on epoxy resin crowns in man. A light and electron microscopic study, *J Periodontol* 46(1):10–26, 1975.)



• **Fig. 2.10** At 3 weeks old, plaque biofilm is almost entirely composed of filamentous bacteria. Heavy plaque formers have spiral bacteria *(a)* associated with subgingival plaque (660×). (From Listgarten MA, Mayo HE, Tremblay R: Development of dental plaque on epoxy resin crowns in man. A light and electron microscopic study, *J Periodontol* 46(1):10–26, 1975.)



• **Fig. 2.11** Approximate proportional distribution of predominant cultivable flora of five oral habitats. (From Simón-Soro Á, Tomás I, Cabrera-Rubio R, et al: Microbial geography of the oral cavity, *J Dent Res* 92:616, 2013. DOI:10.1177/0022034513488119.)

as the mucosa and tooth structure, over 700 different species of bacteria have been identified in the oral biofilm. Oral biofilm from healthy teeth have a higher diversity than from carious teeth.[134]

Significant differences exist in the biofilm communities found in various habitats (ecologic environments) within the oral cavity (Fig. 2.11A and B). The organisms also have unique contributions to the ecosystem (see Fig. 2.11B). Mature biofilm communities have tremendous metabolic potential and are capable of rapid anaerobic metabolism of any available carbohydrate (Fig. 2.12). However, because of the highly structured bacterial microcolonies embedded in an exopolysaccharide (EPS)-rich matrix, there are acidic regions in the biofilm that are not neutralized by saliva buffers.[210]

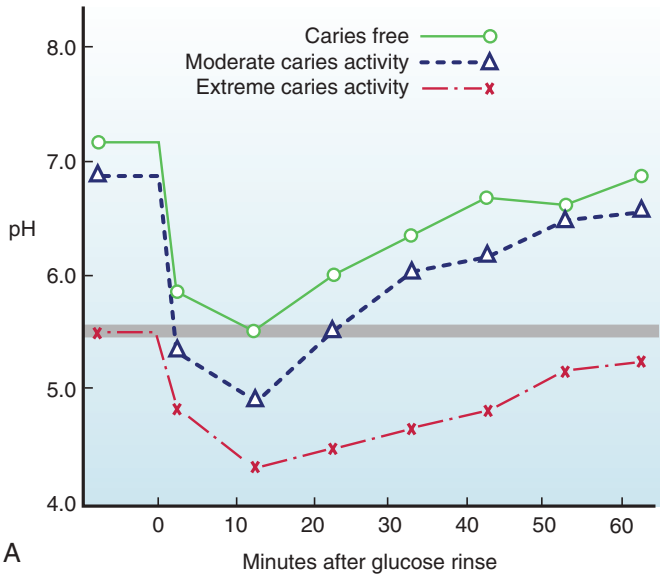
Many distinct habitats may be identified on individual teeth, with each habitat containing a unique biofilm community (Table 2.2; see Fig. 2.11A). Although the pits and fissures on the crown may harbor a relatively simple population of streptococci, the root surface in the gingival sulcus may harbor a complex community dominated by filamentous and spiral bacteria. Even within the same anatomic location there can be a considerable difference in bacterial diversity.[134] For example the mesial surface of a molar may be carious and have a biofilm dominated by large populations of mutans streptococci (MS) and lactobacilli, whereas the distal surface may lack these organisms and be caries free. Generalization about biofilm communities is difficult.

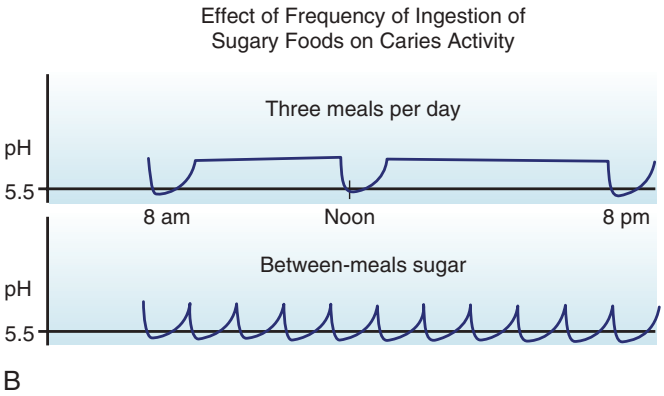
Recent evidence indicates that there are no specific pathogens that correlate with dental caries, but rather microbial communities.[135] Nevertheless, the general activity of biofilm growth and maturation is predictable and sufficiently well known to be of therapeutic importance in the prevention of dental caries.

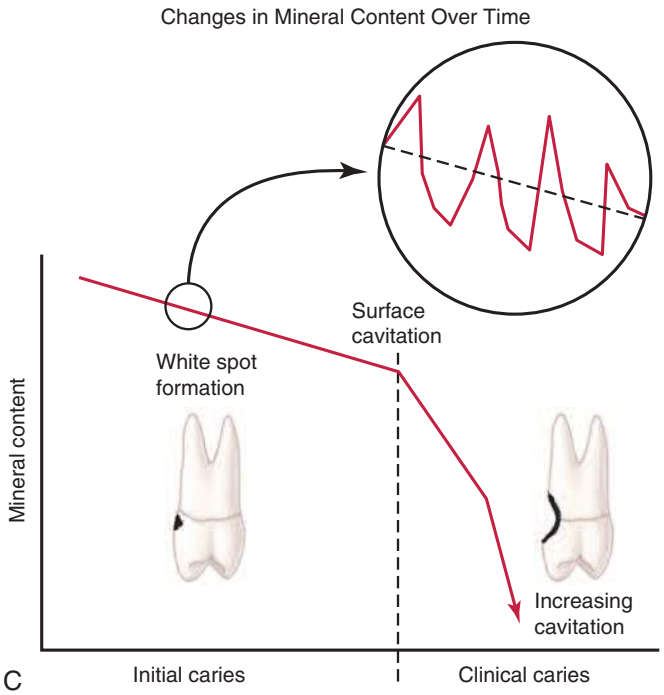
Professional tooth cleanings are intended to control the biofilm (plaque) and prevent caries (and periodontal) disease. However, after professional removal of all organic material and bacteria from the tooth surface, a new coating of organic material begins to accumulate immediately. Within 2 hours, a cell-free, organic film, the acquired enamel pellicle (AEP) (see Fig. 2.6A and C), can cover the previously denuded area completely. The pellicle is formed primarily from the selective precipitation of various components of saliva, particularly selective enzymes. The functions of the pellicle are believed to be (1) to protect the enamel, (2) to reduce friction between teeth, and (3) to provide a matrix for remineralization.[6] Although the pellicle exhibits antibacterial activity due to the presence of several enzymes, it can also function as a facilitator of bacterial cononization.[136]

Tooth Habitats for Cariogenic Biofilm

The tooth surface is unique because it is not protected by the surface-shedding mechanisms (continual replacement of epithelial cells) used throughout the remainder of the digestive tract. The







• **Fig. 2.12** A, Mature biofilm communities have tremendous metabolic potential and are capable of rapid anaerobic metabolism of any available carbohydrates. Classic studies by Stephan show this metabolic potential by severe pH drops at the plaque-enamel interface after glucose rinse. It is generally agreed that a pH of 5.5 is the threshold for enamel demineralization. Exposure to a glucose rinse for an extreme caries activity plaque results in a sustained period of demineralization (pH 5.5). Recording from a slight caries activity biofilm shows a much shorter period of demineralization. B, The frequency of sucrose exposure for cariogenic biofilm greatly influences the progress of tooth demineralization. The top line illustrates pH depression, patterned after Stephan's curves in *A*. Three meals per day results in three exposures of biofilm acids, each lasting approximately 1 hour. The biofilm pH depression is relatively independent of the quantity of sucrose ingested. Between-meal snacks or the use of sweetened breath mints results in many more acid attacks, as illustrated at the bottom. The effect of frequent ingestion of small quantities of sucrose results in a nearly continuous acid attack on the tooth surface. (The clinical consequences of this behavior can be seen in Fig. 2.37.) C, In active caries, a progressive loss of mineral content subjacent to the cariogenic biofilm occurs. Inset illustrates that the loss is not a continuous process. Instead, alternating periods of mineral loss (demineralization) occur, with intervening periods of remineralization. The critical event for the tooth is cavitation of the surface, marked by the vertical dashed line. This event marks an acceleration in caries destruction of the tooth and irreversible loss of tooth structure. An intervention is usually required to arrest the lesion, often of the restorative nature. (A, Adapted and redrawn from Stephan RM: Intra-oral hydrogen-ion concentration associated with dental caries activity, *J Dent Res* 23:257, 1944.)

tooth surface is stable and covered with the pellicle of precipitated salivary glycoproteins, enzymes, and immunoglobulins. It is the ideal surface for the attachment of many oral streptococci. If left undisturbed, biofilm rapidly builds up to sufficient depth to produce an anaerobic environment adjacent to the tooth surface. Given the right conditions (that is, a conducive diet and poor oral hygiene) these tooth habitats will become favorable for harboring pathogenic biofilm and include (1) pits and fissures (Fig. 2.13); (2) the smooth enamel surfaces immediately gingival to the proximal contacts and in the gingival third of the facial and lingual surfaces of the clinical crown (Fig. 2.14); (3) root surfaces, particularly near the cervical line; and (4) subgingival areas (Fig. 2.15). These sites correspond to the locations where caries lesions are most frequently found.

Pits and Fissures

Pits and fissures are particularly susceptible surfaces for caries lesion initiation (Figs. 2.16, 2.17, 2.18, 2.19, 2.20; see also Figs. 2.13 to 2.15). The pits and fissures provide excellent mechanical shelter for organisms and harbor a community dominated by *S. sanguis* and other streptococci.[7] The relative proportion of MS most probably determines the cariogenic potential of the pit-and-fissure community. In susceptible patients, sealing the pits and fissures just after tooth eruption may be the most important strategy in their resistance to dental caries on those surfaces.

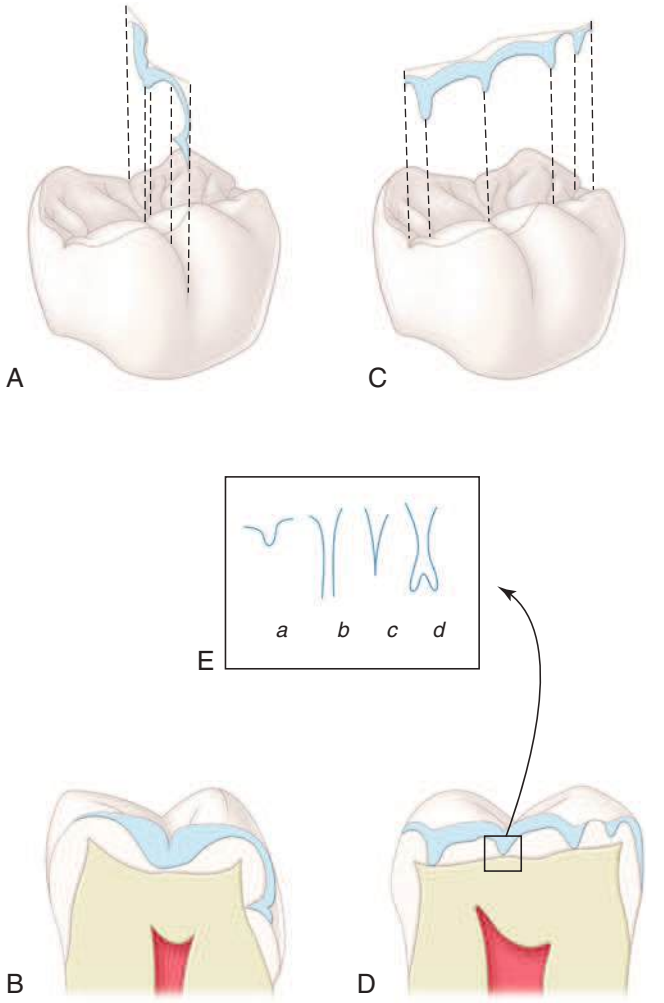
Smooth Enamel Surfaces

The proximal enamel surfaces immediately gingival to the contact area are the second most susceptible areas to dental caries lesions (Figs. 2.21 and 2.22; see also Figs. 2.14 and 2.19). These areas are protected physically and are relatively free from the effects of mastication, tongue movement, and salivary flow. The types and

TABLE 2.2 Oral Habitats[a]

Habitat	Predominant Species	Environmental Conditions Within Biofilm
Mucosa	*S. mitis* *S. sanguis* *S. salivarius*	Aerobic pH approximately 7 Oxidation-reduction potential positive
Tongue	*S. salivarius* *S. mutans* *S. sanguis*	Aerobic pH approximately 7 Oxidation-reduction potential positive
Teeth (noncarious)	*S. sanguis*	Aerobic pH 5.5 Oxidation-reduction negative
Gingival crevice	*Fusobacterium* *Spirochaeta* *Actinomyces* *Veillonella*	Anaerobic pH variable Oxidation-reduction very negative
Enamel caries	*S. mutans*	Anaerobic pH <5.5 Oxidation-reduction negative
Dentin caries	*S. mutans* *Lactobacillus*	Anaerobic pH <5.5 Oxidation-reduction negative
Root caries	*Actinomyces*	Anaerobic pH <5.5 Oxidation-reduction negative

[a]The microenvironmental conditions in the habitats associated with host health are generally aerobic, near neutrality in pH, and positive in oxidation-reduction potential. Significant microenvironmental changes are associated with caries and periodontal disease. The changes are the result of the biofilm community metabolism.



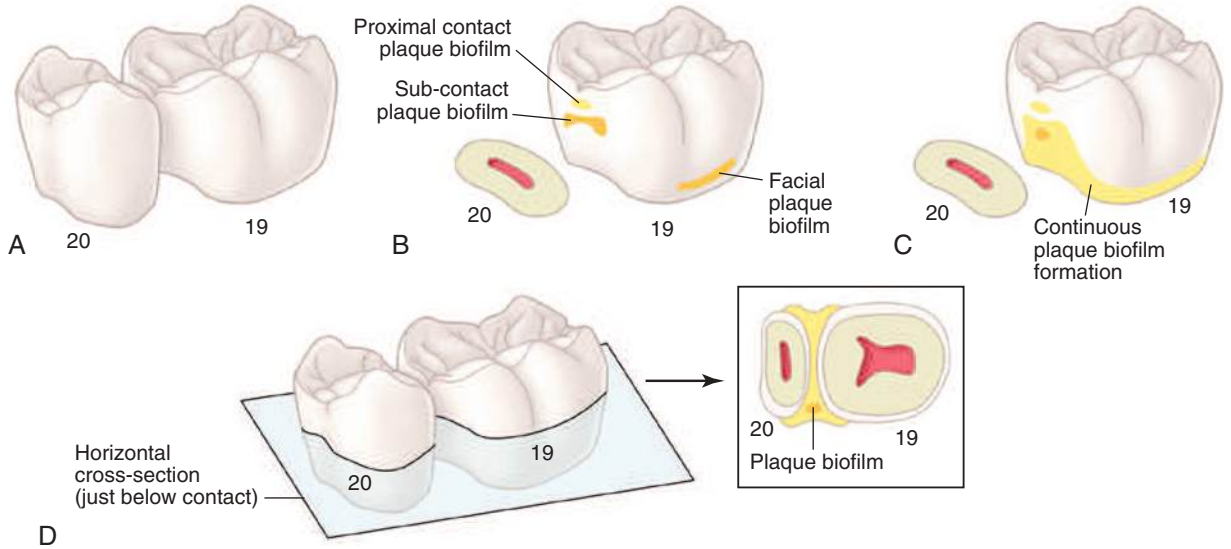
• **Fig. 2.13** Developmental pits, grooves, and fissures on the crowns of the teeth can have complex and varied anatomy. A and B, The facial developmental groove of the lower first molar often terminates in a pit. The depth of the groove and the pit varies. C and D, The central groove extends from the mesial pit to the distal pit. Sometimes grooves extend over the marginal ridges. E, The termination of pits and fissures may vary from a shallow groove *(a)* to complete penetration of the enamel *(b)*. The end of the fissure may end blindly *(c)* or open into an irregular chamber *(d)*. The physical nature of the pit and fissures allows for biofilm to remain stagnated. Additionally, the decreased thickness of enamel (and sometimes absence of enamel) at the bottom of the pit and fissure make these surfaces especially susceptible to dental caries.

numbers of organisms composing the proximal surface biofilm community vary. Important ecologic determinants for the biofilm community on the proximal surfaces are the topography of the tooth surface, the size and shape of the gingival papillae, and the oral hygiene and diet of the patient. A rough surface (caused by caries lesions, a poor-quality restoration, or a structural defect) restricts adequate biofilm removal. This situation favors the occurrence and progression of caries lesions and/or periodontal disease at the site.

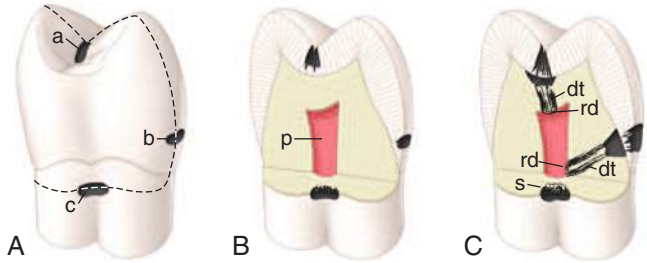
Root Surfaces

The proximal root surface, particularly near the cementoenamel junction (CEJ), often is unaffected by the action of hygiene procedures such as flossing because it may have concave anatomic surface contours (fluting) and occasional roughness at the termination of the enamel. These conditions, when coupled with exposure to the oral environment (as a result of gingival recession), favor the formation of mature, cariogenic biofilm and proximal root-surface caries lesions. Likewise, the facial or lingual root surfaces (particularly near the CEJ), when exposed to the oral environment (because of gingival recession), are often both neglected in hygiene procedures and usually not rubbed by the bolus of food. Consequently, these root surfaces also frequently harbor cariogenic biofilm. Root-surface caries lesions are more common in older patients because of niche availability and other factors sometimes associated with senescence, such as decreased salivary flow due to multiple medications and poor oral hygiene as a result of lowered manual dexterity and decreased motivation. Caries lesions originating on the root are alarming because they (1) have a comparatively rapid progression, (2) are often asymptomatic, (3) are closer to the pulp, and (4) are more difficult to restore.

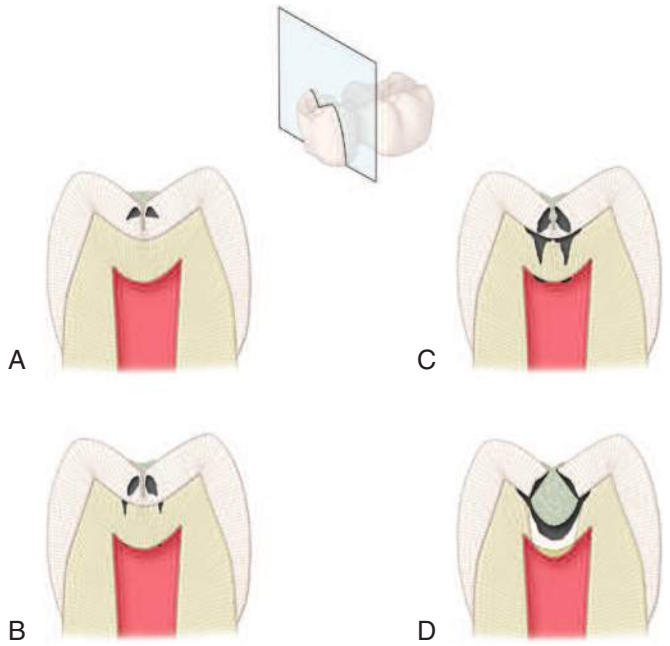
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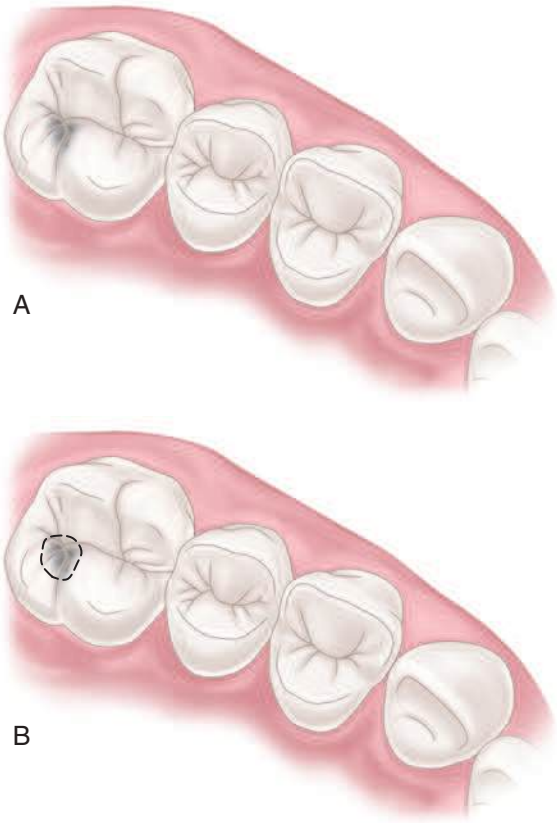
• **Fig. 2.14** Biofilm formation on posterior teeth and associated caries lesions. A, Teeth No. 19 and 20 in contacting relationship. B, The crown of tooth No. 20 has been removed at the cervix. The proximal contact and subcontact biofilm can be seen on the mesial surface of tooth No. 19. The biofilm on the facial surfaces is illustrated. C, During periods of unrestricted growth, the biofilms on the mesial and facial surfaces become part of a continuous ring of biofilm around teeth. D, A horizontal cross section through teeth No. 19 and 20 with heavy biofilm. Inset shows the interproximal space below the contact area filled with gelatinous biofilm. This mass of interproximal biofilm concentrates the effects of biofilm metabolism on the adjacent tooth smooth surfaces. All interproximal surfaces are subject to biofilm accumulation and acid demineralization. In patients exposed to fluoridated water, most interproximal lesions become arrested at a stage before cavitation.



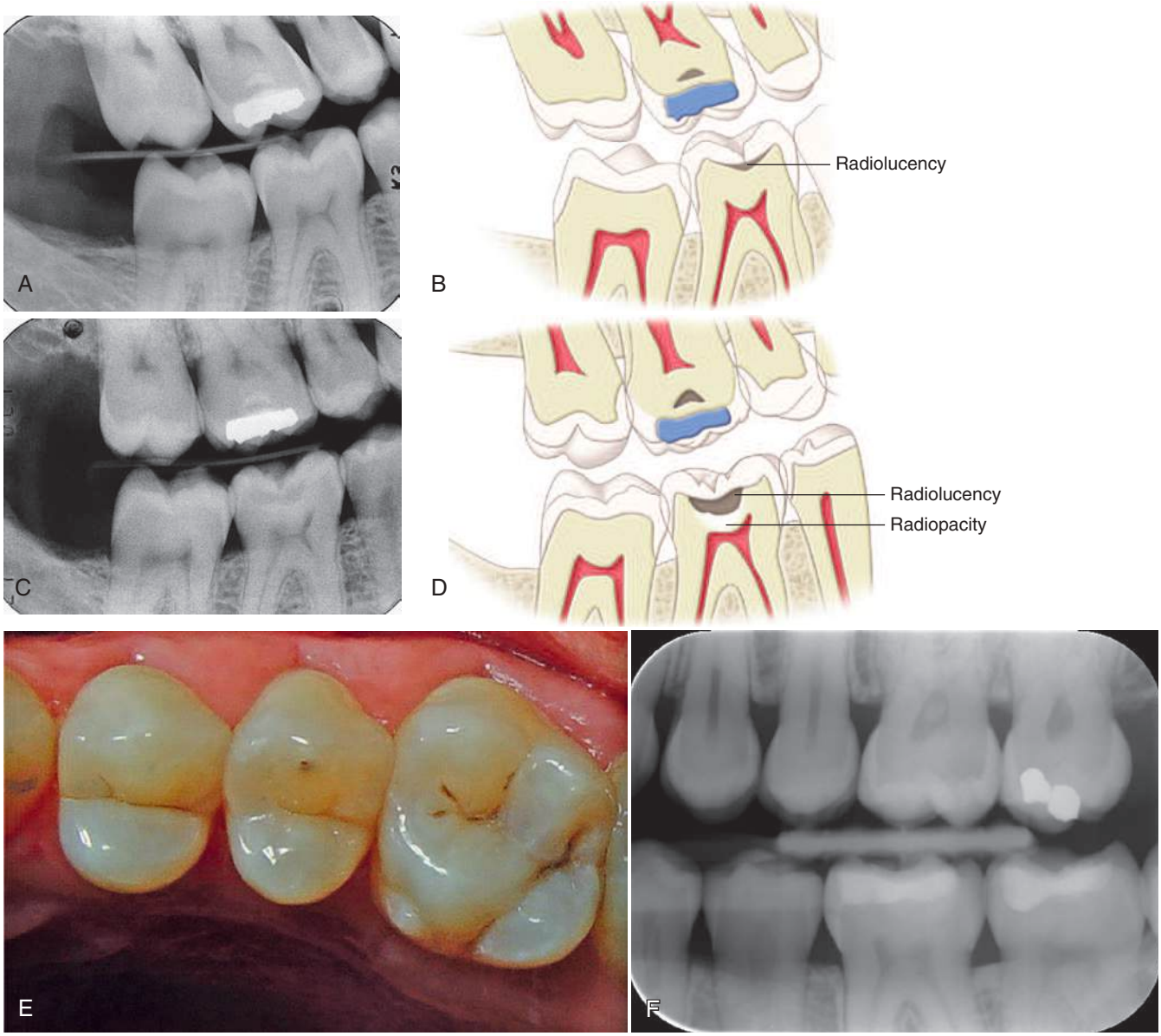
• **Fig. 2.15** A, Caries lesions may originate at many distinct sites: pits and fissures *(a),* smooth surface of crown *(b),* and root surface *(c).* Proximal surface lesion of crown is not illustrated here because it is a special case of smooth-surface lesion. Histopathology and progress of facial (or lingual) and proximal lesions are identical. Dotted line indicates cut used to reveal cross sections illustrated in *B* and *C*. Lesions illustrated here are intended to be representative of each type. No particular association between these lesions is implied. B, In cross section, the three types of lesions show different rates of progression and different morphology. Caries lesion progression and morphology follows the inclination of the enamel rods and/or dentinal tubules. Pit-and-fissure lesions have small sites of origin visible on the occlusal surface but have a wide base. Overall shape of a pit-and-fissure lesion is an inverted V. In contrast, a smooth-surface lesion is V shaped with a wide area of origin and apex of the V directed toward pulp *(p)*. Root caries begins directly on dentin. Root-surface lesions can progress rapidly because dentin is less resistant to caries attack. C, Although advanced caries lesions produce considerable histologic change in enamel, dentin and pulp changes in dentin can be seen even before the lesion reached the dentin. Bacterial invasion of lesion results in extensive demineralization and proteolysis of the dentin. Clinically, this necrotic dentin appears soft, wet, and mushy. Deeper pulpally, dentin is demineralized and is structurally intact. This tissue appears to be dry and leathery in texture. Two types of pulp–dentin response are illustrated. Under pit-and-fissure lesions and smooth-surface lesions, odontoblasts have died, leaving empty tubules called dead tracts *(dt).* New odontoblasts have been differentiated from pulp mesenchymal cells. These new odontoblasts have produced reparative dentin *(rd),* which seals off dead tracts. Another type of pulp–dentin reaction is sclerosis *(s)*—occlusion of the tubules by peritubular dentin. This is illustrated under root-caries lesion.



• **Fig. 2.16** Progression of caries in pits and fissures. A, Initial lesions develop on the lateral walls of the fissure. Demineralization follows the direction of the enamel rods, spreading laterally as it approaches the dentinoenamel junction (DEJ). B, Soon after the initial enamel lesion occurs, a reaction can be seen in the dentin and pulp. Forceful probing of the lesion at this stage can result in damage to the weakened porous enamel and accelerate the progression of the lesion. Clinical detection at this stage should be based on observation of discoloration and opacification of the enamel adjacent to the fissure. These changes can be observed by careful cleaning and drying of the fissure. C, Initial cavitation of the opposing walls of the fissure cannot be seen on the occlusal surface. Opacification can be seen that is similar to the previous stage. Remineralization of the enamel because of trace amounts of fluoride in the saliva may make progression of pit-and-fissure lesions more difficult to detect. D, Extensive cavitation of the dentin and undermining of the covering enamel darken the occlusal surface (see Fig. 2.17).



• **Fig. 2.17** A, Maxillar first molar has undermined discolored enamel owing to extensive pit-and-fissure caries. Note the dark shadow due to the dentin demineralization underneath the enamel. The lesion began as illustrated in Fig. 2.16 and has progressed to the stage illustrated in Fig. 2.16D. B, Discolored enamel is outlined by the broken line in the central fossa region.

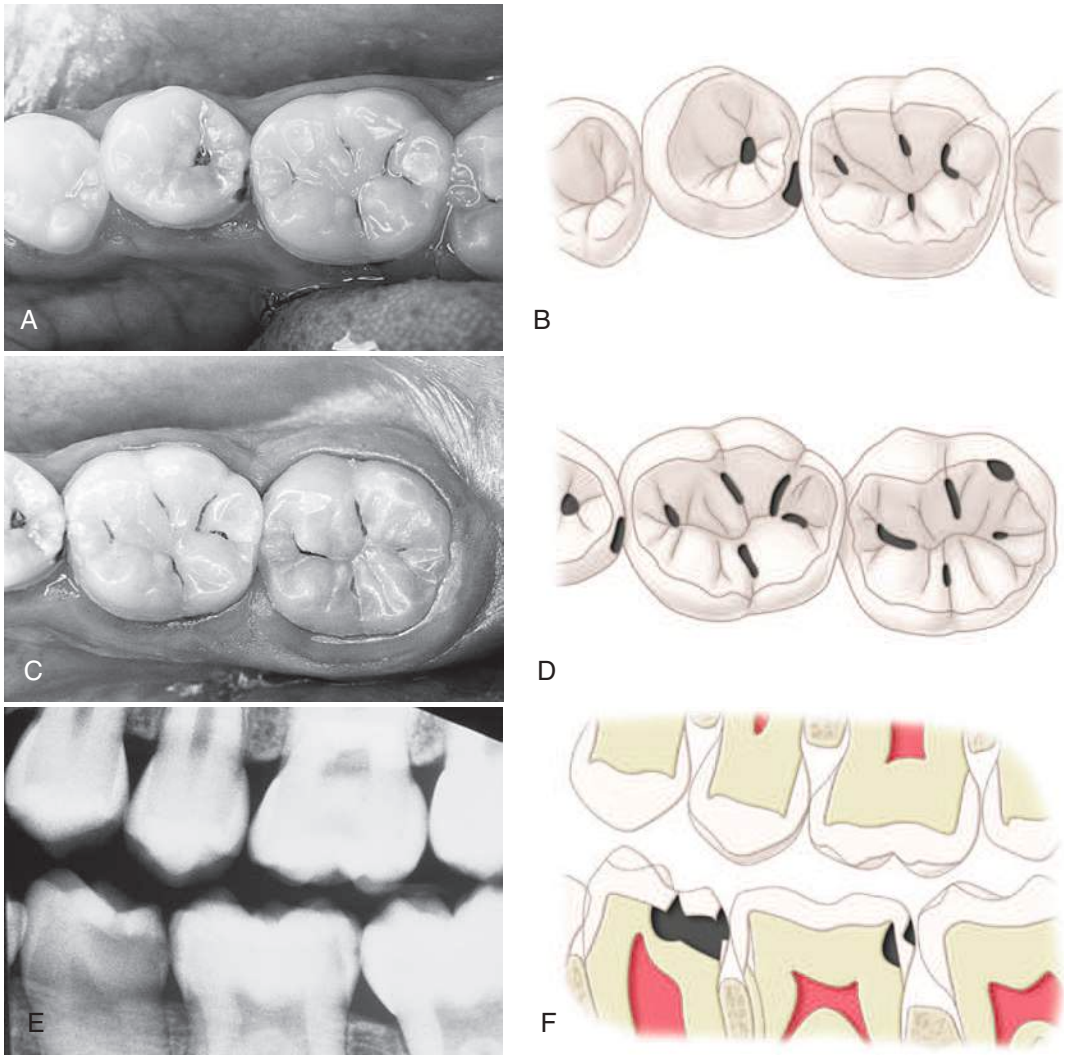


• **Fig. 2.18** Progression of pit-and-fissure caries. A, The mandibular right first molar (tooth No. 30) was sealed. Note radiolucent areas under the occlusal enamel in A and B. The sealant failed, and caries progressed slowly during the next 5 years; the only symptom was occasional biting-force pain. This underscores the need to monitor the sealant for failures. C and D, Note the extensive radiolucency under the enamel and an area of increased radiopacity below the lesion, suggesting sclerosis. Conversely, a well-sealed surface can prevent lesion progression. E and F, This maxillar first molar (tooth No. 14) with a shadow and a radiolucency indicative of dental caries was sealed as part of a clinical study.

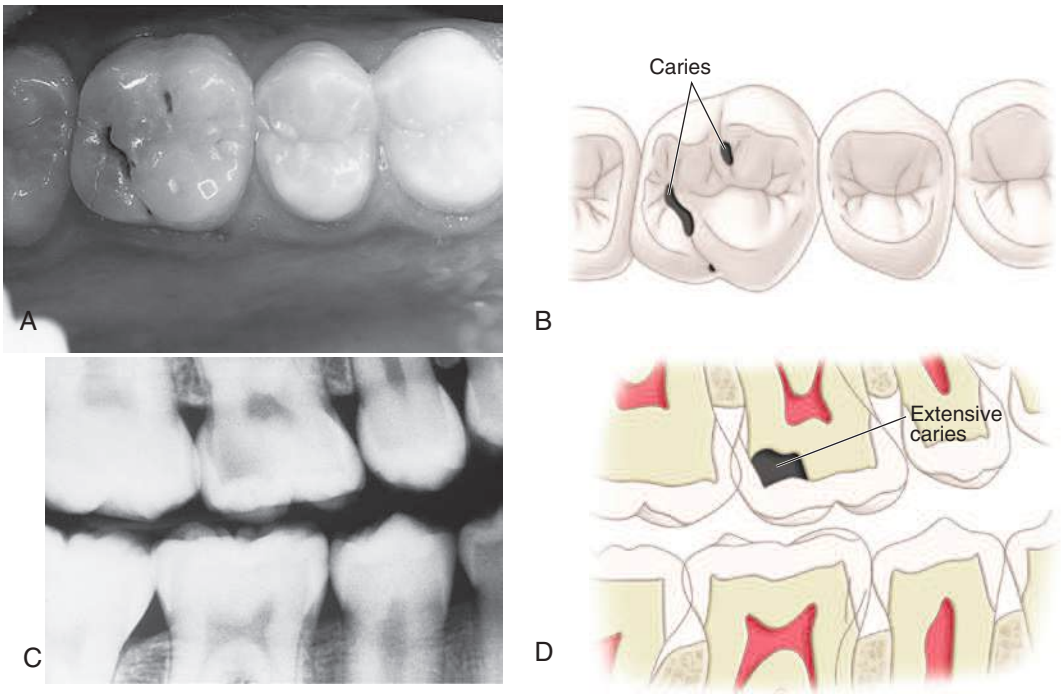
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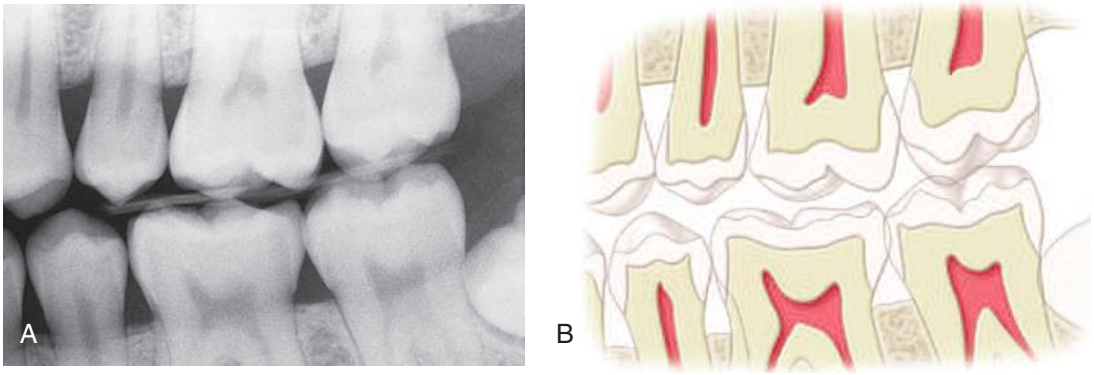
• **Fig. 2.18, cont'd** G–J, The tooth was followed clinically and by bitewing radiographs over the next 8 years. K and L, No progression of the lesion was observed. (E–L, Courtesy Dr. Azam Bakhshandeh, Department of Odontology, Faculty of Health and Medical Sciences, University of Copenhagen, Department of Cariology, Endodontics, and Pediatric Dentistry, Copenhagen, Denmark.)



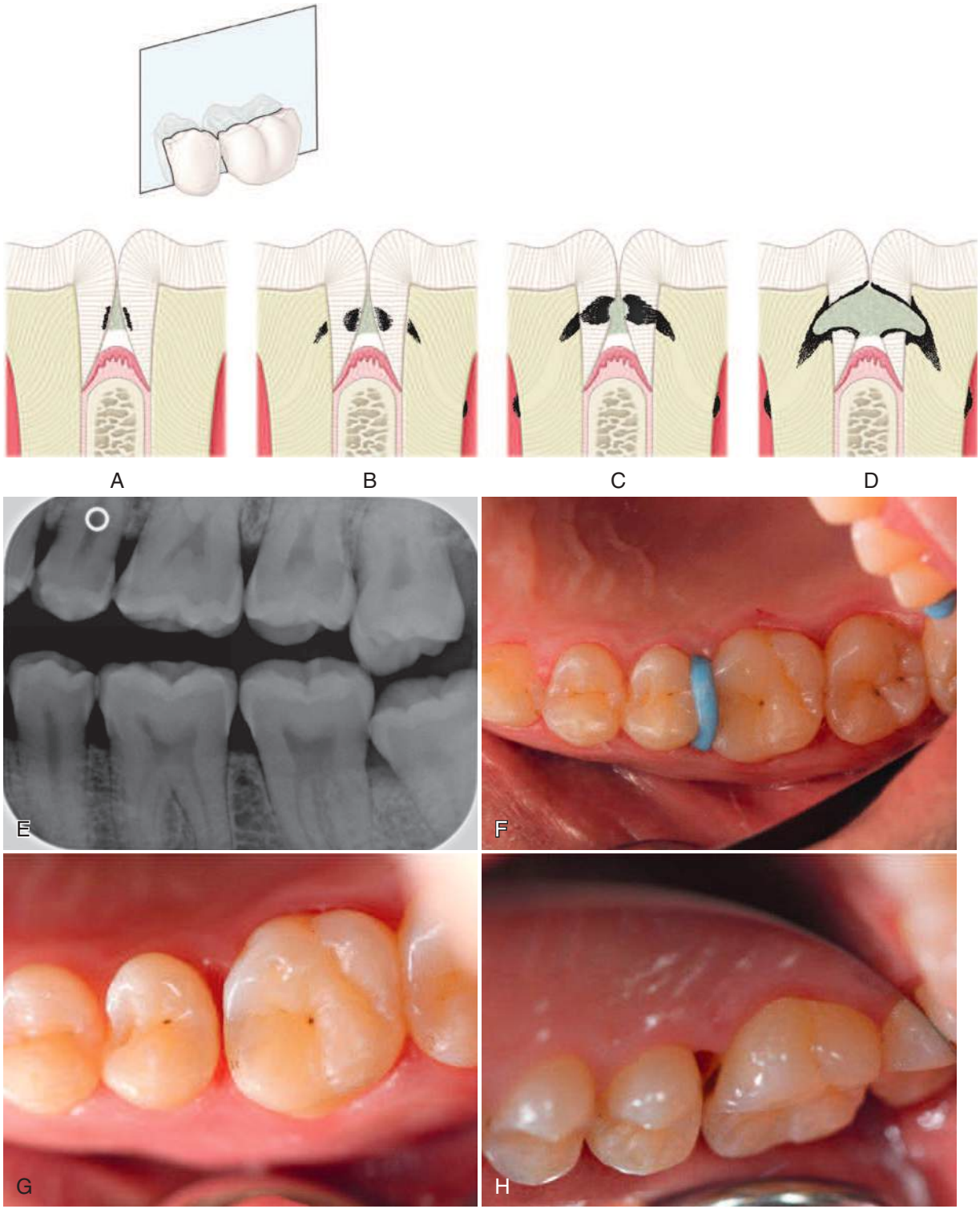
• **Fig. 2.19** A young patient with extensive caries. A and B, The occlusal pits of the first molar and second premolar are carious. An interproximal caries lesion is seen on the second premolar. The second premolar is rotated almost 90 degrees, bringing the lingual surface into contact with the mesial surface of the first molar. Normally, the lingual surfaces of mandibular teeth are rarely affected by caries, but here tooth rotation makes the lingual surface a proximal contact and consequently produces an interproximal habitat, which increases the susceptibility of the surface to caries. C and D, The first and second molars have extensive caries lesions in the pits and fissures. E and F, On the bitewing radiograph, not only can the extensive nature of the caries lesion in the second premolar be seen but also seen is a lesion on the distal aspect of the first molar, which is not visible clinically. (Dark areas in B, D, and F indicate caries.)



• **Fig. 2.20** Example of occlusal caries that has undermined enamel. Note the dark shadow from the dentin next to the stained fissure. A and B, Clinical example. C and D, A bitewing radiograph further confirms the extensive area of demineralization undermining the distofacial cusp.



• **Fig. 2.21** Bitewing radiograph of normal teeth, free from caries. Note the uniform density of the enamel on the interproximal surfaces. A third molar is impacted on the distal aspect of the lower second molar. The interproximal bone levels are uniform and located slightly below the cementoenamel junctions, suggesting a healthy periodontium.



• **Fig. 2.22** Longitudinal sections (see inset for A) showing initiation and progression of caries on interproximal surfaces. A, Initial demineralization (indicated by the shading in the enamel) on the proximal surfaces is difficult to detect clinically or radiographically. The lesions develop right below the contact surface. Flossing of the interproximal area and thorough drying will allow for the best conditions to detect these lesions clinically. B, When proximal caries lesions first become detectable radiographically, the enamel surface is likely still to be intact. An intact surface is essential for successful remineralization and arrest of the lesion. Demineralization of the dentin (indicated by the shading in the dentin) occurs before cavitation of the surface of the enamel. Treatment designed to promote remineralization can be effective up to this stage. C, Cavitation of the enamel surface is a critical event in the caries process in proximal surfaces. After cavitation, lesion arrest becomes more difficult and virtually impossible in an interproximal surface contact. Restorative treatment to repair the damaged tooth surface is required. Cavitation can be diagnosed only by clinical observation. The use of a sharp explorer to detect cavitation is problematic because excessive force in application of the explorer tip during inspection of the proximal surfaces can damage weakened enamel and accelerate the caries process by iatrogenic cavitation. Separation of the teeth using orthodontic separators can be used to provide more direct visual inspection of suspect surfaces. Fiberoptic illumination although not specific for cavitation can facilitate detection; an orange shadow can be seen when lesions reach the dentin. D, Advanced cavitated lesions require prompt restorative intervention to prevent pulpal disease, limit tooth structure loss, and remove the nidus of infection of odontopathic organisms. E, Radiographic image showing radiolucency on distal of tooth No. 13 and mesial of tooth No. 14. F, An orthodontic separator was placed and left in place for 3 days. G, Clinical image of the teeth after separation. H, There was no evidence of cavitation on either surface.

Oral Hygiene and Its Role in the Dental Caries Process

Oral hygiene, accomplished primarily by proper tooth brushing and flossing, is another ecologic determinant of dental caries onset and activity. Careful mechanical cleaning of teeth disrupts the biofilm and leaves a clean enamel surface. The cleaning process does not destroy most of the oral bacteria but merely removes them from the surfaces of teeth. Large numbers of these bacteria subsequently are removed from the oral cavity during rinsing and/or swallowing after flossing and brushing, but sufficient numbers remain to recolonize teeth. Some fastidious organisms and obligate anaerobes may be killed by exposure to oxygen during tooth cleaning; however, no single species is likely to be entirely eliminated. Although all the species that compose mature biofilm continue to be present, most of these are unable to initiate colonization on the clean tooth surface.

Saliva: Nature's Anticaries Agent

Saliva is an extremely important substance for the proper digestion of foods, and it also plays a key role as a natural anticaries agent (Table 2.3). The importance of saliva for oral health is dramatically noted after therapeutic radiation to the head and neck. After radiation, salivary glands become fibrotic and produce little or no saliva, leaving the patient experiencing an extremely dry mouth (xerostomia) (*xero*, dry; *stoma*, mouth), a condition termed *hyposalivation.* Such patients may experience near-total destruction of the teeth in just a few months after radiation treatment.[8,9] Patients that have autoimmune disease (e.g., Sjögren syndrome) also suffer from hyposalivation and experience severe xerostomia, and can experience the same devastating effects of hyposalivation on their dentition. In addition, many common medications are capable of reducing salivary flow and increasing caries risk (Table 2.4). Salivary protective mechanisms that maintain the normal oral flora and tooth surface integrity include bacterial clearance, direct antibacterial activity, buffers, and remineralization.[10]

Bacterial Clearance

Secretions from various salivary glands pool in the mouth to form whole or mixed saliva. The amount of saliva secreted varies greatly over time. When secreted, saliva remains in the mouth for a short time before being swallowed. While in the mouth, saliva lubricates oral tissues and bathes teeth and the biofilm. The secretion rate of saliva may have a bearing on dental caries susceptibility and calculus formation. Adults produce 1 to 1.5 L of saliva a day, very little of which occurs during sleep. The flushing effect of this salivary flow is, by itself, adequate to remove virtually all microorganisms not adherent to an oral surface. The flushing is most effective during mastication or oral stimulation, both of which produce large volumes of saliva. Large volumes of saliva also can dilute and buffer biofilm acids.

Direct Antibacterial Activity

Salivary glands produce an impressive array of antimicrobial products (see Table 2.3). Lysozyme, lactoperoxidase, lactoferrin, and agglutinins possess antibacterial activity. These salivary proteins are not part of the immune system but are part of an overall protection scheme for mucous membranes that occurs in addition to immunologic control. These protective proteins are present continuously at relatively uniform levels, have a broad spectrum of activity, and do not possess the "memory" of immunologic mechanisms. However, it is important to note that the normal resident oral flora apparently has developed resistance to most of these antibacterial mechanisms.

Although the antibacterial proteins in saliva play an important role in the protection of soft tissue in the oral cavity from infection by pathogens, they have little effect on dental caries because similar levels of antibacterial proteins can be found in caries-active and caries-free individuals.[11,12] It is suggested that dental caries susceptibility in healthy individuals is not related to saliva composition.

TABLE 2.3 Elements of Saliva That Control Biofilm Communities

Names	Action	Effects on Biofilm Community
Salivary Enzymes		
Amylase	Cleaves—1,4 glucoside bonds	Increases availability of oligosaccharides
Lactoperoxidase	Catalyzes hydrogen peroxide–mediated oxidation; adsorbs to hydroxyapatite in active form	Lethal to many organisms; suppresses biofilm formation on tooth surfaces
Lysozyme	Lyses cells by degradation of cell walls, releasing peptidoglycans; binds to hydroxyapatite in active conformation	Lethal to many organisms; peptidoglycans activate complement; suppresses biofilm formation on tooth surfaces
Lipases	Hydrolysis of triglycerides to free fatty acids and partial glycerides	Free fatty acids inhibit attachment and growth of some organisms
Nonenzyme Proteins		
Lactoferrin	Ties up free iron	Inhibits growth of some iron-dependent microbes
Secretory immunoglobulin A(IgA) (smaller amounts of IgM, IgG)	Agglutination of bacteria inhibits bacterial enzymes	Reduces numbers in saliva by precipitation; slows bacterial growth
Glycoproteins (mucins)	Agglutination of bacteria	Reduces numbers in saliva by precipitation

TABLE 2.4 Medications With Potential to Cause Hyposalivation or Dry Mouth (Xerostomia)

Action/Medication Group	Medicaments
Sympathomimetic	
Antidepressants	Venlafaxine
	Duloxetine
	Reboxetine
	Bupropion
Anticholinergic	
Tricyclic antidepressants	Amitriptyline
	Clomipramine
	Amoxapine
	Protriptyline
	Doxepin
	Imipramine
	Trimipramine
	Nortriptyline
	Desipramine
Muscarinic receptor antagonists	Oxybutynin
Alpha-receptor antagonists	Tamsulosin
	Terazosin
Antipsychotics	Promazine
	Trifluprom azine
	Mesoridazine
	Thioridazine
	Clozapine
	Olanzapine
Antihistamines	Aziridine
	Brompheniramine
	Chlorpheniramine
	Cyproheptadine
	Dexchlorpheniramine
	Hydroxyzine
	Phenindamine
	Cetirizine
	Loratadine
Anticholinergic, Dehydration	
Diuretics	Furosemide
	Bumetanide
	Torsemide
	Ethacrynic acid
Sympathomimetic	
Antihypertensive agents	Metoprolol
	Moxonidine
	Rilmenidine
Appetite suppressants	Fenfluramine
	Sibutramine
	Phentermine
Decongestants	Pseudoephedrine
Bronchodilators	Tiotropium
Skeletal muscle relaxants	Tizanidine
Antimigraine agents	Rizatriptan
Synergistic Mechanism	
Opioids, hypnotics	Opium
	Cannabis
	Tramadol
	Diazepam
Unknown	
H2 antagonists, proton pump inhibitors	Cimetidine
	Ranitidine
	Famotidine
	Nizatidine
	Omeprazole
Cytotoxic drugs	Fluorouracil
Anti-HIV drugs, protease inhibitors	Didanosine

Adapted from the Kois Center, Support Materials, Always Pages, http://koiscenter.com/store/supmatlist.aspx, accessed January 13, 2012.

Individuals with decreased salivary production (owing to illness, medication, or irradiation) may have significantly higher caries susceptibility (see Table 2.4).

Buffer Capacity

The volume and buffering capacity of saliva available to tooth surfaces have major roles in dental caries protection.[13] The buffering capacity of saliva is determined primarily by the concentration of bicarbonate ion. Buffering capacity can be estimated by titration techniques and may be a useful method for assessment of saliva in caries-active patients. Simple commercial kits are available for chairside tests. The benefit of the buffering is to reduce the potential for acid formation; however, in pathogenic biofilms there are highly compartmentalized clusters of acidic pH and neutral pH that make neutralization by saliva difficult.[210, 211]

In addition to buffers, saliva contains molecules that contribute to increasing biofilm pH. These include urea and sialin, the latter of which is a tetrapeptide that contains lysine and arginine. Hydrolysis of either of these basic compounds results in production of ammonia, causing the pH to increase.

Because saliva is crucial in controlling the oral flora and the mineral content of teeth, salivary testing should be done on patients with signs and symptoms of hyposalivation. Signs include lack of saliva pool on the floor of the mouth, gingivitis and mucositis (including burning mouth syndrome), cheilitis (inflammation and fissuring of the lips and/or the tongue), fissure in the soft mucosa, fungal infections in the mouth such as thrush, glossodynia (painful tongue), sialadenitis (salivary gland infection), saliva that seems thick and stringy, and several caries lesions especially in uncommon areas such as the lower central incisors. Symptoms include complaints of "cotton mouth" (xerostomia); bad breath; difficulty chewing, speaking, and swallowing; a changed sense of taste; and problems wearing dentures.

Remineralization

Saliva and biofilm fluid are supersaturated with calcium and phosphate ions. Without a means to control precipitation of these ions, the teeth literally would become encrusted with mineral deposits. Saliva contains statherin, a proline-rich peptide that stabilizes calcium and phosphate ions and prevents excessive deposition of

these ions on teeth.[14] This supersaturated state of the saliva provides a constant opportunity for remineralizing enamel and can help protect teeth in times of cariogenic challenges.

Diet and Dental Caries

High-frequency exposure to fermentable carbohydrates such as sucrose may be the most important factor in producing cariogenic biofilm and ultimately caries lesions. Frequent ingestion of fermentable carbohydrates begins a series of changes in the local tooth environment, essentially changing the composition of the biofilm, thus favoring the growth of highly acidogenic bacteria that eventually leads to caries lesion formation (see Fig. 2.4). In contrast, when ingestion of fermentable carbohydrates is severely restricted or absent, biofilm growth typically does not lead to caries lesions. Dietary sucrose plays a leading role in the development of pathogenic biofilms and may be the most important factor in disruption of the normal healthy ecology of dental biofilm communities. Sucrose in particular allows the formation of extracellular polysaccharides, which render the biofilm viscous and sticky. Because the eventual metabolic product of cariogenic diet is acid, in addition to caries lesions the exposure to acidity from other sources (e.g., dried fruits, fruit drinks, or other acidic foods and drinks) also may result in dental erosion. The dietary emphasis must include all intakes that result in acidity, not just sucrose.

Clinical Characteristics of the Caries Lesion

As discussed, the caries lesion results from disequilibrium between the demineralization and remineralization processes. However, caries lesions appear in a variety of different clinical presentations depending on lesion location, severity, and progression rate or activity of the demineralization process, among other factors. Clinically, caries lesions can be present on various tooth surfaces, can be cavitated or noncavitated, active or inactive, and vary depending on their severity (initial, moderate, or severe). For a complete description of current caries classification criteria based on lesion severity and activity, see Chapter 3. The following sections will summarize these characteristics.

Clinical Sites for Caries Initiation

The characteristics of a caries lesion vary with the nature of the surface on which the lesion develops. There are three distinctly different clinical sites for caries lesion initiation: (1) developmental pits and fissures of enamel, which are the most susceptible sites; (2) smooth enamel surfaces that shelter cariogenic biofilm; and (3) root surfaces (see Fig. 2.15). Each of these areas has distinct surface morphology and environmental conditions. Consequently each area has a distinct biofilm population. The diagnosis, treatment, and prevention of these different lesion types should take into account the different etiologic factors operating at each site.

Pits and Fissures

Bacteria rapidly colonize the pits and fissures of newly erupted teeth. The type and nature of the organisms prevalent in the oral cavity determine the type of organisms colonizing pits and fissures and are instrumental in determining the outcome of the colonization. Large variations exist in the microflora found in pits and fissures, suggesting that each site can be considered a separate ecologic system. Numerous gram-positive cocci, especially *S. sanguis*, are found in the pits and fissures of newly erupted teeth, whereas large numbers of MS usually are found in carious pits and fissures.

The shape of the pits and fissures contributes to their high susceptibility to caries lesion. The long, narrow fissure prevents adequate biofilm removal (see Fig. 2.13). Considerable morphologic variation exists in these structures. Some pits and fissures end blindly, others open near the dentin, and others penetrate entirely through the enamel.

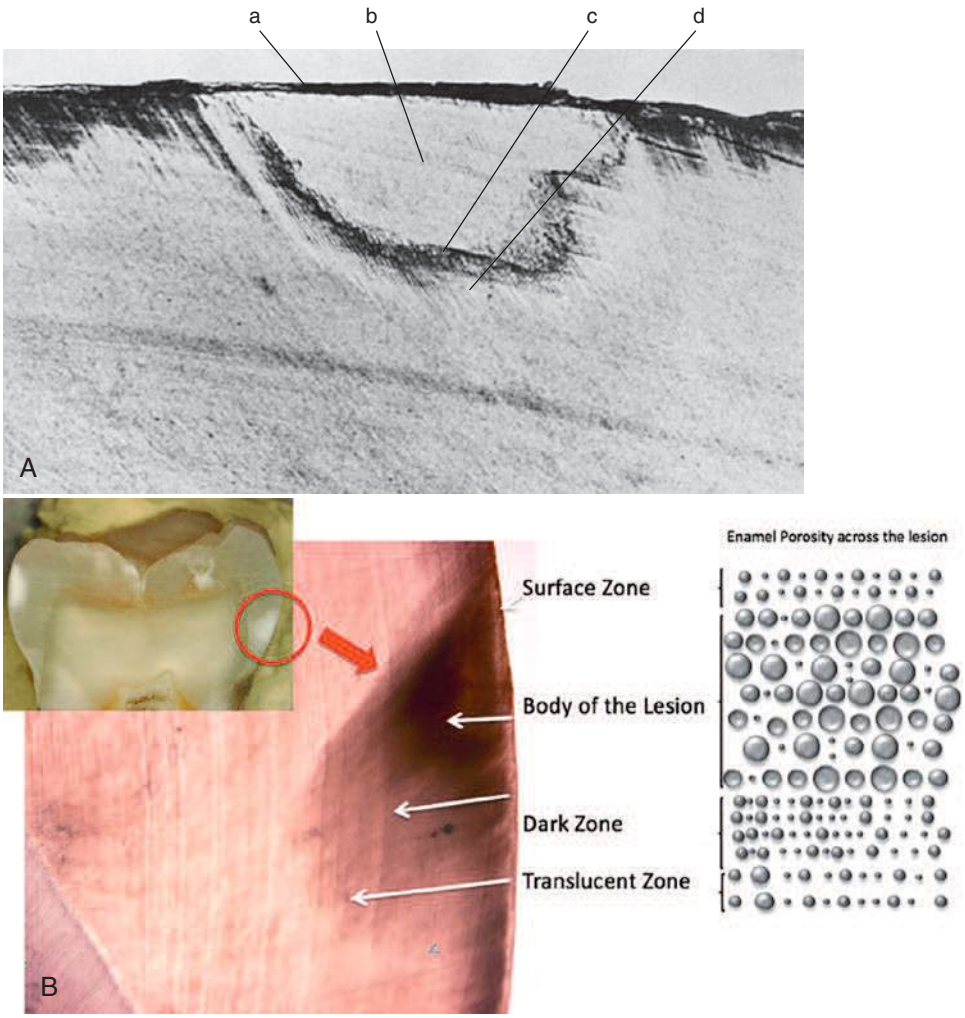
Pit-and-fissure caries lesions typically "expand" as they penetrate into the enamel. The entry site may appear much smaller than the actual lesion, making clinical detection difficult. Caries lesions of pits and fissures develop from demineralization on their walls, known as *wall lesions* (see Fig. 2.16A–C). Progression of the dissolution of the walls of a pit-and-fissure lesion is similar in principle to that of the smooth-surface lesion because a wide susceptible surface area extends inward, paralleling the enamel rods. A lesion originating in a pit or fissure affects a greater area of the dentinoenamel junction (DEJ) than does a comparable smooth-surface lesion. In cross section, the gross appearance of a pit-and-fissure lesion is an inverted V with a narrow entrance and a progressively wider area of involvement closer to the DEJ (see Fig. 2.16D).

Smooth Enamel Surfaces

The smooth enamel surfaces of teeth present a less favorable site for cariogenic biofilm attachment. Cariogenic biofilm usually develops only on the smooth surfaces that are near the gingiva or are under proximal contacts. The proximal surfaces are particularly susceptible to caries lesion formation because of the extra shelter provided to resident cariogenic biofilm owing to the proximal contact area immediately occlusal to it (Fig. 2.22 and 2.23). Lesions starting on smooth enamel surfaces have a broad area of origin and a conical, or pointed, extension toward the DEJ. The path of ingress of the lesion is roughly parallel to the long axis of the enamel rods in the region. A cross section of the enamel portion of a smooth-surface lesion shows a V shape, with a wide area of origin and the apex of the V directed toward the DEJ. After the demineralization reaches the DEJ, when there is enamel cavitation and bacterial contamination, the bacterial metabolic activity within the lesion causes rapid dentin demineralization both laterally and pulpally (see Fig. 2.22).[142,143]



• **Fig. 2.23** Extracted tooth showing extensive caries lesion just gingival to the proximal contact area. (Note the slightly "flat" contact area adjacent to marginal ridge.)



• **Fig. 2.24** A, Cross section of small caries lesion in enamel examined by transmitted light (100×). Surface *(a)* appears to be intact. Body of lesion *(b)* shows enhancement of striae of Retzius. Dark zone *(c)* surrounds body of lesion, whereas translucent zone *(d)* is evident over entire advancing front of lesion. B, The porosity of the lesion changes across the different zones. Thus a sharp explorer can easily penetrate the outer enamel and cause a small cavitation on an active lesion. (A, From Silverstone LM, et al. editors: *Dental caries,* London and Basingstoke, 1981, Macmillan, Ltd. Reproduced with permission of Palgrave Macmillan. B, Courtesy Dr. Masatoshi Ando.)

Root Surfaces

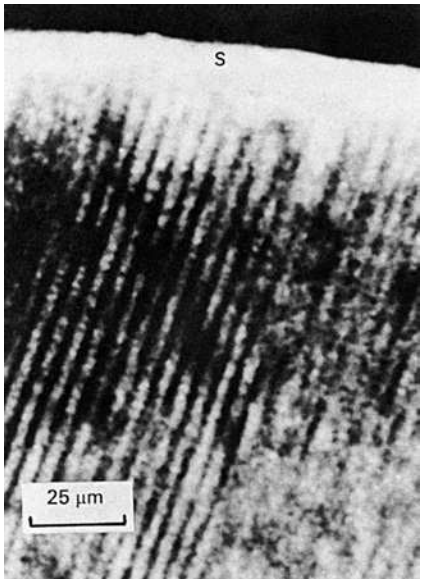
The root surface is rougher than enamel and readily allows cariogenic biofilm formation in the absence of good oral hygiene. The cementum covering the root surface is extremely thin and provides little resistance to caries lesion activity. In addition, the critical pH for dentin is higher than for enamel, so demineralization is likely to start even before the pH reaches the critical level for enamel (pH = 5.5). Root caries lesions have less well-defined margins than coronal caries lesions, tend to be U shaped in cross section, and progress more rapidly than coronal caries lesions because of the lack of protection from an enamel covering. In recent years, the prevalence of root caries has increased significantly because of the increasing number of older persons who retain more teeth, experience gingival recession, and usually have cariogenic biofilm on the exposed root surfaces.[15-18]

Caries Lesion Progression

The progression and morphology of the caries lesion vary depending on the site of origin and the conditions in the mouth (see Figs. 2.15, 2.16, and 2.22). The time for progression from a noncavitated caries lesion to a cavitated caries lesion on smooth surfaces is estimated to be 18 months (± 6 months).[19] Peak rates for the incidence of new lesions occur 3 years after the eruption of the tooth. Occlusal pit-and-fissure lesions develop in less time than smooth-surface caries lesions. Poor oral hygiene and frequent exposures to sucrose-containing or acidic food can produce noncavitated initial ("white spot") lesions (first clinical evidence of demineralization) in 3 weeks. Radiation-induced *hyposalivation* (dry mouth) can lead to development of caries lesions in 3 months from the onset of the radiation. Caries lesion progression in healthy individuals is usually slow compared with the progression in compromised persons.

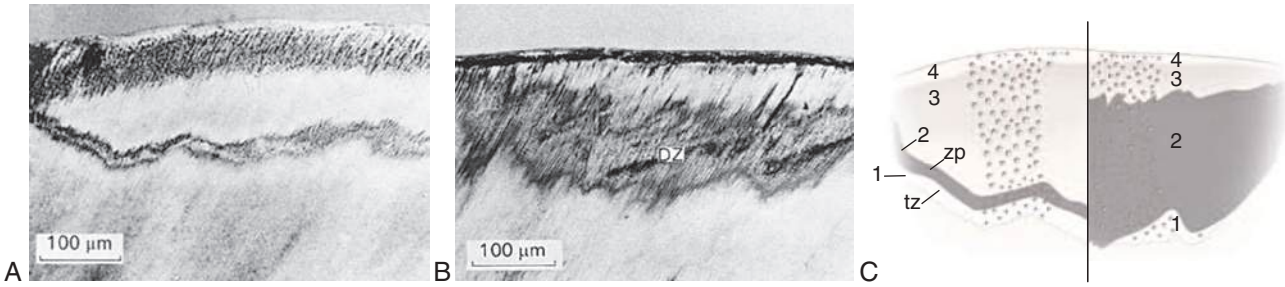
Enamel Caries Lesions

An understanding of the enamel composition and histology is helpful to understand enamel caries histopathology (see Chapter 1; Figs. 2.24, 2.25, and 2.26). On clean, dry teeth the earliest evidence of caries lesion on the smooth enamel surface of a crown

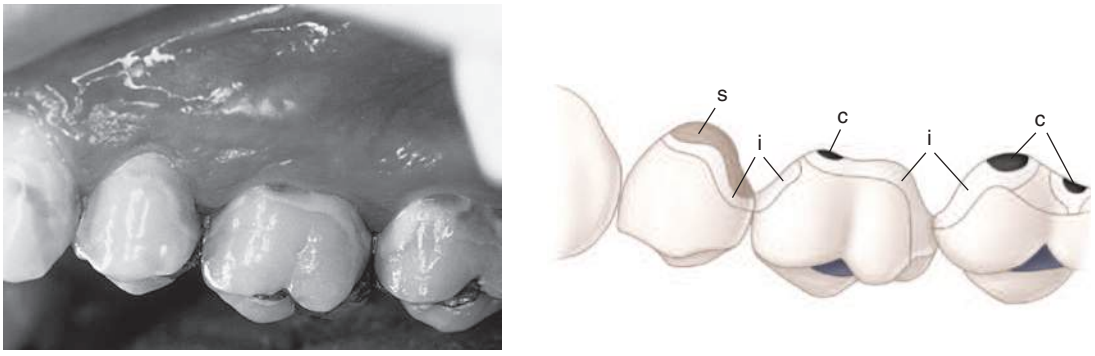


• **Fig. 2.25** Microradiograph (150×) of cross section of small caries lesion in enamel. Well-mineralized surface *(s)* is evident. Alternating radiolucent and radiopaque lines indicate demineralization between enamel rods. (From Silverstone LM, et al, editors: *Dental caries*, London and Basingstoke, 1981, Macmillan, Ltd. Reproduced with permission of Palgrave Macmillan.)

is the initial lesion or a "white spot" (Fig. 2.27; see also Figs. 2.2 and 2.3). These lesions usually are observed on the facial and lingual surfaces of teeth. Initial lesions are chalky white, opaque areas that are revealed only when the tooth surface is desiccated and are termed *noncavitated enamel caries lesions* or *initial lesions*. These areas of enamel lose their translucency because of the extensive subsurface porosity caused by demineralization. Care must be exercised in distinguishing white spots of initial caries lesions from developmental white spot hypocalcifications or other developmental defects of enamel. Initial (white spot) caries lesions partially or totally disappear visually when the enamel is hydrated (wet), whereas hypocalcified enamel is affected less by drying and wetting (Table 2.5). Hypocalcified enamel does not represent a clinical problem except for its potential esthetically objectionable appearance. The surface texture of an initial lesion is unaltered and is undetectable by tactile examination with an explorer. A more advanced lesion develops an irregular surface that is rougher than the unaffected, normal enamel. Softened chalky enamel that can be chipped away with an explorer is a sign of active caries. However, injudicious use of an explorer tip can cause actual cavitation in a previously noncavitated area, requiring, in most cases, restorative intervention. Similar initial lesions occur on the proximal smooth surfaces, although their detection by visual examination is more challenging. Tooth separation with orthodontic separators can facilitate visual



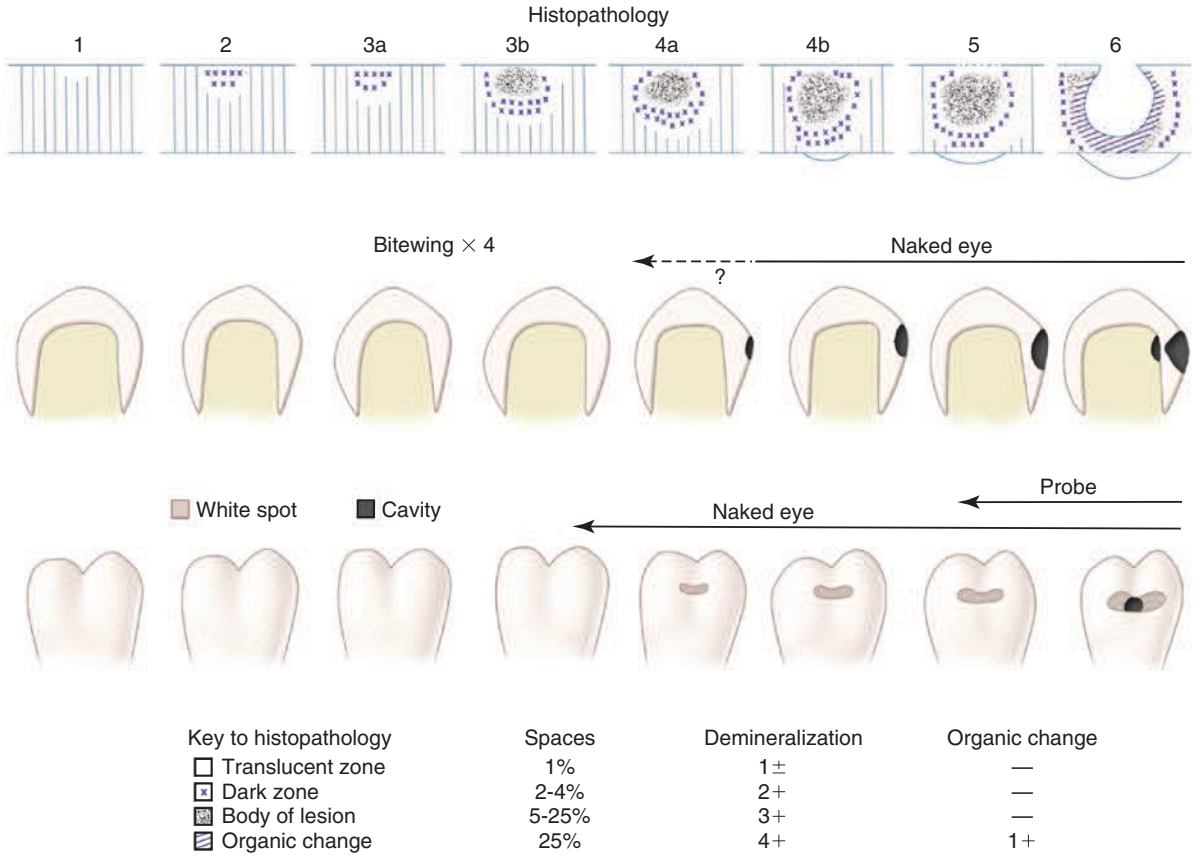
• **Fig. 2.26** A, Cross section of initial caries lesion in enamel examined in quinoline with polarized light (100×). Advancing front of lesion appears as a dark band below body of lesion. B, Same section after exposure to artificial calcifying solution examined in quinoline and polarized light. Dark zone *(DZ)* covers a much greater area after remineralization has occurred (100×). C, Schematic diagram of A and B. Left side indicates small extent of zones 1 and 2 before remineralization. Small circles indicate relative sizes of pores in each zone. Right side indicates increase in zone 2, the dark zone, after remineralization. This micropore system must have been created where previously the pores were much larger. (From Silverstone LM, et al, editors: *Dental caries*, London and Basingstoke, 1981, Macmillan, Ltd. Reproduced with permission of Palgrave Macmillan. *C* was redrawn.)



• **Fig. 2.27** Facial and lingual smooth-surface caries. This patient has high caries activity with rapidly advancing caries lesions. Cariogenic biofilm extends entirely around the cervical areas of the posterior teeth. Several levels of caries involvement can be seen, including cavitation *(c);* initial (noncavitated) white spot lesions *(i);* and stained, roughened, partially remineralized initial (noncavitated) lesions *(s).*

TABLE 2.5 Clinical Characteristics of Normal and Altered Enamel

	Hydrated	Desiccated	Surface Texture	Surface Hardness
Normal Enamel	Translucent	Translucent	Smooth	Hard
Hypocalcified Enamel	Opaque	Opaque	Smooth	Hard
Active Initial Caries	Translucent (early lesions) Opaque (more established initial lesions)	Opaque Opaque	Rough Rough	Softened Softened
Arrested Initial Caries	Shiny and/or dark	Shiny and/or dark	Smooth	Hard
Active Moderate Caries	Opaque	Opaque	Rough	Softened
Arrested Moderate Caries	Shiny and/or dark	Shiny and/or dark	Smooth	Hard
Active Advanced Caries	Opaque	Opaque	Rough	Softened
Arrested Advanced Caries	Shiny and/or dark	Shiny and/or dark	Smooth	Leathery or hard



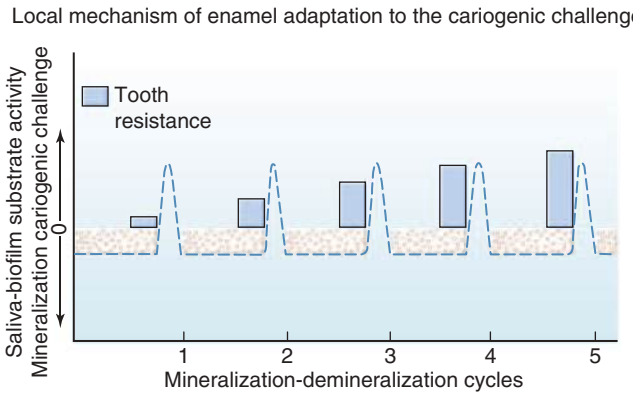
• **Fig. 2.28** Schematic representation of developmental stages of enamel caries lesion correlated with radiographic and clinical examination. Cavitation occurs late in development of the lesion. Before cavitation remineralization is possible, lesion arrest is possible at *any* stage given the right conditions. (Redrawn from Darling AI: The pathology and prevention of caries, *Br Dent J* 107:287–302, 1959. Reproduced by kind permission of the editor.)

examination of proximal tooth surfaces. Initial enamel lesions sometimes may be seen on radiographs as a faint radiolucency that is limited to the superficial enamel. When a proximal lesion is clearly visible radiographically, the lesion may have advanced significantly, and histologic alteration of the underlying dentin probably already has occurred, whether the lesion is cavitated or not (Fig. 2.28).

It has been shown experimentally and clinically that initial enamel caries lesions can remineralize[20,21] whether the dentin is involved or not. Table 2.5 and Table 2.6 list the characteristics of enamel at various stages of demineralization. Initial, noncavitated enamel lesions retain most of the original crystalline framework of the enamel rods, and the etched crystallites serve as nucleating agents for remineralization. Calcium and phosphate ions from

TABLE 2.6 Clinical Significance of Enamel Lesions

	Biofilm	Enamel Structure	Nonrestorative, Therapeutic Treatment (e.g., Remineralization, Antimicrobial, pH control)	Restorative Treatment
Normal Enamel	Normal	Normal	Not indicated	Not indicated
Hypocalcified Enamel	Normal	Abnormal, but not weakened	Not indicated	Only for esthetics
Noncavitated Caries	Cariogenic	Porous, weakened	Yes	Not indicated
Active Caries	Cariogenic	Cavitated, very weak	Yes	Yes
Inactive Caries	Normal	Remineralized, strong	Not indicated	Only for esthetics



• **Fig. 2.29** Diagrammatic representation of enamel adaptation reaction. Enamel interacts with its fluid environment in periods of undersaturation and supersaturation, presented here as periodic cycles. Undersaturation periods dissolve most soluble mineral at the site of cariogenic attack, whereas periods of supersaturation deposit most insoluble minerals if their ionic components are present in immediate fluid environment. As a result, under favorable conditions of remineralization, each cycle could lead toward higher enamel resistance to a subsequent challenge. (Redrawn from Koulouirides T: In Menaker L, editor: *The biologic basis of dental caries*, New York, 1980, Harper & Row.)

saliva can penetrate the enamel surface and precipitate on the highly reactive crystalline surfaces in the enamel lesion. The supersaturation of saliva with calcium and phosphate ions serves as the driving force for the remineralization process. Artificial and natural caries lesions of human enamel have been shown to regress to earlier histologic stages after exposure to conditions that promote remineralization. The presence of trace amounts of fluoride ions during this remineralization process greatly enhances the precipitation of calcium and phosphate, resulting in the remineralized enamel becoming more resistant to subsequent caries activity because of the incorporation of more acid-resistant fluorapatite (Fig. 2.29). Remineralized (arrested) lesions may be observed clinically as either intact, smooth white lesions or discolored, usually brown or black, spots (Fig. 2.30). The change in color is presumably caused by trapped organic debris and metallic ions within the enamel. These discolored, remineralized, arrested caries lesion areas are more resistant to subsequent caries activity than the adjacent unaffected enamel. They should not be restored unless they are esthetically objectionable.

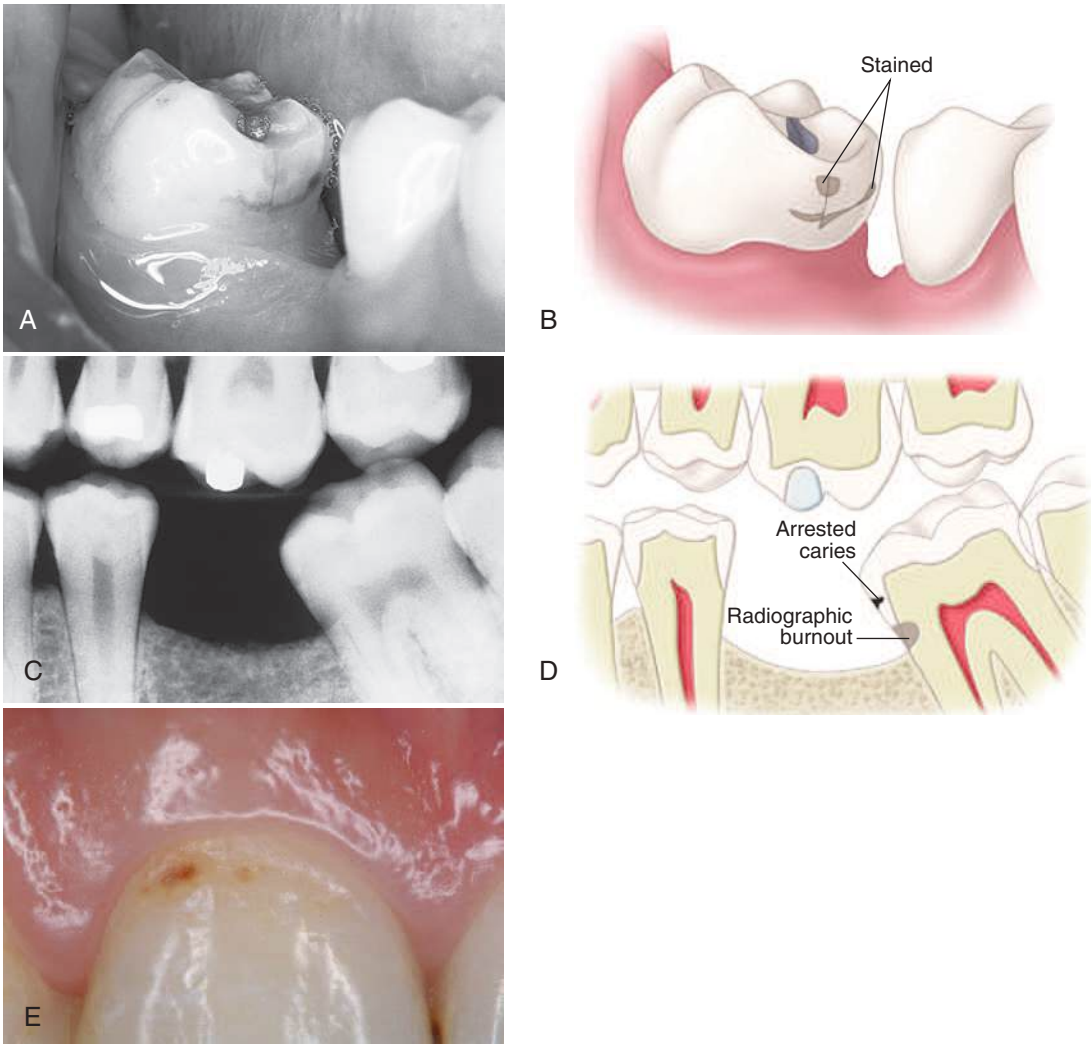
Cavitated enamel lesions can be initially detected as subtle breakdown of the enamel surface. These lesions are very sensitive to probing and can be easily enlarged by using sharp explorers and excessive probing force. More advanced cavitated enamel lesions are more obviously detected as enamel breakdown. Although theoretically any caries lesion given the right conditions can be arrested and have their progression to larger lesions forestalled, once cavitation occurs removal of the biofilm is more difficult and many will require a sealant or restorative intervention. When the tooth surface cavitates, a more retentive site becomes available to the biofilm community. The cavitation of the tooth surface produces a synergistic acceleration of the growth of the cariogenic biofilm community and the expansion of the demineralization with ensuing expanded cavitation. This sheltered, highly acidic, and anaerobic environment provides an ideal niche for cariogenic bacteria. This situation results in a rapid and progressive destruction of the tooth structure.

Dentin Caries Lesions

An understanding of the dentin composition and histology is helpful to understand the histopathology of dentin caries lesions (see Chapter 1; Fig. 2.31). Progression of dental caries in dentin is different from progression in enamel because of the structural differences of dentin (Figs. 2.32, 2.33, and 2.34; see also Fig. 2.31). Dentin contains much less mineral and possesses microscopic tubules that provide a pathway for the ingress of bacteria and egress of minerals.

When enamel demineralization advances to the DEJ, rapid lateral expansion of the caries lesion along the DEJ may occur if there is bacterial contamination. This process is still poorly understood, but it likely occurs only when there is enamel cavitation. For many years this "lateral spread" has been attributed to the DEJ's lower mineral content compared to primary dentin.[137] More recently, it has been shown that a high concentration of MMPs is found at the DEJ. MMPs are implied in dentin caries lesion progression, although this may be a host defense mechanism and not necessarily linked to an increase in caries susceptibility.[138] What is known is that the increased susceptibility to demineralization observed at the DEJ on lesions that have reached dentin is a result of bacterial metabolic activity within the dentinal lesion.[139-141]

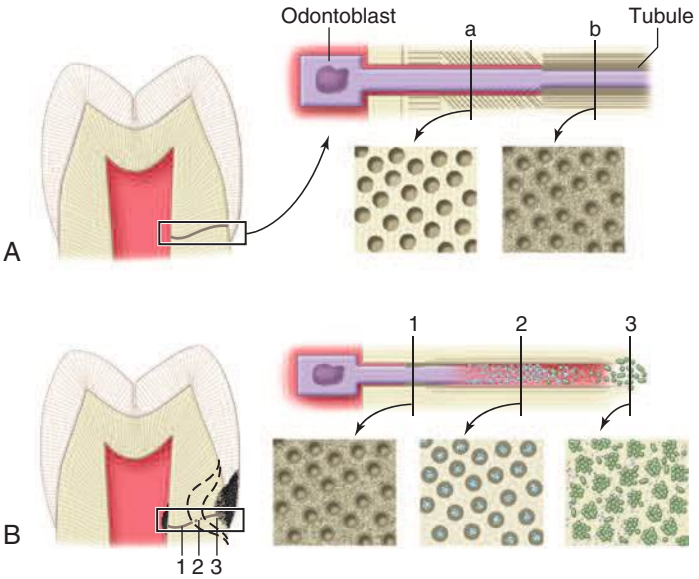
Increasing demineralization of the body of the enamel lesion results in the weakening and eventual collapse of the surface enamel. The resulting cavitation provides an even more protective and retentive habitat for the cariogenic biofilm, accelerating the progression of the lesion. The DEJ provides less resistance to the carious process than either enamel or dentin. The dentin contamination by bacterial metabolites leads to the lateral spread of the lesion at



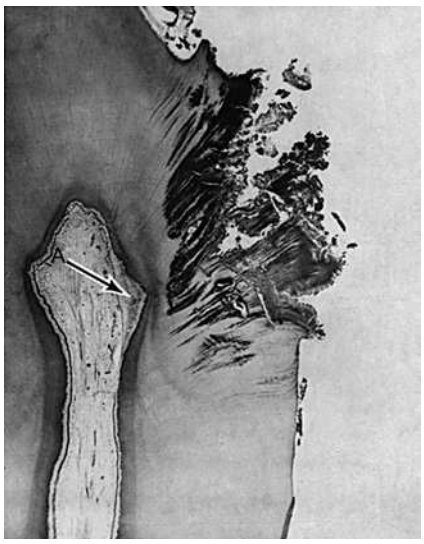
• **Fig. 2.30** A and B, Example of arrested caries lesion on the mesial surface of a mandibular second molar. The area below the proximal contact (A and B, mirror view of tooth No. 18) is partly opaque and stained. Clinically the surface is hard and intact, yet the area is more radiolucent than the enamel above or below the stain. C and D, In a different clinical case, caries diagnosis based only on the radiograph would lead to a false-positive diagnosis (i.e., caries lesion present when it is not). The radiolucency is caused by the broad area of subsurface demineralization that extends from the facial to the lingual line angles. The x-ray beam was directed parallel to the long axis of demineralization and consequently produced a sharply demarcated zone of radiolucency in the enamel. This example illustrates the shortcomings of radiographic diagnosis. Were there not visual access to the mesial surface of the second molar, it would be easy to diagnose active caries incorrectly and consequently restore the tooth. E, Cavitated inactive (arrested) enamel caries lesion on the cervical one third of a central incisor of a 27-year-old patient with low caries risk. This lesion, if not esthetically offensive, does not require a restoration and should be monitored.

the DEJ, producing the characteristic second cone of caries activity in dentin (see Figs. 2.16 and 2.22).[143] Figs. 2.31, 2.35, 2.36, 2.37, and 2.38 illustrate advanced lesions with soft, infected dentin. Because of these characteristics, dentin caries is V shaped in cross section with a wide base at the DEJ and the apex directed pulpally. Caries lesions advance more rapidly in dentin than in enamel because dentin provides much less resistance to acid demineralization owing to less mineralized content and the presence and orientation of the dentinal tubules.

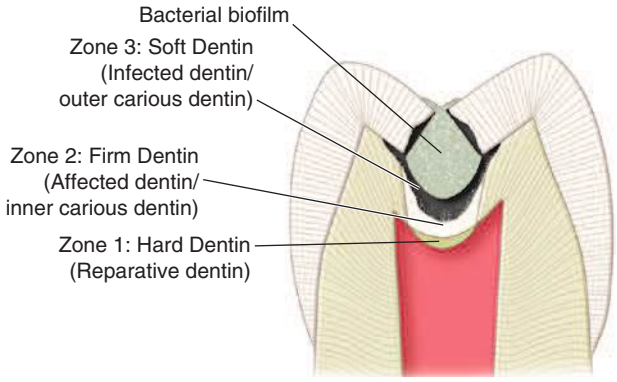
The dental caries process produces a variety of responses in dentin including pain, sensitivity, demineralization, and remineralization. Often pain is not reported even when caries lesions invade dentin except when deep lesions bring the bacterial infection close to the pulp. Episodes of short-duration pain may be felt occasionally during earlier stages of dentin caries lesion progression. The pain is caused by stimulation of pulp tissue by the rapid movement of fluid through the dentinal tubules that have been opened to the oral environment by cavitation. When bacterial invasion of the dentin is close to the pulp, toxins and possibly a few bacteria enter the pulp, resulting in inflammation of the pulpal tissues, increased pulpal pressure, and thus pulpal pain.



• **Fig. 2.31** Normal and carious dentin. A, Normal dentin has characteristic tubules that follow a wavy path from the external surface of dentin, below enamel or cementum, to the inner surface of dentin in the pulp tissue of the pulp chamber or pulp canal. Dentin is formed from the external surface and grows inward. As dentin grows, odontoblasts become increasingly compressed in the shrinking pulp chamber, and the number of associated tubules becomes more concentrated per unit area. The more recently formed dentin near the pulp *(a)* has large tubules with little or no peritubular dentin and calcified intertubular dentin filled with collagen fibers. Older dentin, closer to the external surface *(b),* is characterized by smaller, more widely separated tubules and a greater mineral content in intertubular dentin. The older dentin tubules are lined by a uniform layer of mineral termed *peritubular dentin*. These changes occur gradually from the inner surface to the external surface of the dentin. Horizontal lines indicate predentin; diagonal lines indicate increasing density of minerals; darker horizontal lines indicate densely mineralized dentin and increased thickness of peritubular dentin. The transition in mineral content is gradual, as indicated in Fig. 2.25. B, Carious dentin undergoes several changes. The most superficial infected zone of carious dentin *(3)* is characterized by bacteria filling the tubules and granular material in the intertubular space. The granular material contains very little mineral and lacks characteristic cross-banding of collagen. As bacteria invade dentinal tubules, if carbohydrates are available, they can produce enough lactic acid to remove peritubular dentin. This doubles or triples the outer diameter of the tubules in infected dentin zone. Pulpal to (below) the infected dentin is a zone where the dentin appears transparent in mounted whole specimens. This zone *(2)* is the advancing demineralization front, affected carious dentin, and is characterized by loss of mineral in the intertubular and peritubular dentin. Many crystals can be detected in the lumen of the tubules in this zone. The crystals in the tubule lumen render the refractive index of the lumen similar to that of the intertubular dentin, making the zone transparent. Normal dentin *(1)* is found pulpal to (below) transparent dentin.



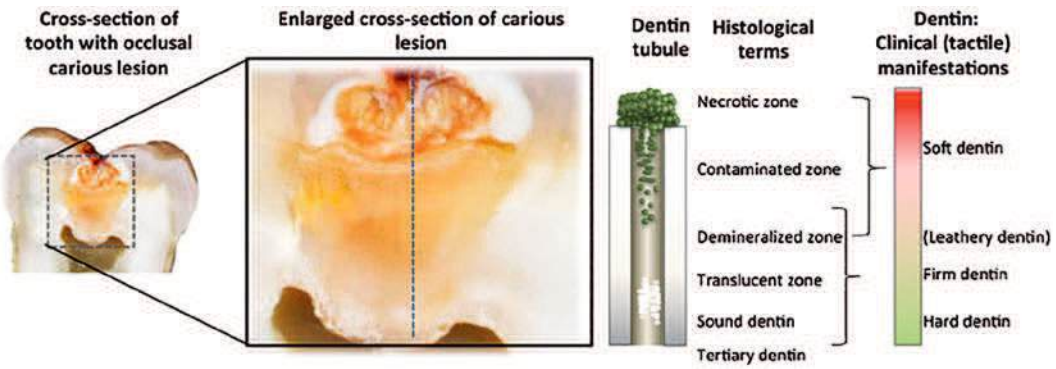
• **Fig. 2.32** Cross section of demineralized specimen of advanced caries in dentin. Reparative dentin *(A)* can be seen adjacent to the most advanced portion of lesion. (From Boyle P: *Kornfeld's histopathology of the teeth and their surrounding structures*, Philadelphia, 1955, Lea & Febiger.)



• **Fig. 2.33** Cross section of occlusal caries. The occlusal enamel appears intact, with a small opening in the occlusal fissure. Enamel is darkened where it is undermined by demineralization. The surface of enamel may not be clinically cavitated, but usually some discoloration is observed at the entrance of the fissure. The lesion is filled with a bacterial plug containing high numbers of MS and lactobacilli. When there is a break in the enamel, the dentin underneath the enamel will be highly contaminated below the plug. Deeper dentin is not as heavily contaminated but is extensively demineralized. Reparative dentin is being formed below the lesion.

The pulp–dentin complex reacts to caries activity by attempting to initiate remineralization and blocking of the open tubules. These reactions result from odontoblastic activity and the physical process of demineralization and remineralization. The pulp–dentin complex reaction occurs even before the lesion has reached the dentin. Dentin can react defensively (by repair) to low-intensity and moderate-intensity caries lesion progression as long as the pulp remains vital and has adequate blood circulation. Dentin reaction can happen through remineralization of intertubular dentin and apposition of peritubular dentin or by reparative dentin, as described in the next paragraphs.

In slowly advancing caries lesions, a vital pulp can repair demineralized dentin by remineralization of the intertubular dentin and by apposition of peritubular dentin. Initial stages of caries lesions or mild caries activity produce long-term, low-level acid demineralization of dentin. Direct exposure of the pulp tissue to microorganisms is not a prerequisite for an inflammatory response. Toxins and other metabolic by-products, especially hydrogen ion,

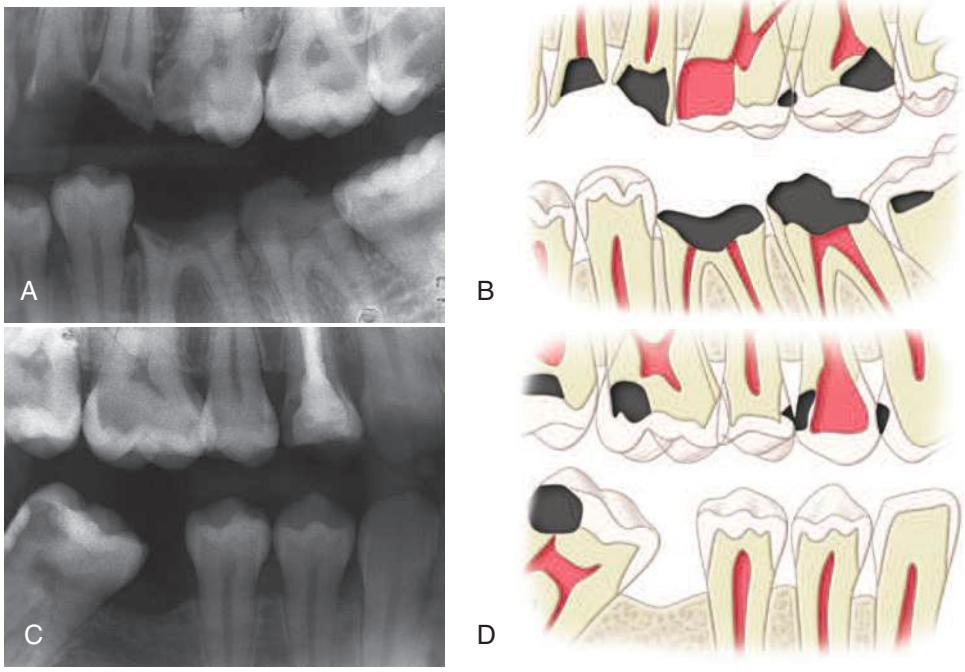


• **Fig. 2.34** A cross section of a caries lesion in dentin and the histologic and clinical manifestations (adapted from Ogawa et al, 1983). The most superficial outer layer of dentin is heavily contaminated, necrotic, with irreversible demineralization and denatured collagen fibers. Clinically this is very soft, wet dentin. This soft, contaminated layer gets progressively harder as the dentin approaches the pulp. The still demineralized dentin (formerly affected dentin) can feel leathery to the touch or firm, but not as hard as sound dentin or tertiary dentin (closest to the pulp, dentin formed in reaction to the caries process). During caries excavation, the goal is to remove only the soft, outer carious dentin (infected dentin), while the inner carious dentin (affected dentin) is remineralizable and can be maintained. (For orientation of layers on tooth, see Fig 2.35.) (Modified from Ogawa K, Yamashita Y, Ichijo T, et al: The ultrastructure and hardness of the transparent layer of human carious dentin. *J Dent Res* 62(1):7–10, 1983.)



• **Fig. 2.35** A, Rampant caries in a preschool child. B, Rampant caries in a school-age child. C, Rampant caries in an adult. *(A,* From Dean JA, Avery DR, McDonald RE: *McDonald and Avery's dentistry for the child and adolescent*, ed 9, St Louis, 2011, Mosby.)

can penetrate via the dentinal tubules to the pulp. Even when the lesion is limited to enamel, the pulp can be shown to respond with inflammatory cells.[22,23] Dentin responds to the stimulus of its first caries demineralization episode by deposition of crystalline material from the intertubular dentin in the lumen of the tubules in the advanced demineralization front (formally called affected dentin) (see Figs. 2.31 and 2.34). The refractive index of the dentin changes and the intertubular dentin with more mineral content than normal dentin is termed *sclerotic dentin*. The apparent function of the sclerotic dentin is to wall off a lesion by blocking (sealing) the tubules. The permeability of sclerotic dentin is greatly reduced compared with that of normal dentin because of the decrease in the tubule lumen diameter.[24] Hypermineralized areas may be seen on radiographs as zones of increased radiopacity (often S shaped, following the course of the tubules) ahead of the advancing, infected portion of the lesion. Sclerotic dentin formation may also be seen



• **Fig. 2.36** Rampant caries in a 21-year-old man. Although occlusal and interproximal lesions exist in the patient, the progress of the occlusal lesions produced the most tooth destruction. The potential for developing occlusal lesions could have been reduced by earlier application of sealants. This extensive amount of caries was the result of the patient's excessive fear of bad breath. In an attempt to keep his breath smelling fresh, he kept sugar-containing breath mints in his mouth most of the day. (Dark areas in B and D indicate caries.)



• **Fig. 2.37** Acute, rampant caries in both anterior (A) and posterior (B) teeth.



• **Fig. 2.38** Advance cavitated active caries lesions on a 35-year-old patient with high caries risk. (Courtesy Dr. Ayesha Swarn, DDS, Operative Dentistry Graduate Program at UNC School of Dentistry.)

under an old restoration. Sclerotic dentin is usually shiny and darker in color but feels hard to the explorer tip. By contrast, normal, freshly cut dentin lacks a shiny, reflective surface and allows some penetration from a sharp explorer tip. This repair by sclerotic dentin occurs only if the tooth pulp is vital.

When these affected tubules become completely occluded by the mineral precipitate, they appear clear when a histologic section of the tooth is evaluated. This portion of dentin has been termed *translucent dentin* (see Zones of Dentinal Caries Lesions) and is the result of mineral loss in the intertubular dentin and precipitation of this mineral in the tubule lumen. Consequently, translucent dentin is softer than normal dentin (see Fig. 2.34)[25] and is called "firm" (affected) dentin, in contrast to sound dentin that is "hard" dentin.

More intense caries activity results in bacterial invasion of dentin. The most superficial (closer to the tooth surface) zone of the carious dentin is the necrotic zone. This very soft (infected) dentin contains a wide variety of pathogenic materials or irritants, including high acid levels, hydrolytic enzymes, bacteria, and bacterial cellular debris. These materials can cause the degeneration and death of odontoblasts and their tubular extensions below the lesion and a mild inflammation of the pulp. The pulp may be irritated sufficiently from high acid levels or bacterial enzyme production to cause the formation (from undifferentiated mesenchymal cells) of replacement odontoblasts (secondary odontoblasts). These cells produce reparative dentin (reactionary or tertiary dentin) on the affected portion of the pulp chamber wall (see Figs. 2.31B and 2.33). This dentin is different from the normal dentinal apposition that occurs throughout the life of the tooth by primary (original) odontoblasts. The structure of reparative dentin varies from well-organized tubular dentin (less often) to very irregular atubular dentin (more often), depending on the severity of the stimulus. Reparative dentin is an effective barrier allowing limited diffusion of material through the tubules and is an important step in the repair of dentin. Severe stimuli also can result in the formation within the pulp chamber of unattached dentin, termed *pulp stones*, in addition to reparative dentin.

The success of dentinal reparative responses, either by remineralization of intertubular dentin and apposition of peritubular dentin or by reparative dentin, depends on the severity of the caries challenge and the ability of the pulp to respond. The pulpal blood supply may be the most important limiting factor to the pulpal responses. Acute, rapidly advancing caries lesions with high levels of acid production overpower host defenses of the pulp and result in infection, abscess, and death of the pulp. Compared with other oral tissues, the pulp is poorly tolerant of inflammation. Small, localized infections in the pulp produce an inflammatory response involving capillary dilation, local edema, and stagnation of blood flow. Because the pulp is contained in a sealed chamber, and its blood is supplied through narrow root canals, any stagnation of blood flow can result in local anoxia and necrosis. The local necrosis leads to more inflammation, edema, and stagnation of blood flow in the immediately adjacent pulp tissue, which becomes necrotic in a cascading process that rapidly spreads to involve the entire pulp.

Maintenance of pulp vitality depends on the adequacy of pulpal blood supply. Recently erupted teeth with large pulp chambers and short, wide canals with large apical foramina have a much more favorable prognosis for surviving pulpal inflammation than fully formed teeth with small pulp chambers and small apical foramina.

Zones of Dentin Caries Lesions

Although enamel and dentin have distinct characteristics, the reaction observed in dentin is a direct response of the pulp to external insults to the enamel.[133] Since enamel is a microporous solid, stimuli in the tooth surface are transmitted to the pulp. The first defense of the pulp is tubular sclerosis, which consists of mineral deposition in the peritubular dentin and in the dentinal tubule's lumen. Wear and age can also cause tubular sclerosis. This process only occurs in vital pulps. Caries progression in dentin proceeds through three changes: (1) Weak organic acids demineralize dentin and expose its organic matrix; (2) the organic matrix of dentin, particularly collagen, is denatured and degraded; and (3) the loss of matrix structural integrity is followed by invasion of bacteria. Three different zones of dentin have been described in moderate and advanced caries lesions (see Figs. 2.33 and 2.35):

- **Soft Dentin (formerly infected dentin).** Also called *outer carious dentin*, soft (infected) dentin is primarily characterized by bacterial contamination. This is the carious dentin closer to the tooth surface, characterized by the presence of bacteria, low mineral content, and irreversibly denatured collagen. Histologically this zone may be referred to as *necrotic* and *contaminated*. While soft dentin typically does not self-repair, the advance front of the soft dentin zone (near firm dentin) is characterized by superficial bacterial invasion and the caries process can still be stalled when a good restorative seal is obtained. Clinically, soft dentin lacks structure and can be easily excavated with hand and rotary instrumentation.
- **Firm Dentin (formerly affected dentin).** Also called *inner carious dentin*, firm (affected) dentin is primarily characterized by demineralization of intertubular dentin and of initial formation of intratubular fine crystals at the advancing front of the caries lesion. As the tubule lumen becomes filled with minerals it will give a "transparent" appearance in a section observed in a light microscope. Histologically this zone may be referred to as *demineralized.* Because of the caries demineralization process, firm dentin is softer than hard, normal dentin. Although organic acids attack the mineral and organic contents of dentin, the collagen cross-linking remains intact in this zone and can serve as a template for remineralization of intertubular dentin. Therefore, provided that the pulp remains vital, firm (affected) dentin is remineralizable. Clinically, firm dentin is resistant to hand excavation and can only be removed by exerting pressure. The transition between soft and firm dentin can have a leathery texture, particularly in slowly advancing lesions, and has been called *leathery dentin*. Clinically, leathery dentin does not deform upon pressure from an instrument but can be excavated with hand instruments such as spoons and curette without much pressure.
- **Hard Dentin.** Hard dentin represents the deepest zone of a caries lesion—assuming the lesion has not yet reached the pulp—and may include *tertiary dentin*, *sclerotic dentin, and normal (or sound) dentin.* Clinically this dentin is hard, cannot be easily penetrated with a blunt explorer, and can only be removed by a bur or a sharp cutting instrument. (See definitions in Box 2.1 for histologic characteristics of hard, tertiary, and sclerotic dentin.)

These dentin caries zones are most clearly distinguished in slowly advancing lesions. In rapidly progressing caries lesions (see Figs. 2.36, 2.37, and 2.38), the difference between the zones becomes less distinct.

Caries Risk Assessment and Management

Although it is very important to detect caries lesions, caries detection and caries diagnosis are not exclusively a tooth-centered process.

It is critical to remember that clinicians treat the entire patient and not just individual teeth and caries lesions. As noted earlier in this chapter, dental caries is a multifactorial medical disease process, and the caries lesions are the expression of a disease process that involves the patient as a whole. Equally important in the management of the caries disease is the ability to individualize caries diagnosis and treatment or interventions for each patient. To do this, the clinician must formulate a caries risk assessment profile that is based on the individual patient's current risk factors and risk indicators. A *risk factor* is defined as an environmental, behavioral, or biologic factor that directly increases the probability that a disease will occur and the absence or removal of which reduces the possibility of disease.[37,38] Risk factors are part of the causal chain of the disease process, or they expose the host to the causal chain; but once the disease occurs, removal of a risk factor may not always result in the disease process being halted. Any definition of risk factor must clearly establish that the exposure has occurred before the outcome or before the conditions are established that make the outcome likely. This means that longitudinal studies are necessary to demonstrate risk factors. Terms such as *risk indicators* and *risk markers* are also used in the caries literature to refer to risk factors, putative risk factors, or something else entirely. *Risk indicators* may refer to existing signs of the disease process, or signs that the disease process has occurred, but are not part of the disease causal chain. For example, existing caries lesions would be risk indicators, as they indicate a risk status, but are not per se part of the causal chain so they are not risk factors. Multiple risk factors (and indicators) have been studied, reviewed, and validated in the assessment of risk for development of future caries disease, but caries risk assessment is not an exact science. Because caries is a complex multifactorial disease, no single risk factor (or indicator, or marker) is highly predictive of future caries. However, caries risk assessment is necessary to identify what (if any) interventions are needed to lower the patient's caries risk and activity—which is the ultimate goal of caries management when using the medical model.[39,40] A discussion of different risk factors will be included in this section of the chapter.

Caries risk assessments are specific for adults and adolescents older than 6 years and for children under 6 years of age. It is very important to spend time with the patient to uncover all relevant risk factors and indicators currently present. Some risk and protective factors can be adjusted and modified by either the patient or the clinician, such as sucrose intake and fluoride exposure; other risk factors are not easily modifiable, such as hyposalivation as the result of a needed medication or an existing condition such as Sjögren syndrome. Understanding and controlling risk factors and protective factors can be very important in the prevention of new caries lesion formation and to slow or arrest the progression of existing caries lesions.

No consensus exists on exactly how to define the risk categories for caries risk assessments. Examples of terms used to describe risk assessment are "at risk," "low risk," "medium risk," "high risk," and "extreme risk." Assignment of these terms is typically based on the subjective judgment of the clinician with general rules applied, based on the clinician's previous clinical experiences and training. If a clinician finds no detectable or active caries lesions and minimum or no identifiable risk factors, that patient would be assigned a *low caries risk*. In this situation, in the current state of the patient's health, the protective factors for not developing caries lesions outweigh the risk factors that could lead to new caries lesions. The strongest predictor for caries risk for patients in the *at-risk* and *high-risk* categories are the number of caries lesions being detected for the patient over the last 2 to 3 years along with past history of caries lesions in the patient's lifetime.[39]

Historically, dentistry has used a surgical model for dental caries management, which mainly contemplated the biomechanical excision (removal, excavation) of caries lesions and the restoration of the resultant *tooth preparation* to form and function with a restorative material. Management of caries disease by a surgical model consisted of detecting cavitations and "treating" them with excavation and restorations. Eventually it became apparent that dealing only with the end result of the disease and not addressing its etiology for each individual patient was not successful in controlling the caries disease process. Since surgical management alone was not successful, a system has been developed using caries management strategies. This system looks at individualized caries risk assessments and uses this information to design treatment plans according to the risk assessment findings. These assessments look at each patient's unique set of pathologic and protective factors. *Caries management by risk assessment* (CAMBRA) represents a management philosophy that manages the caries disease process using a medical model. This process provides an individualized evaluation of a patient's pathologic factors and protective factors and assesses the patient's risk for developing future disease. The risk assessment is then used to develop an *individualized* evidence-based caries management plan that involves aspects of nonsurgical therapeutics and dental surgical interventions. Both risk assessment and patient-centered interventions are based on the concept of caries balance as discussed earlier in this chapter (see Fig. 2.1). The caries balance model is based on minimizing pathologic factors while maximizing protective factors to attain a balance that favors no disease occurring, or health. With the use of CAMBRA for patient management, mounting evidence suggests that early damage to teeth may be reversed and that the incidence of disease manifestations can be significantly reduced or prevented. CAMBRA is a concept that has gained interest in the United States, and is evolving into the standard of care in caries management.[4,41,144,145] Additionally, the International Caries Classification and Management System (ICCMS)[146-148] has endeavored to merge caries detection and assessment with an individualized management plan. Another caries risk assessment method that has been validated for children is the Cariogram.[149,150]

Caries Risk Assessment

The clinical examination process for diagnosis and detection of caries lesions is discussed in detail in Chapter 3. The caries risk assessment is an important part of this overall process of patient care. The clinician must gather all appropriate data from both the interview with the patient and the clinical examination for caries detection to formulate an individualized caries risk assessment. Part of the caries risk assessment identifies the risk (causative) factors, but does not predict the caries outcome. Risk factors can exist for a patient without the disease being expressed at the time of the examination. The predictive model part of the risk assessment looks at the assessment of caries progression in the future. Risk factors are associated with the variables that have value for prediction purposes, which means that the risk factor is present *before* the disease occurs. As discussed, risk indicators are existing signs of the disease process. They are examples of what is happening with the patient's current state of oral health, not how disease occurred. Risk indicators are clinical observations and detection modalities used to identify risk-level status. Examples include visible cavitations

in pits and fissures or in proximal surfaces of teeth, brown spots, active initial lesions, or cavitated lesions on free smooth surfaces, as well as any restorations in the past 3 years. The ideal caries risk assessment method should be inexpensive and easy to use but at the same time have high predictive value. It should be valuable in decision making for caries management in the use of nonsurgical therapeutics and surgical interventions that serve the patient in a cost-effective and health-promoting manner.

One of the roles of caries risk assessment in caries management is to assist in determining the current caries lesion activity. Caries lesions may be detected significantly before frank cavitation occurs. The diagnosis of caries lesions should include whether they are actively progressing. An inactive lesion may be visible clinically or radiographically; however, it may not be active or progressing over time. With a positive shift in protective factors, change in oral hygiene, or reduction of pathologic factors, it is possible for caries lesions to change in density, size, hardness, and clinical appearance. These inactive lesions are arrested and may not require operative intervention. Assessment by the clinician of all the pathologic and protective factors in the patient's current history will greatly aid in the decision regarding current activity. A thorough risk assessment allows for a more predictive analysis of current and future disease activity and assists in deciding on nonsurgical or surgical interventions. Caries risk assessment helps the clinician to identify the etiologic causes of the disease for a specific patient at a specific point in time, and to determine the frequency and treatment protocols for patient follow-up visits. Restorative decisions in terms of material used and cavity preparation design are also influenced by the information gathered in the risk assessment process. The data gathered establish an important baseline for use in future reassessments to help the clinician and the patient measure the effectiveness of the caries management treatment protocol used for the patient. The systematic use of risk assessment profiles is essential in uncovering risk factors that are present before disease manifestation. This information may be useful in the prevention of caries lesions in patients who have risk factors present but no disease expression and then experience a lifestyle change that adds additional risk factors. The new risk factors then become a tipping point for the caries balance equation toward disease expression. For instance, a patient who frequently consumes high-sugar soft drinks during the day (risk factor) suddenly is prescribed a hyposalivation-inducing medication (additional risk factor) that increases caries risk. The informed patient would have the option of making the decision to eliminate a modifiable risk factor (soft drinks) before expression of the disease and before the introduction of the hyposalivation-inducing medication considering this risk assessment.

The incorporation of risk assessments in routine patient care and in each patient's caries management program is necessary because of the multifactorial nature of the caries disease. No one factor is able to predict the probability of a patient developing caries lesions. When developing a preventive and restorative caries management plan, consideration of the whole patient is essential to successful outcomes. In addition, recorded specific risk assessment profiles provide patients with an educational tool that empowers them to be an important part of managing their disease. Finally, risk assessments provide a means for both the clinician and the patient to monitor and measure the proposed caries management protocols over time and evaluate and adjust the protocols as needed. Risk assessments lead to better treatment outcomes for patients.

Knowing certain factors pertaining to the patient's history is key in establishing a caries risk assessment. Factors that have been identified as contributing to caries risk include age, gender, fluoride exposure, home care, smoking habits, alcohol intake, medications, dietary habits, economic and educational status, and general health. Increased smoking, alcohol consumption, use of medications, and sucrose intake result in increased risk for caries development.[42] Children and older adults have increased risks. Decreased fluoride exposure, lower economic status, and lower educational attainment also increase risk. Poor general health also increases the risk. A strong body of evidence suggests that past caries experience is the best predictor of future caries activity.[43] Information that is important and obtained in the patient history interview would be biologic and environmental factors that include, but are not limited to, medical history including current and past diseases, current medications, and history of hyposalivation from medications or conditions; dental history including past history of dental caries, dental phobias, and history of dental conditions; current home care practices and how well this is done; current diet and exposure to sucrose and other fermentable carbohydrates; and current exposure to topical fluoride products in toothpaste, mouth rinses, and fluoridated water supply. Some of these factors are explored in more detail in the following sections.

Fig. 2.39A and B present two examples of caries risk assessment forms that can be used as part of the initial caries diagnosis process; Fig. 2.39C is an example of a caries assessment form that can be used to facilitate communication of the findings with patients. This latter caries assessment form is a useful tool to measure changes and determine the effectiveness of caries management procedures. All of these forms can be incorporated into an electronic patient management system for increased efficiency.

Social, Economic, and Education Status

Social status and economic status are not directly involved in the disease process but are important because they affect the expression and management of the caries disease. The socioeconomic status and educational status of the patient have implications on the necessary compliance and behavioral changes that can decrease risk for caries development. Socioeconomic status and educational status are predictive at the population level but are generally inaccurate at the individual level.

Diet Analysis

Increased frequency of sugar intake in the form of fermentable carbohydrates increases risk for caries by modifying the biofilm to support a lower pH environment. The frequent use of candies and lozenges during the day or night increases the risk. Sugar-containing acidic beverages, including sport drinks, fruit juices, and soft drinks, all contribute to increased risk. Frequent ingestion of fermentable carbohydrates can change the microbiome profile of the biofilm favoring acidogenic and aciduric bacteria and by influencing the pH of the biofilm to support cariogenic bacteria (see Fig. 2.4).

Salivary Analysis

Salivary flow rate, buffering capacity, and pH all can be measured by various methods (tests). The caries risk predictive value for these tests is not fully demonstrated by strong evidence in all circumstances. Patients with normal salivary flow and adequate buffering capacity may still develop and/or have caries. However, when patients have dry mouth symptoms and a salivary flow analysis leads to a diagnosis of hyposalivation, the decreased salivary flow is a predictive risk factor for root caries in older patients with recession and for increased caries in general in other populations. As discussed, saliva has numerous effects in protection against

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Caries Risk Assessment Form (Age >6)

Patient Name:				
Birth Date:			**Date:**	
Age:			**Initials:**	
		Low Risk	**Moderate Risk**	**High Risk**
Contributing Conditions		**Check or Circle the conditions that apply**		
I.	**Fluoride Exposure** (through drinking water, supplements, professional applications, toothpaste)	☐ Yes	☐ No	
II.	**Sugary Foods or Drinks** (including juice, carbonated or non-carbonated soft drinks, energy drinks, medicinal syrups)	Primarily at mealtimes ☐		Frequent or prolonged between meal exposures/day ☐
III.	**Caries Experience of Mother, Caregiver and/or other Siblings** (for patients ages 6-14)	No carious lesions in last 24 months ☐	Carious lesions in last 7-23 months ☐	Carious lesions in last 6 months ☐
IV.	**Dental Home**: established patient of record, receiving regular dental care in a dental office	☐ Yes	☐ No	
General Health Conditions		**Check or Circle the conditions that apply**		
I.	**Special Health Care Needs** (developmental, physical, medical or mental disabilities that prevent or limit performance of adequate oral health care by themselves or caregivers)	☐ No	Yes (over age 14) ☐	Yes (ages 6-14) ☐
II.	**Chemo/Radiation Therapy**	☐ No		☐ Yes
III.	**Eating Disorders**	☐ No	☐ Yes	
IV.	**Medications that Reduce Salivary Flow**	☐ No	☐ Yes	
V.	**Drug/Alcohol Abuse**	☐ No	☐ Yes	
Clinical Conditions		**Check or Circle the conditions that apply**		
I.	**Cavitated or Non-Cavitated** (incipient) **Carious Lesions or Restorations** (visually or radiographically evident)	No new carious lesions or restorations in last 36 months ☐	1 or 2 new carious lesions or restorations in last 36 months ☐	3 or more carious lesions or restorations in last 36 months ☐
II.	**Teeth Missing Due to Caries in past 36 months**	☐ No		☐ Yes
III.	**Visible Plaque**	☐ No	☐ Yes	
IV.	**Unusual Tooth Morphology** that compromises oral hygiene	☐ No	☐ Yes	
V.	**Interproximal Restorations - 1 or more**	☐ No	☐ Yes	
VI.	**Exposed Root Surfaces** Present	☐ No	☐ Yes	
VII.	**Restorations with Overhangs** and/or **Open Margins**; **Open Contacts** with Food Impaction	☐ No	☐ Yes	
VIII.	**Dental/Orthodontic Appliances** (fixed or removable)	☐ No	☐ Yes	
IX.	**Severe Dry Mouth (Xerostomia)**	☐ No		☐ Yes

Overall assessment of dental caries risk: ☐ Low ☐ Moderate ☐ High

Patient Instructions:

A

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• **Fig. 2.39** A, Example of a caries risk assessment form recommended by the American Dental Association. B, Example of a caries risk assessment form used by the University of North Carolina, Department of Operative Dentistry. C, Another caries assessment form used by the University of North Carolina, Department of Operative Dentistry. This form is very useful for patient communication and compliance. (***A,*** *Copyright 2009, 2011 the American Dental Association.)*

Caries Risk Initial Assessment

Name_____

Date_____

Risk	Low Risk	At Risk	High Risk
Risk Rating Score	**≤ 3**	**≥4 and ≤8**	**≥9**

Patient Interview Assessment

Dental History			**Risk Rating**
1. Had non emergency dental care in last year	yes	no	-1
2. Brushes teeth at least twice daily	yes	no	-3
3. Uses fluoridated toothpaste or product daily	yes	no	-3
4. New caries lesions within the last 3 years	yes	no	8
5. Patient has teeth sensitive to hot, cold, sweets	yes	no	3
6. Patient avoids brushing any part of mouth	yes	no	3

Supplemental notes for dental history

Dietary Assessment

1. Water supply currently fluoridated	yes	no	-2
2. Frequent snacking with sugary foods, acidic foods, fermentable carb foods	yes	no	8
3. Sugary drinks including soft drinks, juice, sports drinks, medicinal syrups	yes	no	8
4. Tobacco use of any kind	yes	no	3
5. Excessive alcohol or recreational drug use	yes	no	8
6. Eating disorders	yes	no	5

Supplemental notes for dietary assessment

Xerostomia Assessment

1. Patient is aware of dry mouth or reduced saliva	yes	no	10
2. Medications taken that reduce salivary flow	yes	no	8
3. Medical conditions affecting salivary flow/content	yes	no	8
4. Saliva flow or content visibly abnormal	yes	no	10

Xerostomia Assessments with scores of 8-10 indicate baseline salivary testing is required

Patient Clinical Assessment

Clinical Oral Findings

1. Readily visible biofilm/plaque	yes	no	5
2. Visible cavitated lesions	yes	no	10
3. Interproximal enamel lesions or radiolucencies	yes	no	10
4. Visible white spots	yes	no	5
5. Visible brown spots or non cavitated caries lesions	yes	no	3
6. Deep pits or grooves	yes	no	5
7. Radiographic cavitated lesions	yes	no	10
8. Restorations with overhangs and/or margin concerns or open contacts	yes	no	3
9. Prosthesis ortho, fixed, or removable	yes	no	3

Clinical Assessments with scores of 10 indicate that baseline bacterial testing is required

Risk Rating Total =

10. Clinician's impression of patient's risk	low	at risk	high

This is clinician's impression of patient's risk if different than would be indicated by risk factors marked. Describe in box below.

Patient Compliance Assessment

Patient's attitude and general assessment of patient's ability to comply for each of the following categories:

Oral hygiene compliance patient limitations	yes	no
Dietary recommendation patient limitations	yes	no
Therapeutic homecare products limitations	yes	no
Special needs health care (physical or mental compliance issues)	yes	no

This is clinician's assessment of any perceived limitations for the patient to comply with oral hygiene, dietary, or using home care products. Could be lifestyle, physical, or economic reasons. Describe in box below.

Patient Clinical In Office Tests Indicated From Risk Score

Cariscreen Meter	completed	at risk	low risk	reading _____
GC America Saliva Check	completed	at risk	low risk	
GC America Strep Mutans	completed	at risk	low risk	
GC America Plaque Indicator	completed	at risk	low risk	
UNC Biologic Testing Lab	completed	at risk	low risk	

Notes from oral findings, patient's attitude, office tests, special circumstances that would influence caries risk or management

B

• **Fig. 2.39, cont'd**

Continued

Caries Assessment Testing Results

PREPARED FOR

Name_____

Date_____

PREPARED BY

_____.

Your Risk Scores: Risk predicts your likelihood of developing future disease. Green is low risk and means you are unlikely to develop a cavity whereas red means you are very likely to develop a cavity unless your risk factors are managed. We will use these results to work with you to develop an individualized plan to control your disease.

Existing Dental Conditions



Your dental condition score is based on current areas of decay, number of cavities in the last three years, exposed root surfaces, crowns and fillings that are defective. Your score will also be higher if you wear partial dentures or other appliances.

Saliva Assessment Testing

Saliva pH Testing



>6.8 6.0-6.6 <5.8

Saliva Flow and Buffering



Normal Abnormal

Your saliva is the main protective factor in the caries risk disease state equation. Yellow and Red results can produce an increased incidence of cavities. If the pH of the saliva is low, it sets the stage for the bacteria to grow that cause cavities to form in your mouth. If the quantity of the saliva is below normal, the healing ability of the saliva to remineralize your teeth after acidic food and beverages is greatly reduced. This can, in most incidences, result in a dry mouth that is uncomfortable when eating and lead to an increase in cavities on the roots and other surfaces of your teeth. Any change in saliva content, amount, or pH can increase your risk for cavities to form even if you have not had cavities in the past. New medications and medical conditions can cause your saliva to change rapidly.

Plaque Assessment Testing

Plaque pH Testing



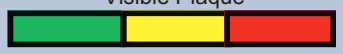
>6.8 6.0-6.6 <5.8

Biofilm Activity



<1500 1500-2000 >2000

Visible Plaque



Low High

Plaque or biofilm is the mass of bacteria that is always changing and clings to the surfaces of your teeth. Plaque is one of the main risk factors that result in cavities. For plaque to produce the acid that dissolves your teeth and form cavities, it has to be in an acidic state, or in other words, have a low pH. The more acidic, the more damage that can result. The type of bacteria in the plaque also influences how easy it is for the damage to occur. The biofilm activity measures the amount of the "bad" bacteria present in the plaque. The higher the number, the more bacteria are present that cause the cavities to form. The amount of plaque on your teeth means more bacteria are present in your mouth. More bacteria produce higher amounts of acid to demineralize your teeth and cause them to decay.

Dietary Habits

Frequency of Snacking



< 1 2-3 >3

Frequency of sweetened beverages and sport drinks



< 1 2-3 >3

A key factor in how you control your disease and prevent cavities is how you eat and what beverages you consume. Snacking with sweet foods or high carbohydrate foods that can form sugars causes the plaque to become acidic. This results in more bacteria forming that produce even more acid. All of this acid dissolves your teeth and forms the cavities. Sweetened drinks or drinks high in acid content also produce low pH plaque and more bacteria. Soft drinks and sports drinks like Gatorade are very low in pH. The more you drink, the more acid the plaque and bacteria produce to cause the cavities in your teeth.

C

• **Fig. 2.39, cont'd**

caries, including inhibition of bacteria, eliminating (clearing of) bacteria from the tooth surface, dilution of bacterial products, buffering of bacterial acids, and offering a reparative environment with necessary calcium and phosphate minerals after bacteria-induced demineralization. Since all of these benefits are missing when patients have salivary hypofunction, patients with "dry mouth" are at higher risk for caries. These patients are more susceptible to dietary changes that are associated with lower pH foods and beverages or foods and beverages containing fermentable carbohydrates, since the protective factors of saliva are diminished in patients with hyposalivation.

Dental Clinical Analysis (Dental Exam)

The dental examination identifies risk indicators more than risk factors. This is important as many of the indicators are directly related to the current caries activity. Risk indicators and current caries activity drive the decision-making process toward the type of intervention that the clinician would prescribe. Visible cavitated caries lesions and active initial lesions on teeth are both indicators of caries risk. Visible biofilm may also be considered a risk factor for caries development. Other examination findings that would influence increased risk for caries are exposed root surfaces; deep pits or grooves; fixed, removable prosthesis or orthodontic appliances used; and existing restorations with open contacts, open margins, rough surfaces, or overhangs.

Bacterial Biofilm Analysis

Use of supplemental tests to analyze the bacterial component of the biofilm has little value to determine the patient's risk level. For example, the presence of *MS* or lactobacilli in saliva or plaque as a sole predictor for caries in primary teeth has been shown to have low sensitivity but high specificity. The measurement of adenosine triphosphate (ATP) activity of the biofilm bacteria as a surrogate measure of caries activity has also not been identified to be a strong predictor of risk. Although these bacterial tests may be useful for patient communication by providing insight into the specifics of the biofilm (bacterial types and resultant environment), the predictive value (for caries risk) of the evidence of the tests requires further validation by controlled research studies.

Risk Assessment Considerations for Children Under 6 Years Old

In addition to all the above mentioned risk factors for adults and adolescents, there are age-specific risk factors and indicators that should be considered for children under 6 years. These include presence of active caries in the primary caregiver in the past year, feeding on demand past 1 year of age, bedtime bottle or sippy cup with anything other than water, no supervised brushing, and severe enamel hypoplasia.

Caries Management and Protocols or Strategies for Prevention

Caries risk assessment is only meaningful if used in conjunction with a corresponding caries management program. The caries prevention or management program should comprise a menu of prevention therapies and interventions that should be recommended on the basis of the level of caries risk (Table 2.7 and Box 2.2) and address the patient's risk factors. However, as discussed in the previous section, no caries risk assessment system is perfect. Therefore, in addition to the outcome of the caries risk assessment tool, the clinician needs to use his or her best clinical judgment, coupled with the best available research evidence, to design a preventive or therapeutic program that works for the patient. Needless to say, this is a dynamic process, so monitoring and

TABLE 2.7 Suggested Risk-Based Interventions for Adults[a]

Caries Risk Category	Office-Based Interventions	Home-Based Interventions
High		
	3-month recare examination and oral prophylaxis Fluoride varnish at each recare visit Individualized oral hygiene instructions and use of specialized cleaning aids (e.g., powered toothbrush, Waterpik) Dietary counseling Bitewing radiographs every 6–12 months[b]	Brush with prescription fluoride dentifrice (e.g., 1/1%/5000 ppm NaF) Use sugar substitutes (e.g., xylitol, erythritol) Apply calcium-phosphate compounds (e.g., MI Paste) On selected cases use agents to modulate a dysbiotic microbiome (e.g., xylitol gum or lozenge, chlorhexidine rinse). Use agents to encourage a healthy microbiome (probiotics, prebiotics, like arginine) If xerostomic, increase salivary function (e.g., xylitol gum, rinses, oral moisturizers)
Moderate		
	4–6 month recare examination and oral prophylaxis Fluoride varnish at each recall Reinforce proper oral hygiene Dietary counseling	Brush with fluoride dentifrice (e.g., 1450 ppm fluoride) OTC fluoride rinse (e.g., 0.05% NaF)
Low		
	9–12 month recare examination and oral prophylaxis Reinforce good oral hygiene	Brush with fluoride dentifrice

NaF, Sodium fluoride; *OTC,* over the counter; *ppm,* parts per million.

[a]These are general guidelines and should be customized based on the specific patient's needs and on weight of individual risk factors uncovered with a caries risk assessment instrument.

[b]Data from US Department of Health and Human Services, Public Health Service, Food and Drug Administration; and American Dental Association, Council on Dental Benefit Programs, Council on Scientific Affairs. The selection of patients for dental radiographic examinations. Rev. ed. 2004.
Available at www.ada.org/prof/resources/topics/radiography.asp. Accessed January 20, 2012.

Modified from Shugars DA, Bader JD: *MetLife quality resource guide: risk-based management of dental caries in adults,* ed 3, Bridgewater, NJ, 2009–2012, Metropolitan Life Insurance Co., p. 6.

• BOX 2.2 Sample Preventive Protocol for a High-Risk Patient With Cavitated Caries Lesions

1. A comprehensive oral and radiographic evaluation is conducted charting caries lesions, periodontal pocket probing, existing restorations, and oral cancer exam; the medical and dental histories are reviewed. In this hypothetical patient, multiple caries lesions, poor oral hygiene, and generalized marginal gingivitis are noted.
2. A caries risk assessment is completed with emphasis on discovering the risk factors that are contributing to the causative aspect of the caries problem and discovering the risk factors for the patient that are predictors of future caries risk. A discussion of these risk factors with the patient is essential for understanding of the caries disease process and the patient's role and the provider's role in controlling the disease.
3. Diagnosis and treatment planning procedures are completed and discussed with the patient.
4. Nonoperative and operative treatments will be completed in three phases: (i) a control phase, (ii) a definitive treatment phase, and (iii) a maintenance reassessment phase.
5. CONTROL PHASE (2–4 weeks):
   a. Oral hygiene procedures are explained and reviewed at each patient visit. The frequency of the visits in this initial phase is determined by severity of the disease. This could be weekly or more frequent, depending on the provider's evaluation. When attempting behavior modification, repetition is essential. Review of patient's current home care techniques and frequency of home care are reviewed along with lifestyle issues that might impede compliance. Use of a powered toothbrush and an oral irrigator for possibly improving patient compliance and technique are discussed. Evaluation of the patient's motivation along with the person's mental and physical capacity to comply with recommendations for home care must be noted and considered by the provider. Specific recommendations are listed for the patient to use at home, and the patient agrees that this is practical for his or her life situation.
   b. Prescription fluoride toothpaste is prescribed (5000 parts per million [ppm]) and the patient is instructed to brush with it three times per day and to use according to given instructions (do not rinse after use, only expectorate excess). Any products used for home care should be carefully reviewed for the pH of the product. Products with pH lower than 6 should be carefully considered whether their use would contribute to the pH shift of the biofilm to pathologic for caries lesion formation.
   c. A diet analysis is completed, analyzed, and reviewed with the patient. Cariogenic foods and beverages are identified and alternatives suggested. Also acidic foods and beverages that are contributing to the pH shift to a lower oral pH environment are identified and discussed. Options for foods that have impact to raise the oral pH are suggested, such as foods rich in arginine. Again the patient's motivation and abilities to fit the necessary diet modifications into his or her lifestyle must be evaluated and discussed.
   d. A microbiologic survey may be completed for patient motivation. This can be accomplished using either the adenosine triphosphate (ATP) chairside device or formal saliva samples to identify specific mutans streptococci (MS) and lactobacilli counts.
   e. A saliva analysis is conducted to determine stimulated flow rate, salivary pH and buffering capacity, and viscosity. Treatment protocols for hyposalivation are recommended for patients with an analysis that indicates deficiencies in the above areas. Patients with low salivary pH would need to use baking soda rinses during the day and use xylitol gum or other recommended products to raise pH levels, increase flow, and increase buffering capacity.
   f. Caries control (described elsewhere in this chapter) is completed. This may involve use of silver diamine fluoride (SDF) for cavitated lesions or partial caries removal and interim restoration of all cavitated caries lesions (usually during one appointment and restoring with glass ionomer) or a combination of both. This is critical to reduce the bacterial load and prevent further progression on the lesion.
   g. A prophylaxis is performed and 5% sodium fluoride (NaF) varnish is applied to all teeth.
   h. If SDF is used, the lesion should be reassessed 1 to 2 weeks after first application to determine if it has arrested. If the lesion has not arrested a new application of SDF is advised. Lesions that will not be restored need to have a new SDF application every 6 months.
   i. The microbiologic testing, if done, may be repeated 2 to 4 weeks after initiation of treatment. A reduction in counts can be a great motivational tool for patients to continue with their modifications in behavior (diet or oral hygiene). Diet analysis is followed up and reviewed again. Successes in diet modifications are positively reinforced. Shortcomings in diet modifications are discussed and options explored with the patient to rectify diet issues, where needed. Home care regimens are reviewed again with patients and refined. It is important to listen to the patient and work to incorporate diet and home care into the patient's lifestyle and abilities to mentally and physically comply with the recommendations.
6. DEFINITIVE TREATMENT PHASE:
   a. The glass ionomer provisional restorations are replaced (usually by quadrants) with definitive restorations.
   b. Oral hygiene procedures are reinforced at each visit. Flossing and brushing three times per day with prescription toothpaste is recommended.
   c. Patients with root caries may benefit from chewing xylitol chewing gum with at least 1 gram of xylitol per piece three to six times per day, preferably after meals and snacks.
   d. Some patients with low salivary flow or low salivary buffering capacity may benefit from applying casein phosphopeptide–amorphous calcium phosphates (CPP-ACP) to teeth after brushing and flossing prior to retiring to bed.
   e. If reduced salivary flow rates are considered to be a major etiologic factor, the patient should be instructed to chew sugar-free mints several times a day or use other recommended products for xerostomia treatment. Prescribing pilocarpine or other salivary stimulant should be considered.
   f. When all definitive restorations are completed, the patient then enters the maintenance reassessment phase.
7. MAINTENANCE REASSESSMENT PHASE:
   a. The patient should be recalled every 3 months. Oral hygiene and home care procedures are reviewed and evaluated. Recommendations for improvement and modifications to home care are evaluated and discussed.
   b. Prophylaxis followed by fluoride varnish application is accomplished.
   c. Caries risk assessment is completed again; changes are noted in risk factors that have been controlled and those risk factors still listed as causative and predictive factors.
   d. Diet analysis and recommendations from previous visits are reviewed and evaluated.
   e. Patient continues use of prescription 5000-ppm toothpaste, CPP-ACP paste, and xylitol chewing gum as advised. Any other recommendations to changes or additions to the product protocols are reviewed, discussed, and implemented.
   f. Every 6 months, salivary evaluations are repeated. Microbiologic evaluations may also be repeated to keep patient motivation.
   g. Bitewing radiographs are taken on an annual basis or more frequently if new lesions continue to be detected.
   h. It is critical for the patient to understand that caries is a disease that is only controlled and not "cured." The protocol that is determined to be currently successful may have to be periodically reviewed, updated, and changed. More importantly, the patient will be much like a patient with diabetes, requiring lifetime medication and therapy, diet control, and lifestyle management for disease stability, and will need to be dedicated to a lifetime of careful management of caries risk factors to keep the disease controlled.

TABLE 2.8 Health History Factors Associated With Increased Caries Risk

History Factor	Risk-Increasing Observations
Age	Childhood, adolescence, senescence
Fluoride exposure	No fluoride in public water supply No fluoride toothpaste
Smoking	Risk increases with amount smoked
Alcohol	Risk increases with amount consumed
General health	Chronic illness and debilitation decreases ability to give self-care
Medication	Medications that reduce salivary flow

TABLE 2.9 Clinical Examination Findings Associated With Increased Caries Risk

Clinical Examination	Risk-Increasing Findings
General appearance	Appearance: sick, obese, or malnourished
Mental or physical disability	Inability or unwillingness to comply with dietary and oral hygiene instruction
Mucosal membranes	Dry, red, glossy mucosa, suggesting decreased salivary flow
Active caries lesions	Cavitation and softening of enamel and dentin; circumferential chalky opacity at gingival margins
Plaque biofilm	High plaque scores
Gingiva	Puffy, swollen, and inflamed gingiva; bleeds easily
Existing restorations	Large numbers, indicating past high caries rate; poor quality, indicating increased habitat for cariogenic organisms

periodic reassessment and reevaluation of the disease activity and prevention or management program are critical.

Management of dental caries and its consequences remains the predominant activity of dentists; however, preventive and diagnostic services are increasing.[44,45] Although these activities relate to a variety of dental problems, diagnosis and prevention of caries are major parts of these increases. In a modern practice model, the restoration of a caries lesion can no longer be considered a cure for dental caries. Rather, the practitioner must identify patients who have active caries lesions and patients at high risk for caries and institute appropriate preventive and treatment measures. This section presents some measures that may reduce the likelihood of a patient developing caries lesions. Depending on the risk status of the patient and the risk factors at play for the specific patient, the dentist must decide which of these to institute. With this paradigm shift, dentistry is focusing increasingly on limiting the need for restorative treatment.

Preventive treatment methods are designed to limit tooth demineralization caused by cariogenic bacteria, preventing initial lesions and progression of initial lesions into cavitated lesions. These methods include (1) limiting pathogen growth and altering metabolism of the biofilm, (2) increasing the resistance of the tooth surface to demineralization, and (3) increasing biofilm pH. A caries prevention and management program is a complex process involving multiple interrelated factors (Tables 2.8, 2.9, 2.10, and 2.11; see also Table 2.7 and Box 2.2). The primary goals of a caries prevention program are (1) to modulate the biofilm into a nonpathogenic biofilm and (2) to create an environment conducive to remineralization. Prevention should start with a consideration to maintain the patient with a noncariogenic biofilm. Although the general health of the patient, fluoride exposure history, and function of the immune system and salivary glands have a significant impact on the patient's caries risk, the patient may have little control over these factors. The patient usually is capable of controlling other factors such as diet, oral hygiene, use of agents to modulate the biofilm, and dental care (which may include use of sealants and restorations). This section presents a variety of factors that may have an impact on caries prevention.

General Health

The patient's general health has a significant impact on overall caries risk. Declining health signals the need for increased preventive measures, including more frequent recalls. Every patient has an effective surveillance and destruction system for "foreign" bacteria. The effectiveness of a patient's immune system depends on overall health status. Patients undergoing radiation or chemotherapy treatment have significantly decreased immunocompetence and are at risk for increased caries.

Medically compromised patients should be examined for changes in the following: plaque index, salivary analysis, oral mucosa, gingiva, and teeth. Early signs of increased risk include increased biofilm; puffy, bleeding gingiva; dry mouth with red, glossy mucosa; and initial caries lesions on teeth. Decreased saliva flow is common during acute and chronic systemic illnesses and is responsible for the dramatic increase in biofilm. Ambulatory patients with chronic illnesses often take multiple medications, which individually or in combination may reduce salivary flow significantly (see Table 2.4). Saliva should be tested for flow and buffering capacities when signs or symptoms are detected from an oral examination.

Diet

Dietary sucrose has two important detrimental effects on how biofilm affects caries. First, frequent ingestion of foods containing sucrose provides a change in the biofilm profile from a noncariogenic biofilm to a cariogenic biofilm.[151] Second, mature biofilm exposed frequently to sucrose rapidly metabolizes it into organic acids, resulting in a profound and prolonged decline in pH. Caries activity is most strongly stimulated by the frequency, rather than the quantity, of sucrose ingested. The message that excessive and frequent sucrose intake can cause caries has been widely disseminated and is well known by laypeople. Despite this knowledge, dietary modification for the purpose of caries control has failed as a public health measure. For an individual patient, dietary modification may be effective if the patient is properly motivated and supervised. Evidence of new caries activity in adolescent and adult patients indicates the need for dietary counseling. The goals of dietary counseling should be to identify the sources of sucrose and acidic foods in the diet and to reduce the frequency of ingestion of both. Minor dietary changes such as substitution of sugar-free foods for snacks are more likely to be accepted than more dramatic changes. Rampant (or acute) caries (a rapidly developing process usually involving several teeth) is a sign of gross dietary inadequacy, a

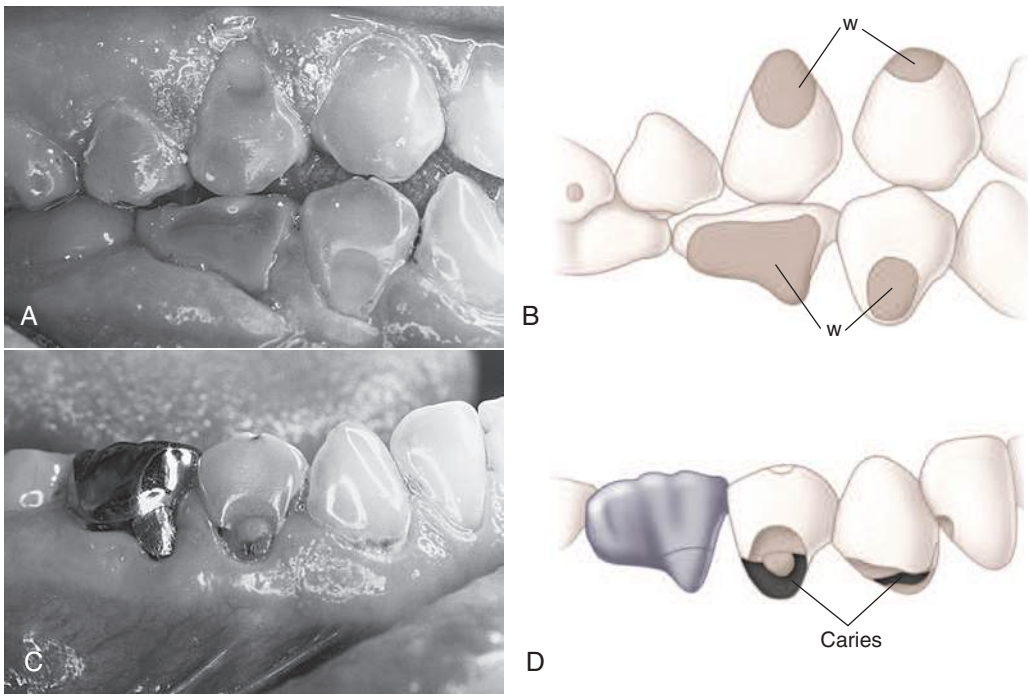
TABLE 2.10 Methods of Caries Treatment by the Medical Model

Method and Indications	Rationale	Techniques or Material
A. Limit Cariogenic Substrate		
Indications: Frequent sucrose exposure Poor-quality diet	Reduce number, duration, and intensity of acid attacks Reduce selection pressure for mutans streptococci (MS)	Diet diary Eliminate sucrose from between-meal snacks Substantially reduce or eliminate sucrose from meals
B. Modify Microflora		
Indications: High MS counts High lactobacilli counts	Modify a dysbiotic biofilm and encourage a healthy biofilm Reduce overall biofilm burden to improve gingival health	Bactericidal mouthrinse (chlorhexidine) Topical fluoride treatments; probiotics and prebiotics
C. Disorganize Plaque Biofilm		
Indications: High plaque biofilm scores Puffy red gingiva High bleeding point score	Prevents plaque biofilm succession Decreases plaque biofilm mass Promotes buffering	Brushing Flossing Other oral hygiene aids as necessary (e.g., electric toothbrush)
D. Modify Tooth Surface		
Indications: Noncavitated lesions Surface roughening	Increase resistance to demineralization Decrease plaque biofilm retention	Systemic fluorides Topical fluorides Smooth surface
E. Stimulate Saliva Flow		
Indications: Dry mouth with little saliva Red mucosa Medication that reduces salivary flow	Increases clearance of substrate and acids Promotes buffering	Eat noncariogenic foods that require lots of chewing Sugarless chewing gum Medications to stimulate salivary flow Use dry mouth topical agents, oral lubricants, etc.
F. Seal Susceptible Surfaces		
Indications: Moderate and high caries risk individuals Teeth with susceptible anatomy (deep grooves) Initial noncavitated enamel lesions in high-risk patients (smooth-surface sealants)	Prevents colonization (infection) of pit-and-fissure system with cariogenic plaque biofilm Inhibits progression of smooth-surface lesion	Use pit-and-fissure and smooth-surface resin sealants
G. Restore Active Cavitated Surfaces		
Indications: Cavitated lesions Defective restorations	Eliminate nidus of cariogenic biofilm Deny habitat for cariogenic biofilm reinfection	Apply SDF to caries lesions as indicated Restore all cavitated lesions Correct all defects (e.g., marginal crevices, proximal overhangs)

TABLE 2.11 Treatment Strategies

Examination Findings	Nonrestorative, Therapeutic Treatment	Restorative Treatment	Follow-up[a]
Normal (no lesions)	None	None	1-year clinical examination
Hypocalcified enamel (developmental white spot)	None for nonhereditary lesions; hereditary lesions (dentinogenesis imperfecta) may require special management	Treatment is elective; esthetics (restore defects)	1-year clinical examination
Noncavitated enamel lesions only; bitewing radiographs indicated (demineralized white spot)	Techniques A–E in Table 2.9, as indicated	Seal defective pits and fissures as indicated	3 months; evaluate: oral flora, mutans streptococci (MS) counts, progression of white spots, presence of cavitations
Possible cavitated lesions (active caries) and other noncavitated lesions present; bitewing radiographs indicated	Techniques A–E in Table 2.9, as indicated	Techniques F and G (restorations, sealants) in Table 2.9 as indicated	3 months; evaluate: oral flora, MS counts, progression of white spots, presence of new cavitations, pulpal response
Inactive caries; no active (new cavitations) or noncavitated lesions	None	Treatment is elective; esthetics (restore defects)	1-year clinical examination

[a]These are only generalized follow-up times. Particular circumstances may dictate shorter or longer follow-up intervals, depending on presence of primary and secondary modifying risk factors (see Fig. 2.1).



• **Fig. 2.40** Erosion wear and poor home care leading to caries. A young female patient with severe wear on the facial surfaces of posterior teeth. This patient subsequently was found to have a hiatal hernia with frequent regurgitation of stomach acids. Too vigorous brushing and acid demineralization of teeth accelerated the loss of tooth structure (*A* and *B*). Areas of severe wear *(w)* exhibit dentin hypersensitivity. Dentin pain was the symptom that caused the patient to seek dental care. Advising the patient to reduce vigorous tooth brushing resulted in cessation of all brushing. Caries activity rapidly occurred (*C* and *D*).

complete absence of oral hygiene practice, systemic illness, or a combination of these pathologic factors. Rampant caries that is present primarily on proximal surfaces may point more to diet as the main driving factor, whereas rampant caries in the cervical and interproximal areas may point to diet and hygiene as the driving factors. The presence of rampant caries indicates the need for comprehensive patient evaluation. Textbooks on dental caries, nutrition, and medicine should be consulted as needed.

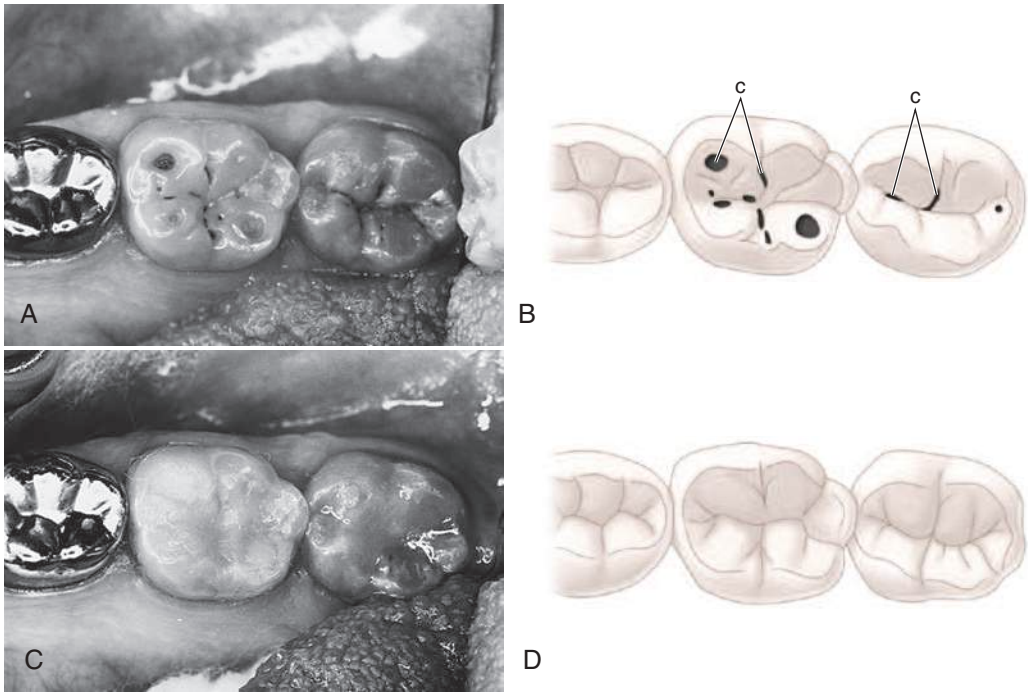
For high-risk patients, a formal diet analysis should routinely be undertaken to identify cariogenic foods and beverages that are frequently ingested. This analysis may take place by asking the patient to recall everything that was ingested in the past 24 hours (the "24-hour diet interview"). Alternatively, a more detailed diet analysis may be facilitated by requesting the patient to record a diet diary, which consists usually of a 5- to 7-consecutive-day period with 2 days surveyed being weekend days as patients' diets frequently change considerably on weekends. A form should be provided to the patient that divides each day into six segments (breakfast, morning, lunch, afternoon, dinner, and evening), and the patient should be instructed to write down everything ingested, including medications, the amount, and the time of consumption. The diary is then analyzed by the dentist, and a discussion is held with the patient to suggest appropriate alternatives.[46] The dentist should focus on type of cariogenic foods and time intervals between their consumption as well as length of exposure of each cariogenic episode. For example, if a patient were to record "10:00 AM 1 soda," the dentist should clarify the type of soda consumed, the quantity of soda, and the duration of time to finish the soda. An 8-oz glass of sugar-containing soda consumed over a 5-minute period will have a different impact than a 20-oz glass sipped over a 2-hour period.

Oral Hygiene

Biofilm-free tooth surfaces do not decay. Daily removal of biofilm by dental flossing and tooth brushing with a fluoridated toothpaste is the best patient-based measure for preventing caries and periodontal disease (Figs. 2.40 and 2.41). Löe established supragingival plaque as the etiologic agent of gingivitis.[47] Longstanding gingivitis may lead to damage of the epithelial attachment and progression to more serious periodontal disease. Although plaque scores are not reliable indicators of caries risk, the presence of biofilm over a lesion may help assess lesion activity.[152,153] Effective biofilm control by oral hygiene measures using a fluoridated toothpaste results in resolution of the gingival inflammation and remineralization of any initially demineralized enamel surface.

Mechanical biofilm disorganization by brushing and flossing has the advantage of not eliminating the normal oral flora. Topical antibiotics may control oral biofilm, but long-term use predisposes the host to infection by antibiotic-resistant pathogens such as *Candida albicans*. Frequent mechanical biofilm removal does not engender the risk of infection by opportunistic organisms; rather it changes the species composition of biofilm in the selection for pioneering organisms and the denial of habitat to potential pathogens. The oral flora on the teeth of patients with good biofilm control has a high percentage of *S. sanguis* or *S. mitis* and is much less cariogenic than older, mature biofilm communities, which have a significantly higher percentage of MS.

Krasse showed that a combination of oral hygiene and diet counseling is effective in children.[48] In this classic study, children in two schools were monitored for *Lactobacillus* levels. The children in one school were given feedback about their *Lactobacillus* levels and proper preventive oral hygiene and dietary instruction. After



• **Fig. 2.41** A and B, Photograph of the occlusal surfaces of the teeth illustrated in Fig. 2.40. C and D, After cessation of oral hygiene procedures, caries *(c)* rapidly developed in the exposed dentin and fissures on the occlusal surfaces. Caries was treated conservatively by excavation of softened dentin and restoration of the excavations and fissures with composite.

18 months, the children in the school receiving preventive counseling had an average of 3.3 new restorations, whereas the control schoolchildren, who received no counseling, averaged 8.2 new restorations. This study is an excellent demonstration that good oral hygiene and dietary improvements may be effective when using microbiologic testing as a motivational tool.

Rigid oral hygiene programs should be prescribed only to high-risk persons with evidence of active disease. Overzealous, universal application of oral hygiene training programs is frustrating to dentists and their patients. High-risk patients should receive intensive oral hygiene training usually with a prescription-strength fluoridated toothpaste (5000 parts per million [ppm] F), dietary instruction, and preventive dental treatment, as necessary, to control the progress of disease. Adults with a low caries experience probably require less frequency of daily flossing and brushing.

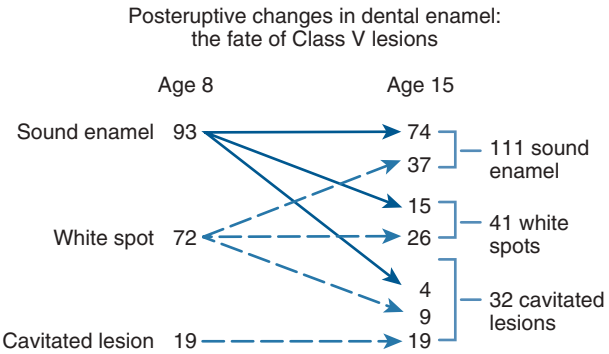
Biofilm control requires a little dexterity and a lot of motivation. Some knowledge of tooth contours, embrasure form, proximal contacts, and tooth alignment is helpful in optimal biofilm control. Instruction should include the selection and use of mechanical aids based on patient needs. Professional tooth cleanings can have an important effect on caries reduction. One study divided grade school students into three treatment groups: control, monthly professional cleaning, and twice-a-month professional cleaning.[49] In students with low MS levels, the once-a-month cleaning group showed ~50% fewer new carious surfaces (0.8 surfaces/student) than the control group (1.8 surfaces/student). In the high MS group, the control group had the most new caries (2.5 surfaces/student), whereas the once-a-month cleaning group had similar levels (0.96 surfaces/student) to the low MS group, and the twice-a-month cleaning group had almost one tenth the number of new lesions (0.34 surfaces/student) as the control group. This study showed that professional biofilm removal on grade school students, even only once every 2 weeks, dramatically reduces the development of new caries lesions. There is little evidence that professional prophylaxis at the usual 6-month intervals reduces dental caries levels; therefore it is important to customize recall intervals to the patient risk. Equal or greater reductions may be expected in patients who practice proper oral hygiene methods for biofilm removal.

Another adjunct to regular brushing and flossing is the regular use of electric toothbrushes and oral irrigation devices. A recent study has demonstrated these devices are effective in the removal of biofilm and, more importantly, change the composition of the biofilm in a favorable direction when used regularly.[50] Since most patients are not very efficacious on judicious removal of biofilm, brushing may be viewed more as a delivery mechanism for fluoride. Recommendations of brushing for at least 2 minutes and not rinsing may increase compliance and improve outcomes. Additionally, simple suggestions, like feeling tooth surfaces with the tongue to assess if the biofilm was removed, may improve awareness and assist in efficacy.

Fluoride Exposure

The highest level of evidence for caries prevention and reduction supports the exposure of teeth to fluoride. Fluoride in trace amounts increases the resistance of tooth structure to demineralization and is particularly important for caries prevention (Fig. 2.42). When fluoride is available during cycles of tooth demineralization, it is a major factor in reduced caries activity.[51] Fluoride seems to be an essential nutrient for humans and is required only in very small quantities. Laboratory animals fed a completely fluoride-free diet develop anemia and reduced reproduction after four generations. When available to humans, fluoride produces spectacular decreases in caries incidence. The availability of fluoride for caries risk reduction has primarily been achieved through the fluoridation of

A



B

	Baseline			Cavitation at Follow-up		
	Backer Dirks, 1966	Ferreira Zandona et al, 2012		Backer Dirks, 1966	Ferreira Zandona et al, 2012	
	Class V lesions	Class V lesions	All surfaces	Class V lesions	Class V lesions	All surfaces
Sound	93	7905	59335	4%	1%	2%
Initial	72	737	3207	13%	10%	24%
Enamel Cavities		10	275		20%	67%

• **Fig. 2.42** A, Initial (white spot) lesions of enamel (stage 3 in Fig. 2.28) may remineralize, remain unchanged, or progress to cavitated lesions. In this study, done in a community with a fluoridated public water supply, only 9 of 72 noncavitated lesions became cavitated. More than half of noncavitated lesions (37 of 72) regressed to become indistinguishable from normal enamel. B, White spot lesions of enamel (stage 3 in Fig. 2.28) may remineralize, remain unchanged, or progress to cavitated lesions. A comparison of a study done in a community with a fluoridated public water supply published in 1966 (Backer Dirks, 1966) versus a study completed in a nonfluoridated community published in 2012 (Ferreira Zandona et al, 2012). The 1966 study followed 184 smooth surfaces (buccal) over a 7-year period. The 2012 followed 62,812 surfaces over a 4-year period, of those 8652 were buccal surfaces. In both studies only a very small portion of sound surfaces cavitate (4% versus 1%–2%). Most initial lesions in enamel on smooth surfaces will not progress to cavitation (only 13% progressed to cavitation in 1966 and only 10% in 2012). When other surfaces are included the number of lesions that progress to cavitation more than double (24%) but it still represents only one fourth of all initial lesions. On smooth surfaces even when enamel cavities are considered, only 20% progress to cavitation exposing the dentin; and, when all surfaces are included, 30% of enamel cavities do not progress to cavitations exposing the dentin. (A, Redrawn from Backer Dirks O: Posteruptive changes in dental enamel. *J Dent Res* 45:503–511, 1966.)

community water systems. Fluoride exposure may occur by means of diet, toothpastes, mouth rinses, and professional topical applications. The optimal fluoride level for public water systems is 0.7 mg per liter of water.[52] The percentage of the U.S. population with public fluoridated community water systems has increased from 62% (140 million) in 1999 to 66% (162 million) in 2000, to 69% (184 million) in 2006,[23,54] to 74% (284 million) in 2014. Public water fluoridation has been one of the most successful public health measures instituted in the United States. For communities that have fluoridated water systems, the annual cost averages about $0.70 per person.[53] For every $1 spent on water fluoridation, $6 of health savings are realized. The U.S. Public Health Service recommends 0.7 ppm of fluoride in public water programs to reap the benefits of water fluoridation and reduce the risk of fluorosis. Excessive fluoride exposure (≥10 ppm) results in fluorosis, which initially causes enamel to become white but may eventually cause a brownish discoloration, a condition termed *mottled enamel*.

Fluoride exerts its anticaries effect by three different mechanisms. First, the presence of fluoride ion greatly enhances the precipitation into tooth structure of fluorapatite from calcium and phosphate ions present in saliva. This insoluble precipitate replaces the soluble salts containing manganese and carbonate that were lost because of bacteria-mediated demineralization. This exchange process results in the enamel becoming more acid resistant (see Fig. 2.29). Second, initial caries lesions are remineralized by the same process. Third, fluoride has antimicrobial activity. In low concentrations, fluoride ion inhibits the enzymatic production of glucosyltransferase. Glucosyltransferase promotes glucose to form extracellular polysaccharides, which increases bacterial adhesion. Intracellular polysaccharide formation also is inhibited, preventing storage of carbohydrates by limiting microbial metabolism between the host's meals. In high concentrations (12,000 ppm) used in topical fluoride treatments, fluoride ion is directly toxic to some oral microorganisms, including MS. Suppression of growth of MS after a single topical fluoride treatment may last several weeks.[55] It is possible to lengthen this suppression greatly by a change in dietary habits (especially eliminating sucrose) and by the patient's conscientious application of a good oral hygiene program.

All fluoride exposure methods (Table 2.12) are effective to some degree. The clinician's goal is to choose the most effective combination for each patient. This choice must be based on the patient's age, caries experience, general health, and oral hygiene. Children with developing permanent teeth benefit most from systemic fluoride treatments via the public water supply. In regions without adequate fluoride in the water supply, dietary supplementation of fluoride is indicated for children and sometimes for adults. The amount of fluoride supplement must be determined individually. This is

TABLE 2.12 Fluoride Treatment Modalities[a]

Route	Method of Delivery	Concentration (ppm)	Caries Reduction (%)
Systemic Topical	Public water supply	0.7	50–60
	Self-application		
	Low-dose/high-frequency rinses (0.05% sodium fluoride daily)	225	30–40
	High-potency–low-frequency rinses (0.2% sodium fluoride weekly)	900	30–40 after 2 years
	Fluoridated dentifrices (daily)	1000–1450	20
	Prescription-strength fluoridated dentifrices (daily)	4950	32
	Professional application		
	Acidulated phosphate fluoride gel (1.23%) annually or semiannually	12,300	40–50
	Sodium fluoride solution (2%)	20,000	40–50
	Sodium fluoride varnish (5%)	22,500	30
	Stannous fluoride solution (8%)	80,000	40–50
	Silver diamine fluoride (SDF) (38%)	44,800	~96.1 for caries arrest ~70.3 for caries prevention

ppm, Parts per million.
[a]Caries reduction estimates for topically administered fluorides indicate their effectiveness when used individually. When they are combined with systemic fluoride treatment, they can provide some additional caries protection.

of particular importance in rural areas with individual wells because the fluoride content of well water can vary greatly within short distances.

Topical application of fluoride should be done periodically for children and adults who are at high risk for caries development. The periodicity varies with the case. Teeth can be cleaned free of biofilm before the application of topical fluorides. Flossing followed by tooth brushing is recommended for this purpose. Pumicing of teeth (professional prophylaxis) is able to remove a considerable amount of the fluoride-rich surface layer of enamel and may be counterproductive. Acidulated phosphate fluoride gel is effective, but a potential risk of swallowing excessive amounts of fluoride exists, particularly in young children. Acidulated phosphate fluoride is available in thixotropic gels and has a long shelf life. Stannous fluoride (8% F), another option, has a bitter, metallic taste; may burn the mucosa; and has a short shelf life. Although the tin ion in stannous fluoride may be responsible for staining the teeth, it may also be beneficial in arresting root caries. Topical fluoride agents should be applied according to the manufacturer's recommendations and always under supervision so as to limit ingestion.

Various fluoride varnishes are available and are successful in preventing caries. Varnishes provide a high uptake of the fluoride ion into enamel and are widely accepted as the vehicle of choice for fluoride delivery to young adults and older adults alike. Fluoride varnishes are professionally applied and may provide the most cost-effective means of delivery of fluoride to teeth. These varnishes are effective bactericidal and caries-preventive agents. Fluoride varnishes were developed several decades ago in an attempt to improve fluoride application techniques and benefits. European countries have used fluoride varnishes for many years. Numerous randomized clinical trials conducted outside the United States point to the efficacy and safety of fluoride varnishes as caries-preventive agents.[56-64] Fluoride varnish enables the deposition of large amounts of fluoride on an enamel surface, especially on a demineralized enamel surface. Calcium fluoride precipitates on the surface, and often fluorapatite is formed. The high concentration of surface fluoride also may provide a reservoir for fluoride, which promotes remineralization. Although additional research on fluoride varnishes is needed, the use of a fluoride varnish as a caries-preventive agent should be expanded because it has advantages over other topical fluoride vehicles in terms of safety, ease of application, and fluoride concentration at the enamel surface.[6] The American Dental Association (ADA) Council on Scientific Affairs recently endorsed the use of fluoride-containing varnishes as caries prevention agents.[65]

Current evidence indicates that fluoride varnishes with the concentration of 5% sodium fluoride are the most efficacious of all topically applied fluoride products.[58,66] For patients with a high risk of caries, fluoride varnish should be applied every 3 months. For moderate-risk patients, application every 6 months is indicated. Fluoride varnish is not indicated for low-risk patients.

When applying fluoride varnish, the clinician dries off saliva from teeth and applies a thin layer of fluoride varnish directly onto teeth. Because the fluoride varnish sets when contacting moisture, thorough isolation of the area is not required. Only tooth brushing, rather than prophylaxis, is necessary before application. The main disadvantage of fluoride varnish is that a temporary change in tooth color may occur. Patients should avoid eating for several hours and avoid brushing until the next morning after the varnish has been applied.

Self-administered fluoride rinses have an additive effect (about 20% reduction) when used in conjunction with topical or systemic fluoride treatment. Fluoride rinses are indicated in high-risk patients and patients exhibiting a recent increase in caries activity. Two varieties of fluoride rinses have similar effectiveness: (1) high dose–low frequency and (2) low dose–high frequency. High-dose (0.2% F)–low-frequency rinses are best used in supervised weekly rinsing programs based in public schools. Low-dose (0.05% F)–high-frequency rinses are best used by individual patients at home. A high-risk or caries-active patient should be advised to use the rinse daily. The optimal application time is in the evening. The rinse should be forced between teeth many times and then expectorated, not swallowed. Eating and drinking should be avoided after the rinse.

Routine use of over-the-counter fluoride containing dentifrice three times per day is recommended for all patients. These toothpastes generally contain 0.32% sodium fluoride (1450 ppm). For moderate-risk and high-risk patients 6 years or older, prescription dentifrices containing higher concentrations of fluoride are recommended. These products typically contain 1.1% sodium fluoride

(5000 ppm) and can be safely used up to three times per day in this age group.[67] For most benefit, patients should be instructed to not rinse after brushing and avoid eating or drinking for 30 minutes after use.

Silver Diamine Fluoride

Silver diamine fluoride (SDF) is a topical solution used as a caries-arresting and antihypersensitivity agent.[154,155]

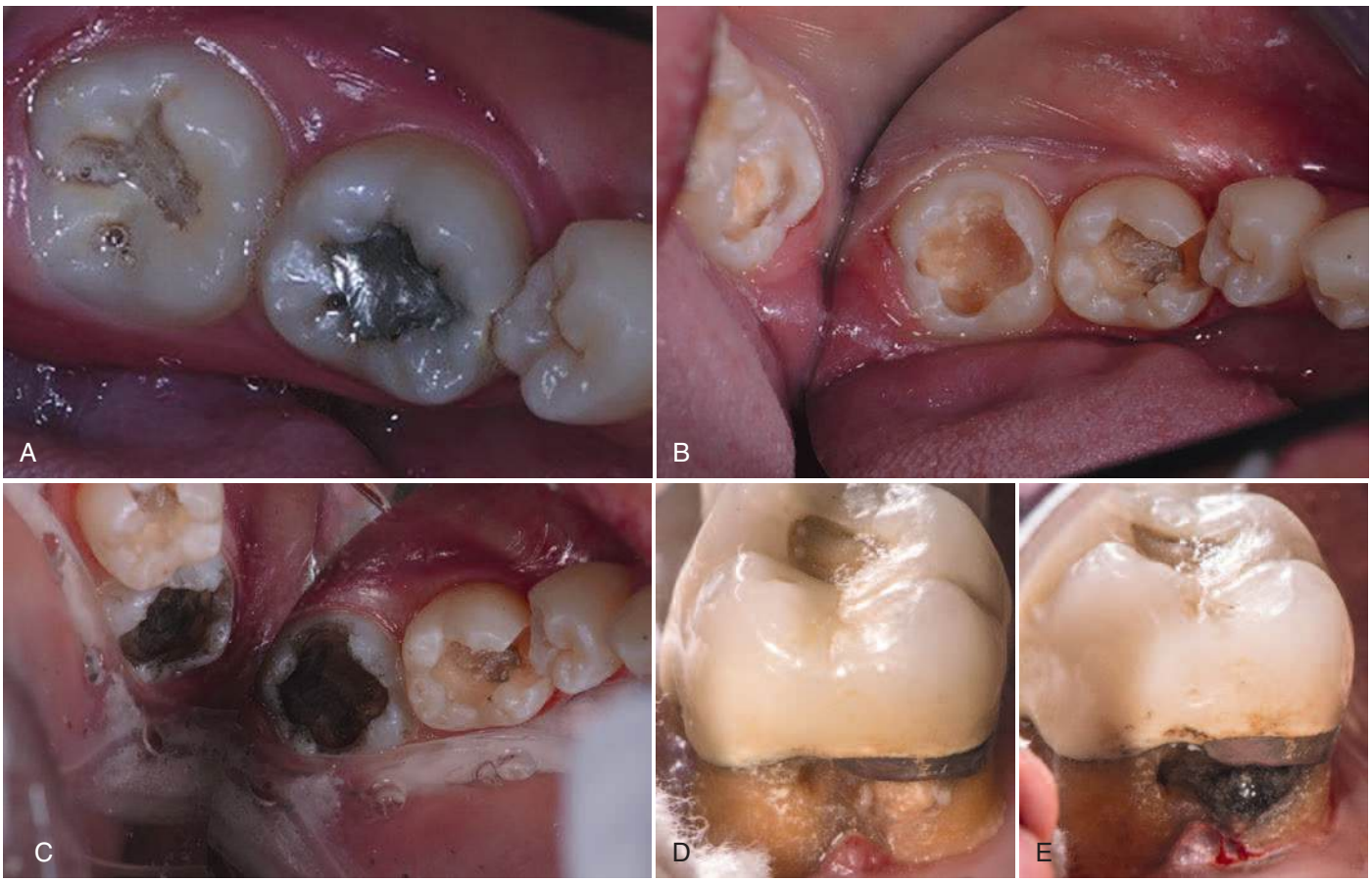
SDF was recently cleared by the US Food and Drug Administration (FDA) as an antihypersensitivity agent and is used off-label to arrest caries lesions. Both silver and fluoride play active roles in their mechanisms of arresting caries development and treatment of tooth hypersensitivity.[156-158] Silver has an antibacterial action that slows demineralization and enhances remineralization.[159] SDF, due to its ease of use, has been recommended to arrest large cavitated lesions, allowing not only conservation of tooth structure, but delivery of treatment to groups of patients that either do not have access to traditional restorative care or to whom delivery of the standard treatment is challenging. The main drawback of SDF is that with the precipitation of silver, the carious dentin becomes stained black. Some researchers have studied the use of SDF on root caries and have found it to be effective at preventing root caries lesions.[160]

Application of SDF usually only requires removal of the biofilm and application of the product with a microbrush for 3 minutes, then either rinsing the area with water or covering the lesion with fluoride varnish. Gingival tissues and lips should be covered with petroleum jelly to avoid staining. This approach may give patients who would otherwise be unable to receive treatment a low-cost alternative to arrest caries lesions and preserve their dentition (Fig. 2.43). However, the anticaries effect of SDF may be reduced over time. It has been reported that up to 50% of arrested lesions were "reactivated" by 24 months after one application of SDF.[161] The authors reported that this was likely due to the biofilm-retaining nature of these cavitated lesions. Thus there is a rationale for restoring these arrested lesions from both the caries management approach and for function and esthetics.

Immunization

For many years, investigators have been trying to develop an effective anticaries vaccine. Several prototypes have been tested in animals, but at this time neither the safety nor the efficacy of such a vaccine has been demonstrated in humans.[68,69]

Even if an anticaries vaccine was developed, some concerns remain, which may affect its widespread use. First, the potential adverse effects of a vaccine must be identified. The safety of such



• **Fig. 2.43** A, Young high caries risk patient. Tooth No. 17 with extensive caries lesion. Tooth was asymptomatic and roots were still developing. B, Unsupported enamel was removed, soft caries was left on the pulpal floor to avoid exposing the pulp. C, SDF was applied and left for 3 minutes. Darkened dentin can be observed. Tooth was restored with amalgam and remained asymptomatic. D, Tooth with PFM crown and secondary root caries on distal surface with difficult access for restoration. E, SDF applied and darkened dentin can be observed. (A–C, Courtesy Dr. Nguyen Ngoc, resident Graduate Operative Program at UNC; D and E, Courtesy Dr. Epure, resident Graduate Operative Dentistry Program at UNC.)

a vaccine has not yet been shown; concerns about a possible cross-reaction with human heart tissue remain. Second, its cost must be compared with that of public water fluoridation, which is inexpensive and already effective at reducing caries. Vaccination may be no more effective than fluoride therapy, which has a proven safety record. It may however be practical to use a caries vaccine when public water fluoridation is impractical in developed countries, and it may be useful in developing countries. Third, limitations imposed by governmental regulatory agencies may affect the widespread use of an anticaries vaccine.

Saliva

Saliva, as noted earlier in this chapter, provides an effective first line of defense against dental caries. Saliva works by diluting acid produced in biofilm, washing the acid away (swallowing), buffering the produced acid (bicarbonate + phosphate), and assisting in remineralization (calcium + phosphate). Saliva also works by forming a pellicle. When normal salivary flow rates are compromised, patients are usually at high risk for developing caries.

Normal aging does not result in reduced salivary flow rates; however, many older patients have compromised salivary flow rates as a result of medications taken for systemic conditions. Many commonly prescribed medications have hyposalivation as a potential side effect. A recent study has indicated that 63% of the 200 most commonly prescribed drugs in the United States have the adverse effect of reduced salivary flow rates.[70]

One important strategy for the prevention of caries in such patients is to increase the salivary flow rate and, by this, the concomitant buffering capacity. For these patients, gathering initial baseline data on salivary flow rates is critical. Commercial kits are available for the assessment of salivary flow. These kits provide data on simulated flow rates, the pH of saliva, and the level of buffering capacity. If saliva samples are sent for microbiologic testing, specific MS and lactobacilli counts may be determined. Specific strategies for improving flow rates and reducing bacterial counts may then be initiated. Subsequent retesting is necessary to identify the relative efficacy of the strategies.

When attempting to improve salivary flow rates, a consultation with the patient's physician may be in order. It may be possible that a less xerogenic drug be prescribed or the dose of a current drug reduced. Changing the time of taking the medication(s) may be useful. Prescribing salivary stimulants such as pilocarpine or cevimeline may be very beneficial in patients with functioning salivary glands but who have medication-induced hyposalivation.

Other strategies to improve salivary flow rates include increased water intake, use of sugar-free candies/mints several times a day, and/or use of xylitol chewing gum. Xylitol will be discussed in the next section.

Chemical Agents

Chemical agents to help modulate the biofilm have been proposed. These agents have been shown to reduce MS levels but this alone has not been shown to modify caries outcomes.[71] Prior to initiating procedures to reduce the numbers of cariogenic bacteria in the oral cavity, bacterial testing should be conducted to determine baseline microbiologic variables. Saliva samples may be tested for specific MS and lactobacilli levels; commercial devices may help evaluate the level of ATP in the biofilm. Bacterial counts or ATP levels are not reliable risk indicators but can be useful as motivational tools for patients.

Various antimicrobial agents are available (Table 2.13). As discussed, fluoride has antimicrobial effects. Two differing strategies have been suggested for reducing bacterial counts. The traditional approach is the use of chlorhexidine (CHX) mouthwash, varnish, or both, along with prescription fluoride toothpaste. When using this approach, it may be prudent to use toothpaste free from sodium laurel sulfate (SLS), which causes the foaming action in dentifrices. Although data are equivocal, evidence demonstrates that SLS reduces the ability of CHX to reduce biofilm formation.[72] Although CHX decreases MS, there is no evidence that it decreases caries incidence in the absence of fluoride.

CHX was first available in the United States as a rinse and was first used for periodontal therapy. It was prescribed as a 0.12%

TABLE 2.13 Antimicrobial Agents

	Mechanism of Action	Spectrum of Antibacterial Activity	Persistence in Mouth	Adverse Effects
Antibiotics				
Vancomycin	Blocks cell wall synthesis	Narrow	Short	Increases gram-negative flora
Kanamycin	Blocks protein synthesis	Broad	Short	Can increase caries activity
Actinobolin	Blocks protein synthesis	Streptococci	Long	Unknown
Bis Biguanides				
Alexidine	Antiseptic; prevents bacterial adherence	Broad	Long	Bitter taste; stains teeth and tongue brown; mucosal irritation
Chlorhexidine	Antiseptic; prevents bacterial adherence	Broad	Long	Bitter taste; stains teeth and tongue brown; mucosal irritation
Halogens				
Iodine	Bactericidal	Broad	Short	Metallic taste
Fluoride	1–10 parts per million (ppm) reduces acid production; 250 ppm bacteriostatic; 1000 ppm bactericidal	Broad	Long	Increases enamel resistance to caries attack; fluorosis in developing teeth with chronic high doses

rinse to high-risk patients for short-term use. It is also available as a varnish, and the most effective mode of varnish use is as a professionally applied material.[73] Chlorhexidine gluconate (0.12%) solution is effective in its ability to chelate cations and, as a result, disrupts cell adhesion, cell membrane function, and the ability of MS to uptake glucose and produce glucans and its metabolism resulting in a decrease in MS counts. Emilson concluded that CHX varnishes provide effective reduction in MS, although recent evidence is not as conclusive in favor of a CHX rinse.[74,75] Despite some favorable evidence, because there is only a weak correlation with a reduction in caries, CHX currently has very limited application as a regularly used caries management antimicrobial agent.

Xylitol is a natural five-carbon sugar obtained from birch trees. MS cannot ferment (metabolize) xylitol, so no acid is produced. Over time xylitol reduces the number of MS in the biofilm. It is usually recommended that patients chew two pieces of xylitol gum containing a total of 1 g xylitol 3 to 6 minutes after eating or snacking. Chewing any sugar-free gum after meals reduces the acidogenicity of the biofilm because chewing stimulates salivary flow, which improves the buffering of the pH drop that occurs after eating.[80] Reductions in caries rates are greater however when xylitol is used as the sugar substitute.[81]Although there is some suggestion that xylitol may reduce caries rate,[82] enhance remineralization, and help arrest dentinal caries,[78,79,162] studies in adults have not been able to confirm its role on caries control[163] except for root caries.[164] The evidence supporting the use of xylitol products for the purpose of caries reduction in both adults and children is very weak.[165]

A myriad of other chemical agents have been suggested to selectively kill gram-positive cariogenic coaggregates of bacteria. Among these are propolis, nutraceutical phenols from licorice root, arginine, and bactericidal products such as sodium hypochlorite 0.2% in the form of an oral rinse.[166-168] However, despite the effect of these strategies on decreasing cariogenic bacteria counts, they have not been shown to actually decrease caries incidence in the absence of fluoride. Caries is a biofilm disease modulated by diet; therefore changes to bacterial counts will have a short-term impact on the caries process. In the absence of other changes, for example a significant reduction in fermentable carbohydrate intake, the biofilm microbiome will continue to adapt to the acidogenic and aciduric environment caused by a highly cariogenic diet; and these antimicrobial strategies will have little impact on caries outcomes.

Calcium and Phosphate Compounds

Amorphous calcium-phosphate (ACP) products have become commercially available and reportedly have the potential to remineralize tooth structure.[83] ACP is a reactive and soluble calcium phosphate compound that releases calcium and phosphate ions to convert to apatite and remineralize the enamel when it comes in contact with saliva. Forming on the tooth enamel and within the dentinal tubules, ACP provides a reservoir of calcium and phosphate ions in the saliva.[84] Casein phosphopeptide (CPP) is a milk-derived protein that binds to the tooth's biofilm and is used to stabilize ACP. Remineralization products that use CPP as a vehicle to deliver and maintain a supersaturation state of ACP at or near the tooth surface have recently been introduced. Some of these products contain other caries-preventive agents such as fluoride and xylitol (e.g., MI Paste Plus, GC America, Alsip, IL). Gum, lozenges, and topically applied solutions containing CPP-ACP have also been reported to remineralize white spots.[85,86] Mounting evidence indicates that CPP-ACP complexes, when used regularly, are effective in enamel remineralization[87-90]; however, most of these studies have not looked at its effect independent of fluoride. The CPP-ACP products must not be prescribed to patients with allergies to dairy (milk) proteins.

Probiotics

One novel approach for reducing dental caries that has emerged in recent years is that of probiotics. The fundamental concept is to inoculate the oral cavity with bacteria that will compete with cariogenic bacteria and eventually replace them. Obviously the probiotic bacteria must not produce significant adverse effects.

A number of commercial products have been introduced and have been demonstrated to be safe in short-term studies. However, their relative level of efficacy remains unknown. It has been speculated that for the probiotic microorganisms to gain dominance, existing pathogens must first be eliminated. The concept of probiotics holds significant promise but considerably more research is required.

Sealants

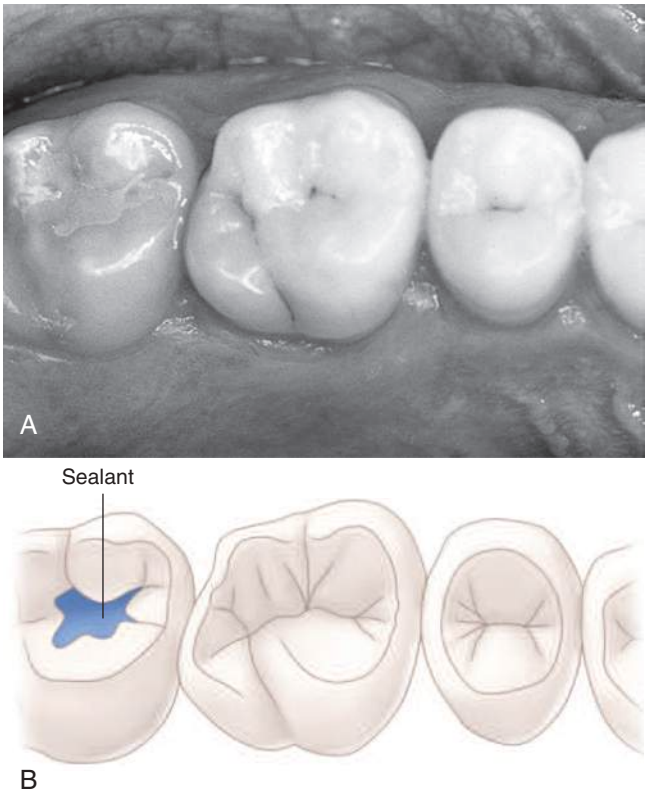
Although fluoride treatments are most effective in preventing smooth-surface caries, they are less effective in preventing pit-and-fissure caries. The use of sealants is an effective preventive treatment for caries in pit-and-fissure areas.[91] Sealants have three important preventive effects. First, sealants mechanically fill pits and fissures with a resin-based polymer. Second, because the pits and fissures are physically closed off from the oral environment with the sealant resin, MS and other cariogenic organisms no longer have access to their preferred habitat. Third, sealants render the surface of the tooth, where the pits and fissures are located, easier to clean by toothbrushing and mastication (Figs. 2.44 and 2.45).

Based on the available scientific literature, pit-and-fissure sealants are appropriate for prevention of caries in susceptible teeth and, within limits, for arresting initial caries lesions.[92-96] There is mounting evidence that sealing initial lesions is an effective means to arrest dental caries.[169]

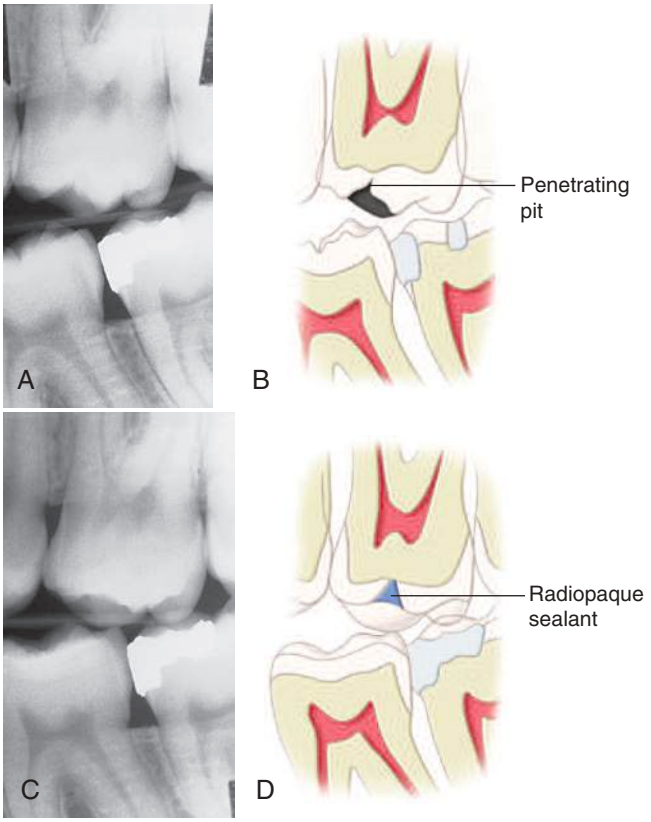
Another strategy recently introduced is the use of extremely low-viscosity resin sealants for the infiltration of white-spot caries lesions on smooth surfaces[97] (Fig. 2.46). In situ studies demonstrate the ability of resin sealants, also called *infiltrants*, to prevent further demineralization under cariogenic conditions.[98] Resin infiltrants have reportedly been used in free (i.e., facial and lingual) as well as in interproximal enamel surfaces. However, similar to the sealant technique used in pit-and-fissure areas, the technique requires careful attention to detail. A practice-based randomized clinical research trial has indicated that infiltration was more effective at preventing lesion progression than instructions on oral hygiene, flossing, and fluoride use.[170]

Restorations

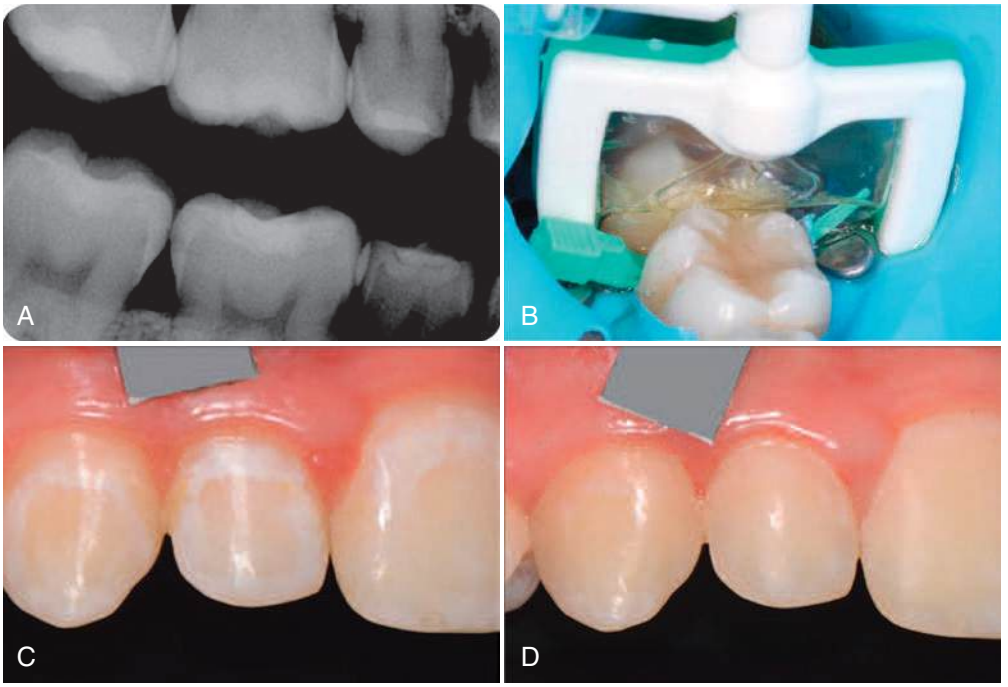
The status of a patient's existing restorations may have an important bearing on the outcome of preventive measures and caries treatment. Old restorations that are rough and plaque retentive should be smoothed and polished. If the marginal integrity is inadequate, then the restoration should be repaired or replaced. Restoration defects such as overhangs, open proximal contacts, and defective contours contribute to plaque formation and retention. These defects should be corrected, usually by replacement of the defective restoration. Detection of caries around restorations and sealants (CARS) ("secondary caries") may be difficult around old restorations. Discoloration of the enamel adjacent to a restoration suggests the potential for CARS, but is not a definitive diagnosis. This condition



• **Fig. 2.44** A and B, Sealant applied to the central fossa of a maxillary second molar. This tooth was treated because of the appearance of chalky enamel and softening in the central fossa. A highly filled sealant was used (see Fig. 2.43).



• **Fig. 2.45** A and B, Radiograph of a maxillary first molar with a deep central fossa pit that appears to penetrate to the dentin. C and D, The central pit was sealed with a highly filled, radiopaque sealant. The sealant is readily visible on the radiograph.



• **Fig. 2.46** A, Radiographic image of first mandibular molar with a proximal lesion on distal surface. B, Tooth was infiltrated according to manufacturer's instructions (Icon, DMG America). C and D, Clinical images of smooth-surface lesions on anterior teeth before and after infiltration, with Icon. (A, Courtesy Dr. Andrea Cortes, Operative Dentistry Specialization Program, El Bosque University. B, Courtesy Dr. Olga Lucia Zarta (chair) and Dr. Ainimsay Benitez (resident), Operative Dentistry Specialization Program, Dental School, El Bosque University, Bogotá, Colombia. C and D, Reprinted with permission from Meyer-Lueckel H, Paris, *Zahnmedizin up2date* 267–290, 2017.)

appears as a localized opalescent area next to the restoration margins. (*Exception:* A bluish color of facial or lingual enamel that directly overlies an old, otherwise acceptable amalgam restoration does not indicate replacement unless there is a need for improvement of esthetics. Such a discoloration may be caused by the amalgam itself.) Because metallic restorations are radiopaque, the radiolucency of CARS may be masked. The restoration of cavitated active caries lesions is preventive only because it removes large numbers of cariogenic organisms and some of the sites in which they may be protected, and it restores adequate tooth contours and surface smoothness that may facilitate oral hygiene. As emphasized earlier, the placement of a restoration into a cavitated carious tooth does not cure the caries process.

Although caries diagnostic and preventive measures have been improved and are more widely used, the repair of destruction caused by the caries process still is necessary for many patients. The treatment regimen is dictated by the patient's caries status. If the patient is at high risk for caries development, treatment should consist of restorative procedures concomitant with many of the preventive measures described previously. The damage done by caries may be repaired while at the same time seeking to lower the patient's risk status for further caries development. Sometimes, patients present with active caries lesions in numerous teeth. Because these teeth may be in jeopardy and because of the large numbers and sites of cariogenic bacteria, arresting medication for caries control (e.g., SDF) or restorative treatment for caries control may be indicated, as described later in this section. This procedure rapidly eliminates the caries lesions, providing better assessment of the pulpal responses of some teeth and greater success of the preventive measures instituted. Later, teeth may be restored with more definitive restorations.

Once caries has produced cavitation of the tooth surface, unless the cariogenic biofilm can be removed from a cavitated caries lesion or the lesion has been treated with SDF, preventive measures will be inadequate to prevent further progression of the lesion. Excision of the lesion (tooth preparation) and proper tooth restoration will not only stop the progression of active, cavitated lesions but also restore tooth function and esthetics.

Clinical Considerations for Caries Removal

Under normal operative clinical circumstances, in a vital permanent tooth with no symptoms or signs of pulpal pathology, the extent of the dentin caries removal (excavation) is dictated primarily by lesion severity and depth:

- Moderate lesions (lesions not reaching the inner one third of dentin and with no anticipated risk of pulp exposure) should be excavated to a caries-free DEJ and firm dentin.
- Advanced (deep) lesions (lesions reaching the inner one third of dentin and with anticipated risk of pulp exposure) should be excavated to a caries-free DEJ and soft dentin, following a selective caries removal (SCR) protocol.[171,172]

For a complete description of current caries classification criteria based on lesion severity and activity, please see Chapter 3. The SCR protocol is a professionally recognized and accepted tooth-level caries control treatment,[173-176] and it can be used on any tooth with an advanced (deep) caries lesion that is deemed restorable, and for which the pulpal and periapical areas are deemed healthy. Selective caries removal consists of complete caries removal peripherally to a sound, caries-free DEJ; axially and pulpally, caries is removed to within approximately 1 mm of the pulp within soft dentin; a glass ionomer (e.g., Fuji IX, GC America, Alsip, IL) temporary restoration or a definitive restoration is then placed. Growing evidence suggests that temporization followed by reentry does not contribute to improved clinical outcomes[27, 28]; therefore current research supports the placement of a definitive restoration.[30,177-181] Selective caries removal allows a restoration to be placed while avoiding pulpal exposure.[26] Avoiding a pulpal exposure has a great impact on the lifetime prognosis of the tooth and long-term treatment costs.[182] Although the residual dentin thickness cannot be accurately assessed clinically, its preservation is a significant factor in avoiding pulpal distress.[183,184]

In slowly advancing lesions, it is expedient to remove the soft (infected) dentin until the readily identifiable hard (sclerotic) dentin is reached. In rapidly advancing lesions (see Figs. 2.36, 2.37, and 2.43A), little clinical evidence (as determined by texture or color change) exists to indicate the extent of the soft (infected) dentin. For deep lesions, this lack of clinical evidence may result in an excavation that risks pulp exposure. In a tooth with a deep advanced caries lesion, no history of spontaneous pain, normal responses to thermal stimuli, and a vital pulp, a deliberate, selective caries lesion removal (as noted) may be indicated. This procedure is supported by a large body of evidence.[26-34,99-123,185-192]

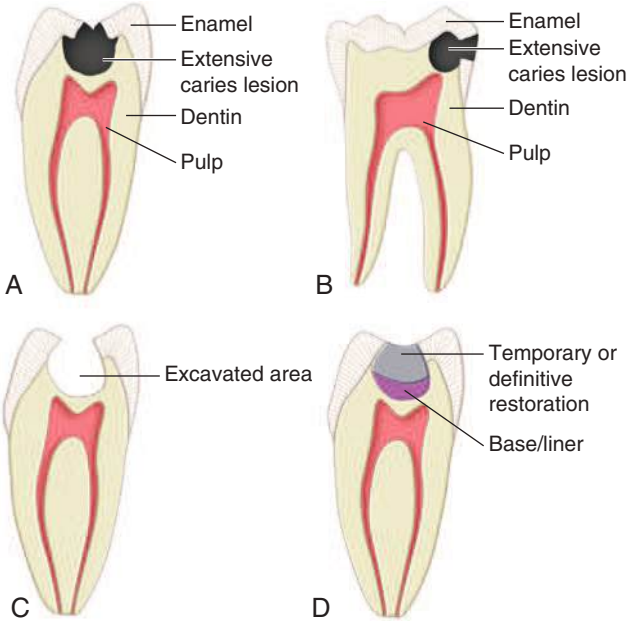
Removal of the bacterial infection has been seen as an essential part of all operative procedures; however, even removal of dentin up to hard dentin in deep, advanced caries lesions does not ensure a "sterile" dentin as bacteria have been found to be present in all dentinal layers in deep caries lesions. Nevertheless, even when bacteria are present, compounding evidence indicates that when a good seal is present the lesion will arrest[193-196] (see Fig. 2.18E–L). Therefore it is not necessary to remove all of the dentin that has been compromised by the caries process. Although caries detection solutions such as 1% acid red 52 (acid rhodamine B or food red 106) in propylene glycol[35] have been developed to help stain the infected layer, these dyes bind and stain the demineralized dentin matrix and do not stain bacteria exclusively. Complete removal of all stained tooth structure in the preparation therefore ultimately leads to significantly larger preparations than the traditional visual-tactile method of evaluating for caries removal, so their use is no longer recommended.[197-199]

The following are additional clinical considerations for caries lesion removal:

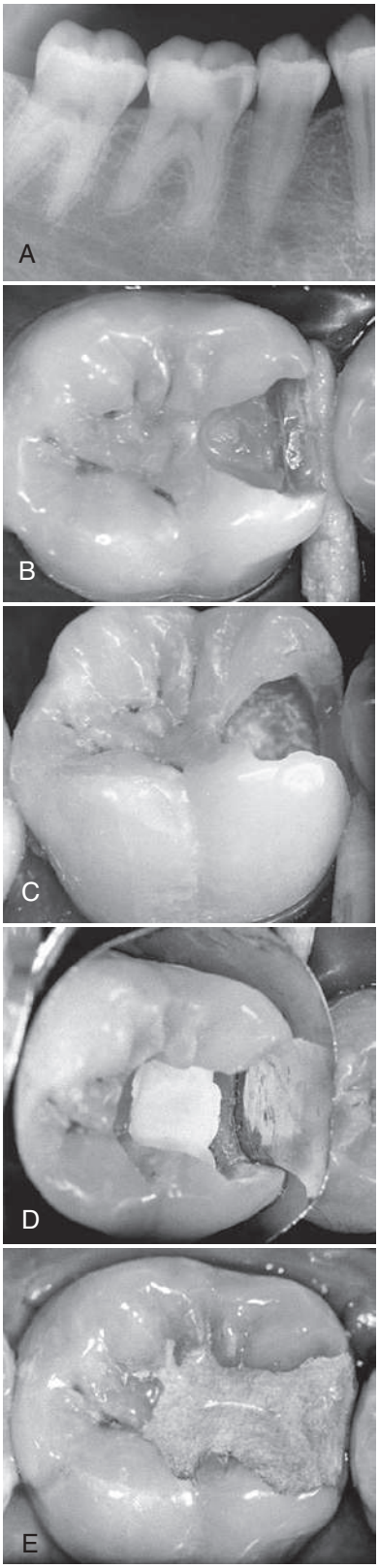
a. Regardless of which protocol is used for caries removal (caries removal to firm dentin or selective caries removal to soft dentin), the tooth may be restored with a final restoration (Figs. 2.47 and 2.48). However, the patient should be clearly informed that in the treatment of advanced (deep) dentin caries some leathery and soft dentin may remain under the restoration. This remaining caries-affected dentin has many implications, including higher risk for endodontic complications[205] (not because "caries was left under the restoration" but because deep caries was present to begin with) and a radiographic presentation that may suggest secondary or residual caries.
b. Although there is overwhelming scientific evidence in support of the SCR protocol, this is still a controversial clinical procedure in certain countries[32] with potential liability implications as well as implications for clinical board examinations, most of which still require complete caries removal (excavation) to hard dentin as a criteria for a successful clinical board examination.
c. The patient should be clearly informed of the risks and benefits of the selective caries removal to soft dentin protocol, and provide informed consent. If the patient is not willing to accept the risks, then the alternative, either complete caries removal with a higher risk of complications (endodontic therapy,

postoperative sensitivity) or tooth extraction, should be presented.

d. A sealed tooth-restoration interface is critical for the success of the restorative procedure (see Fig. 2.18E–L). When a proper seal is NOT obtained, or when/if the seal is compromised, marginal leakage and ingress of bacteria/fluids/nutrients will allow lesion progression. This is particularly critical when soft dentin remains under the restoration.
e. When excavating advanced (deep) dentin caries, if a caries-free DEJ cannot be obtained, a protective restoration (sedative filling, temporary restoration) should be used and tooth restorability should be reassessed. A permanent restoration should not be used unless a sound DEJ (or peripheral dentin margin) is reached (see Figs. 2.47 and 2.48).
f. In slowly advancing lesions, it is practical to remove the soft (infected) dentin until firm (demineralized, affected) dentin is reached. Despite being demineralized, with adequate marginal seal, the lesion may arrest and the affected dentin may remineralize. This allows for the most conservative tooth preparation without jeopardizing the restoration.
g. The pulpal diagnoses outlined here rely on signs and symptoms of pulp pathology using the best diagnostic tools available. However, actual pulpal status is difficult to determine



• **Fig. 2.47** Schematic representation of caries-control procedure. A and B, Faciolingual (A) and mesiodistal (B) cross sections of mandibular first molar show extensive preoperative occlusal and proximal caries lesions. C, Tooth after excavation of extensive caries. Note remaining unsupported enamel. D, Temporary amalgam restoration inserted after appropriate liner or base.



• **Fig. 2.48** A, Preoperative clinical radiograph illustrating extensive caries lesion in the proximal and occlusal regions of the mandibular right first molar. Initial caries excavation of tooth. B, Remaining caries requires further excavation. Also, note the wedge in place, protecting rubber dam and soft tissue; it has been lightly shaved by a bur. C, Remaining unsupported enamel under mesiolingual cusp. D, Tooth ready for placement of temporary restoration. Carious involvement required further extension than that seen in *B* and *C*. Liner or base material has been applied to deepest excavated areas, and matrix, appropriately wedged, has been placed. E, Temporary restoration completed for caries-control procedure. Caries has been eliminated, the pulp adequately protected, and interarch and intraarch positions of tooth maintained by caries-control procedure.

clinically—bacteria and toxins progressing ahead of caries may cause areas of undetectable and asymptomatic pulpal necrosis or irreversible pulpitis.

h. Use of calcium hydroxide (or other liner/base materials) after caries removal and before use of a protective or final restoration does not appear to improve outcomes.[200]
i. Teeth that are restorable only by use of full cuspal coverage restoration generally are not appropriate for the SCR technique because of the difficulty of evaluating the tooth for possible failures such as continuing caries activity under the full coverage restoration. Another concern is the cost of rectifying failures.
j. This protocol recognizes that exposures will occur despite admonitions to leave soft and leathery (infected) dentin rather than expose the pulp. Whenever there is a pulp exposure, the treatment should be consistent with existing endodontic guidelines for vital pulp exposures.
k. Advances in vital pulp therapy may allow successful endodontic therapy of pulp exposures without the need for root canal treatment. With the advent of tricalcium silicates, the prognosis of pulp capping (partial and full pulpotomy of permanent mature teeth) has tremendously improved. Recent observational and randomized trials, as well as systematic reviews and meta-analyses, have shown that these procedures have success for pulp vitality that exceeds 90%.[201-203] Complete caries removal with the option of vital pulp therapy should be one of the options offered to the patient during the informed consent process.
l. The SCR protocol can be applied either on a single tooth being treated or to several teeth in a caries control phase treatment. High caries risk patients with multiple cavitated caries lesions can benefit from a treatment plan designed specifically to gain control over the rapid progression of their disease. The term *caries control* refers to a procedure in which multiple teeth with acute threatening caries are treated quickly by (1) applying a medication (e.g., SDF) to arrest the caries lesions or (2) removing the infected tooth structure, medicating the pulp if necessary, and restoring the defect(s) with a temporary or definite material. The intent of caries-control procedures is immediate, corrective intervention of advanced caries lesions so as to prevent pulpal disease and avoid possible sequelae such as toothache, root canal therapy, or more complex ultimate restorations.

   This allows the caries process in these cavitated lesions to be essentially arrested and the overall bacterial load to be reduced. Caries-control procedures must be accompanied by other preventive measures (Table 2.14). Teeth rapidly treated by caries-control procedures subsequently may be treated by using routine restorative techniques if appropriate pulpal responses are obtained.

TABLE 2.14 Caries-Control Restoration

Initial treatment	Caries risk assessment Education and motivation Thorough evaluation and documentation of lesions Temporization of all large cavitated lesions by caries-control restorations Specific nonrestorative, therapeutic treatment (see Table 2.10) Plaque control (see Table 2.10, technique C) Dietary control (see Table 2.10, technique A)
Preliminary assessment	Gingival response as a marker of plaque biofilm control effectiveness Arrest of cavitated lesions with SDF as indicated Pulpal response of teeth with selective caries-excavation and restoration Assessment of patient compliance with medications, oral hygiene, and dietary control measures
Follow-up care	Careful clinical evaluation of teeth Replacement of caries-control restorations with permanent restorations Monitoring of plaque biofilm and mutans streptococci (MS) levels Further antimicrobial treatment and dietary reassessment as indicated by new cavitations, noncavitated lesions, or high MS levels

Root Caries Management

It is clear that the baby boom generation of North America is aging. In the year 1900, 3% of the US population was over 60 years of age, whereas in the year 2000, 13% of the population was over 60 years.[124] In the year 2030, it is estimated that at least 20% of the population will be 60 years or older. Root caries is a pervasive problem in a high percentage of older patients.[125,126] Many of these patients have had extensive restorative dentistry done in their lifetimes. Approximately 38% of patients between the ages of 55 and 64 years have root caries, and 47% of those between 65 and 74 years have experienced root caries.[127] The incidence of root caries in old-older adults (over 75 years) is even higher.[128]

One of the primary etiologic factors for these patients is their use of prescription drugs for a wide variety of systemic medical problems. It has been estimated that 63% of the 200 most commonly prescribed medications have dry mouth as an adverse effect. It is the subsequent reduction in salivary flow rates and concomitant diminished buffering capacity resulting from use of these medications that is primarily responsible for the increase in root caries in older patients.

The critical pH of dentin (pH at which dentin begins to demineralize) is between 6.2 and 6.7, whereas that of enamel is about 5.5.[129] As a result, root dentin will demineralize in very weak acids, and root caries progresses at about twice the rate of coronal caries. Thus it is critical that all older patients receive thorough clinical and radiographic examinations on a regular basis.

As described previously in this chapter, a caries risk assessment should be carried out for all older patients. Factors that increase the risk for root caries include the following:

1. Gingival recession
2. Poor oral hygiene
3. Cariogenic diet
4. Presence of multiple restorations or multiple missing teeth
5. Existing caries
6. Xerogenic medications
7. Compromised salivary flow rates

Once it has been determined that a patient is at high risk for root caries, an aggressive preventive protocol should be considered. This protocol is based upon four primary strategies for the prevention of root caries. The *first* strategy is to improve salivary flow rates and increase the buffering capacity. The *second* strategy is to modulate a cariogenic biofilm in the oral cavity. The *third* strategy is to reduce the quantity and numbers of exposures of ingested refined

carbohydrates, and the *fourth* is to attempt to remineralize initial lesions and prevent new lesions from developing. In addition to following the aforementioned protocol, three additional considerations are important:

1. Use of powered toothbrushes and irrigation devices. It is critical that patients susceptible to root caries practice meticulous oral hygiene. However, many of these patients have physical and visual deficiencies, which makes it difficult for them to adequately cleanse the mouth. For these patients, a powered toothbrush may be advantageous.[207] Additionally, daily use of a water irrigation device (Waterpik, Water Pik Inc., Fort Collins, CO) may be beneficial. Although the device will not remove biofilm, studies have shown that daily use will change the composition of the biofilm in a beneficial way. However, since the toothbrush essentially provides the means of fluoride delivery, use of a water irrigation device alone is not indicated unless used with a fluoridated rinse (e.g., ACT).
2. Restoration of all root caries lesions with a fluoride-releasing material. Resin-modified glass ionomer materials are preferred for definitive restorations of active root caries lesions primarily because they bond effectively to both enamel and dentin and they act as reservoirs for fluoride, which may be re-released into the oral cavity.[130,131] These materials may be effective as anticaries materials only if patients reload the material a minimum of three times a day by brushing with fluoride-containing toothpaste or by using other fluoride-containing products.[208] Alternatively these same root caries lesions can be arrested by application of SDF (see Fig. 2.43D and E).
3. Educating patients of the necessity for three exposures to fluoride per day and for reloading the fluoride-releasing materials can assist in motivating them to improved levels of compliance.[208]

In summary, many older patients are experiencing an epidemic of root caries, primarily as a result of the hyposalivatory effects of medications prescribed for systemic illnesses. Many root caries lesions occur in locations that make them difficult if not impossible to restore. The dental profession has a strong track record of prevention, and it is clear that with root caries, prevention is much better than restoration.

Summary

Caries management efforts must be directed not at the tooth level (traditional or surgical treatment) but at the total-patient level (medical model of treatment). Restorative treatment does not cure the caries process. Instead, identifying and eliminating the causative factors for caries must be the primary focus, in addition to the restorative repair of damage caused by caries.

Much of the remainder of this textbook presents information on when and how to restore tooth defects. Many tooth defects are the result of caries activity. Only implementation of appropriate caries-preventive measures will lower the probability that caries lesions will recur. Tooth restorations are preventive in the sense that they remove numerous cariogenic organisms in the affected site and eliminate a protected habitat for other cariogenic bacteria; however, they primarily repair the tooth damage caused by caries and have only a limited impact on the patient's overall caries risk.

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